

Epoxy mortar FIS EM Plus

The powerful injection mortar for rebar connections and cracked concrete

2



Rail fastenings



Rebar connections

Applications

- Post-installed rebar connections
- Joints for concrete layers
- Rim beam anchorings
- Seismic applications
- Anchorings in diamond-drilled or in waterfilled drill holes
- Heavy steel constructions
- Silo installations
- Tall shelving
- Sound barriers
- Temporary or removable fixings (with internal threaded anchor RG M I)

Advantages

- The ETA assessment guarantees a service life of 100 years. The expert report of IEA Stuttgart even confirms a working life up to 120 years and thus underlines the reliability and durability of the FIS EM Plus.
- The optimised formulation of the epoxy resin mortar FIS EM Plus leads to improved load values in cracked and non-cracked concrete.
- The mortar can be used for rebar connections from diameter 8 to 40, including

seismic applications.

- With the threaded rod FIS A, the loads to be applied can be designed variably by selecting the anchorage depth. The mortar is approved for diamond drilled and water-filled drillholes as well as seismic applications and thus offers safety under extreme conditions.
- For practical use on the building site, FIS EM Plus can be processed at low temperatures down to -5 °C.

Certificates



ETA-17/0979, for cracked concrete
ETA-17/1056, Post-installed rebar connections



See ICC-ES Evaluation Report at www.icc-es.org



Fire resistance classification R240



Building materials

Approved for anchorings in:

- Concrete C20/25 to C50/60, cracked and non-cracked

Also suitable for:

- Natural stone with dense structure

Functioning

- The epoxy mortar FIS EM Plus combined with the FIS A threaded rod, is suitable for pre-positioned and push-through installation.
- Resin and hardener are stored in two separate chambers and are not mixed and activated until extrusion through the injection capsule in the static mixer.
- The mortar is injected bubble-free from the drill hole base.
- The mortar bonds the entire surface of the anchor with the drill hole wall and seals off the drill hole.
- The anchor is set manually by lightly rotating it until it reaches the drill hole base.
- During push-through installation, the annular gap between the threaded rod and attachment is filled with FIS EM Plus.

For use with

Anchors & sleeves page

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Dispenser page 87

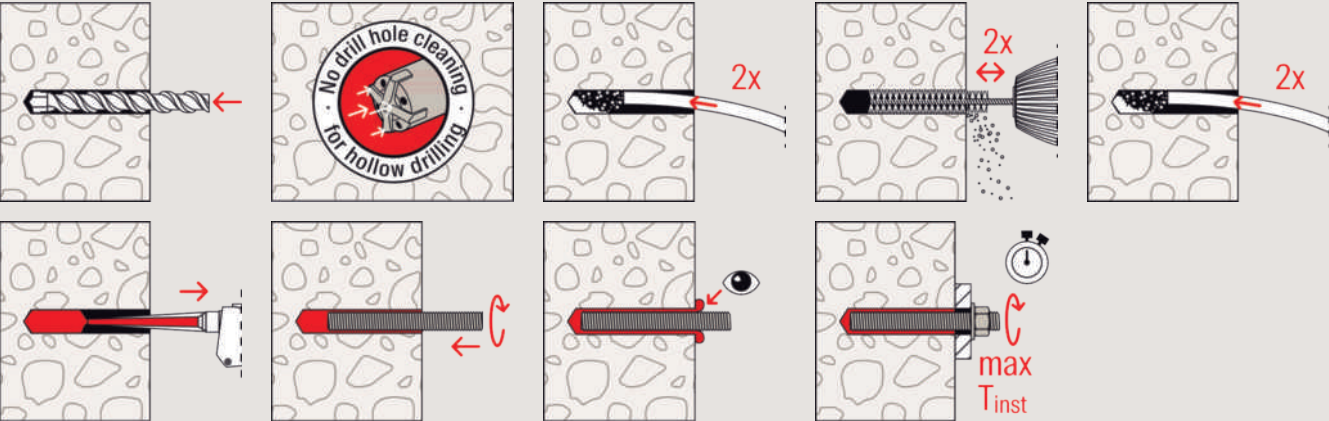


Accessories page

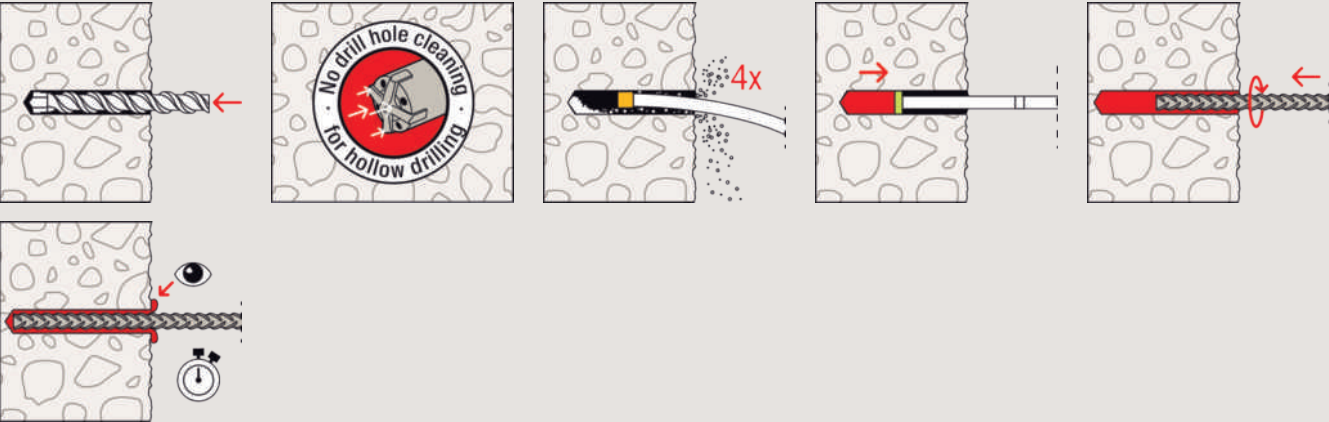
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Installation in concrete with FIS EM Plus and FIS A / RG M



Installation of post-installed rebar connections in concrete with FIS EM Plus in hammer drilled holes



Technical data

Epoxy mortar FIS EM Plus



FIS EM Plus 390 S

FIS EM Plus 585 S

FIS EM Plus 1500 S

Item	Item No.	Approval		Languages on the cartridge	Scale unit	Contents	Sales unit [pcs]
		ETA	ICC				
FIS EM Plus 390 S	544154 ¹⁾	●	●	DE, EN, FR, NL, ES, PT	180	1 cartridge 390 ml, 2 x FIS MR Plus	6
FIS EM Plus 585 S	544165 ¹⁾	●	●	EN, ZH, RU, KO, CS, PL	270	1 cartridge 585 ml + 2 x FIS UMR	6
FIS EM Plus 1500 S	544173 ¹⁾	●	●	EN, ES, PT, ZH, RU, PL	700	1 cartridge 1500 ml, 2 x FIS UMR	4

1) Dangerous goods - no express shipping possible.

Technical data

Static mixer



FIS MR Plus

FIS UMR

Item	Item No.	Contents	Sales unit [pcs]
FIS MR Plus	545853	10 static mixer FIS MR Plus	10
FIS UMR	520593	10 static mixer for 585 ml and 1500 ml cartridges	10

Technical data

Epoxy mortar FIS EM Plus



FIS EM Plus 390 S in bucket

Item	Item No.	Approval		Languages on the cartridge	Contents	Sales unit [pcs]
		ETA	ICC			
FIS EM Plus 390 S in bucket	544172 ¹⁾	●	●	DE, EN, FR, NL, ES, PT	20 cartridges 390 ml, 20 x FIS MR Plus	1

1) Dangerous goods - no express shipping possible.

Curing times

FIS EM Plus Temperature in anchoring base [°C]	Maximum processing time t_{work} [min.]	Minimum curing time ¹⁾ t_{cure} [hrs.]
-5 – 0	240	200
> 0 – +5	150	90
> +5 – +10	120	40
> -10 – +20	30	18
> +20 – +30	14	10
> +30 – +40	7	5

1) In wet concrete or water filled holes the curing times must be doubled.

Loads

Injection system FIS EM Plus with internal threaded anchor RG M I

Permissible loads of a single anchor^{1) 2)} in normal concrete of strength class C20/25.
For the design the complete current assessment ETA-17/0979 has to be considered.

Type	Screw material ³⁾	Effective anchorage depth h_{ef} [mm]	Minimum member thickness h_{min} [mm]	Maximum installation-torque $T_{inst,max}$ [Nm]	Cracked concrete				Non-cracked concrete			
					Permissible tension (N_{perm}) and shear loads (V_{perm}); minimum spacing (s_{min}) and edge distances (c_{min}) with reduced loads				Permissible tension (N_{perm}) and shear loads (V_{perm}); minimum spacing (s_{min}) and edge distances (c_{min}) with reduced loads			
					$N_{perm}^{4)}$ [kN]	$V_{perm}^{4)}$ [kN]	$s_{min}^{4)}$ [mm]	$c_{min}^{4)}$ [mm]	$N_{perm}^{4)}$ [kN]	$V_{perm}^{4)}$ [kN]	$s_{min}^{4)}$ [mm]	$c_{min}^{4)}$ [mm]
RG M8 I	5.8	90	120	10	9.0	5.3	55	55	9.0	5.3	55	55
	8.8	90	120	10	11.3	8.3	55	55	13.8	8.3	55	55
	R-70	90	120	10	9.9	5.9	55	55	9.9	5.9	55	55
RG M10 I	5.8	90	130	20	12.9	8.3	65	65	13.8	8.3	65	65
	8.8	90	130	20	12.9	13.3	65	65	20.0	13.3	65	65
	R-70	90	130	20	12.9	9.3	65	65	15.7	9.3	65	65
RG M12 I	5.8	125	170	40	20.2	12.1	75	75	20.5	12.1	75	75
	8.8	125	170	40	20.2	19.3	75	75	32.4	19.3	75	75
	R-70	125	170	40	20.2	13.5	75	75	22.5	13.5	75	75
RG M16 I	5.8	160	210	80	33.2	22.4	95	95	37.6	22.4	95	95
	8.8	160	210	80	33.2	30.9	95	95	47.4	30.9	95	95
	R-70	160	210	80	33.2	25.1	95	95	42.0	25.1	95	95
RG M20 I	5.8	200	260	120	46.4	35.4	125	125	58.6	35.4	125	125
	8.8	200	260	120	46.4	51.4	125	125	66.3	51.4	125	125
	R-70	200	260	120	46.4	39.4	125	125	66.7	39.4	125	125

¹⁾ Design according to EN 1992-4:2018 (for static resp. quasi-static loads). The partial safety factors for material resistance as regulated in the ETA as well as a partial safety factor for load actions of $\gamma_L = 1.4$ are considered. As a single anchor counts e.g. an anchor with a spacing $s \geq 3 \times h_{ef}$ and an edge distance $c \geq 1.5 \times h_{ef}$. Accurate data see ETA.

²⁾ The specified loads are valid for anchorages in dry and damp concrete. For temperatures in the anchoring substrate up to 50 °C (resp. short term up to 72 °C). Drill hole cleaning as per specification in the ETA. The factor Ψ_{sus} for sustained load was taken into account with 1.0.

³⁾ Further steel grades, versions and technical data see ETA, e.g. for dry internal conditions, galvanised steel (gvz); for damp interiors and for outdoor use, stainless steel (R).

⁴⁾ In the case of combinations of tension and shear loads, bending moments with reduced or minimum spacing and edge distances (anchor groups), the design must be carried out in accordance with the provisions of the complete ETA and the provisions of the EN 1992-4:2018. We recommend using our anchor design software C-FIX.

Loads

Injection system FIS EM Plus with threaded rod FIS A resp. RG M

Permissible loads of a single anchor^{1) 2)} in normal concrete of strength class C20/25.
For the design the complete current assessment ETA-17/0979 has to be considered.

Type	Material/ surface ³⁾	Effective anchorage depth h_{ef} [mm]	Minimum member thickness h_{min} [mm]	Maximum installation- torque $T_{inst,max}$ [Nm]	Cracked concrete				Non-cracked concrete			
					Permissible tension (N_{perm}) and shear loads (V_{perm}); minimum spacing (s_{min}) and edge distances (c_{min}) with reduced loads				Permissible tension (N_{perm}) and shear loads (V_{perm}); minimum spacing (s_{min}) and edge distances (c_{min}) with reduced loads			
					$N_{perm}^{4)}$ [kN]	$V_{perm}^{4)}$ [kN]	$s_{min}^{4)}$ [mm]	$c_{min}^{4)}$ [mm]	$N_{perm}^{4)}$ [kN]	$V_{perm}^{4)}$ [kN]	$s_{min}^{4)}$ [mm]	$c_{min}^{4)}$ [mm]
FIS A M 8	5.8	60	100	10	5.4	6.3	40	40	9.0	6.3	40	40
	5.8	80	110	10	7.2	6.3	40	40	9.0	6.3	40	40
	5.8	160	190	10	9.0	6.3	40	40	9.0	6.3	40	40
	R-70	60	100	10	5.4	6.0	40	40	9.9	6.0	40	40
	R-70	80	110	10	7.2	6.0	40	40	9.9	6.0	40	40
	R-70	160	190	10	9.9	6.0	40	40	9.9	6.0	40	40
FIS A M 10	5.8	60	100	20	6.7	9.7	45	45	10.9	9.7	45	45
	5.8	90	120	20	10.1	9.7	45	45	13.8	9.7	45	45
	5.8	200	230	20	13.8	9.7	45	45	13.8	9.7	45	45
	R-70	60	100	20	6.7	9.2	45	45	10.9	9.2	45	45
	R-70	90	120	20	10.1	9.2	45	45	15.7	9.2	45	45
	R-70	200	230	20	15.7	9.2	45	45	15.7	9.2	45	45
FIS A M 12	5.8	70	100	40	9.6	14.3	55	45	13.7	14.3	55	45
	5.8	110	140	40	17.8	14.3	55	45	20.5	14.3	55	45
	5.8	240	270	40	20.5	14.3	55	45	20.5	14.3	55	45
	R-70	70	100	40	9.6	13.7	55	45	13.7	13.7	55	45
	R-70	110	140	40	17.8	13.7	55	45	22.5	13.7	55	45
	R-70	240	270	40	22.5	13.7	55	45	22.5	13.7	55	45
FIS A M 16	5.8	80	120	60	11.7	23.5	65	50	16.8	26.9	65	50
	5.8	125	170	60	22.9	26.9	65	50	32.7	26.9	65	50
	5.8	320	360	60	37.6	26.9	65	50	37.6	26.9	65	50
	R-70	80	120	60	11.7	23.5	65	50	16.8	25.2	65	50
	R-70	125	170	60	22.9	25.2	65	50	32.7	25.2	65	50
	R-70	320	360	60	42.0	25.2	65	50	42.0	25.2	65	50
FIS A M 20	5.8	90	140	120	14.0	28.0	85	55	20.0	40.0	85	55
	5.8	170	220	120	36.3	42.3	85	55	51.9	42.3	85	55
	5.8	400	450	120	58.6	42.3	85	55	58.6	42.3	85	55
	R-70	90	140	120	14.0	28.0	85	55	20.0	39.4	85	55
	R-70	170	220	120	36.3	39.4	85	55	51.9	39.4	85	55
	R-70	400	450	120	65.7	39.4	85	55	65.7	39.4	85	55
FIS A M 24	5.8	96	160	150	15.4	30.8	105	60	22.0	44.1	105	60
	5.8	210	270	150	49.9	60.6	105	60	71.3	60.6	105	60
	5.8	480	540	150	84.3	60.6	105	60	84.3	60.6	105	60
	R-70	96	160	150	15.4	30.8	105	60	22.0	44.1	105	60
	R-70	210	270	150	49.9	56.8	105	60	71.3	56.8	105	60
	R-70	480	540	150	94.3	56.8	105	60	97.3	56.8	105	60
FIS A M 30	5.8	120	190	300	21.6	43.1	140	80	30.8	61.6	140	80
	5.8	280	350	300	76.8	96.0	140	80	109.8	96.0	140	80
	5.8	600	670	300	133.8	96.0	140	80	133.8	96.0	140	80
	R-70	120	190	300	21.6	43.1	140	80	30.8	61.6	140	80
	R-70	280	350	300	76.8	90.2	140	80	109.8	90.2	140	80
	R-70	600	670	300	150.1	90.2	140	80	150.1	90.2	140	80

¹⁾ Design according to EN 1992-4:2018 (for static resp. quasi-static loads). The partial safety factors for material resistance as regulated in the ETA as well as a partial safety factor for load actions of $\gamma_L = 1.4$ are considered. As a single anchor counts e.g. an anchor with a spacing $s \geq 3 \times h_{ef}$ and an edge distance $c \geq 1.5 \times h_{ef}$. Accurate data see ETA.

²⁾ The specified loads are valid for anchorages in dry and damp concrete. For temperatures in the anchoring substrate up to 50 °C (resp. short term up to 72 °C). Drill hole cleaning as per specification in the ETA. The factor ψ_{sus} for sustained load was taken into account with 1.0.

³⁾ Further steel grades, versions and technical data see ETA, e.g. for dry internal conditions, galvanised steel (gvz); for damp interiors and for outdoor use, stainless steel (R).

⁴⁾ In the case of combinations of tension and shear loads, bending moments with reduced or minimum spacing and edge distances (anchor groups), the design must be carried out in accordance with the provisions of the complete ETA and the provisions of the EN 1992-4:2018. We recommend using our anchor design software C-FIX.

Approval body for construction products
and types of construction

Bautechnisches Prüfamt

An institution established by the Federal and
Laender Governments



European Technical Assessment

ETA-22/0001
of 31 July 2023

English translation prepared by DIBt - Original version in German language

General Part

Technical Assessment Body issuing the
European Technical Assessment:

Deutsches Institut für Bautechnik

Trade name of the construction product

fischer injection system FIS EM Plus

Product family
to which the construction product belongs

Post-installed reinforcing bar (rebar) connections with
improved bond-splitting behaviour under static loading
and seismic action

Manufacturer

fischerwerke GmbH & Co. KG
Otto-Hahn-Straße 15
79211 Denzlingen
DEUTSCHLAND

Manufacturing plant

fischerwerke

This European Technical Assessment
contains

20 pages including 3 annexes which form an integral part
of this assessment

This European Technical Assessment is
issued in accordance with Regulation (EU)
No 305/2011, on the basis of

332402-00-0601-v02, Edition 10/2022

This version replaces

ETA-22/0001 issued on 8 June 2022

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Specific Part

1 Technical description of the product

The subject of this European technical assessment is the post-installed connection, by anchoring or overlap connection joint, of reinforcing bars (rebars) in existing structures made of normal weight concrete, using the fischer injection system FIS EM Plus in accordance with the regulations for reinforced concrete construction.

Reinforcing bars with a diameter ϕ from 8 to 40 mm according to Annex A and the injection mortar FIS EM Plus are used for the post-installed rebar connection. The rebar is placed into a drilled hole filled with injection mortar and is anchored via the bond between embedded reinforcing bar, injection mortar and concrete.

The product description is given in Annex A.

2 Specification of the intended use in accordance with the applicable European assessment Document

The performances given in Section 3 are only valid if the rebar connection is used in compliance with the specifications and conditions given in Annex B.

The verifications and assessment methods on which this European Technical Assessment is based lead to the assumption of a working life of the rebar connections of at least 50 and/or 100 years. The indications given on the working life cannot be interpreted as a guarantee given by the producer, but are to be regarded only as a means for choosing the right products in relation to the expected economically reasonable working life of the works.

3 Performance of the product and references to the methods used for its assessment

3.1 Mechanical resistance and stability (BWR 1)

Essential characteristic	Performance
Characteristic resistance to tension load (static and quasi-static loading)	See Annex C 1 to C 3
Characteristic resistance to tension load (seismic loading)	See Annex C 4

3.2 Safety in case of fire (BWR 2)

Essential characteristic	Performance
Reaction to fire	Class A1

4 Assessment and verification of constancy of performance (AVCP) system applied, with reference to its legal base

In accordance with European Assessment Document EAD No. 332402-00-0601-v02, the applicable European legal act is: [96/582/EC].

The system to be applied is: 1

5 Technical details necessary for the implementation of the AVCP system, as provided for in the applicable European Assessment Document

Technical details necessary for the implementation of the AVCP system are laid down in the control plan deposited with Deutsches Institut für Bautechnik.

Issued in Berlin on 31 July 2023 by Deutsches Institut für Bautechnik

Dipl.-Ing. Beatrix Wittstock
Head of Section

beglaubigt:
Stiller

Installation conditions and application examples reinforcing bars

Figure A1.1:

Column / wall to foundation / slab

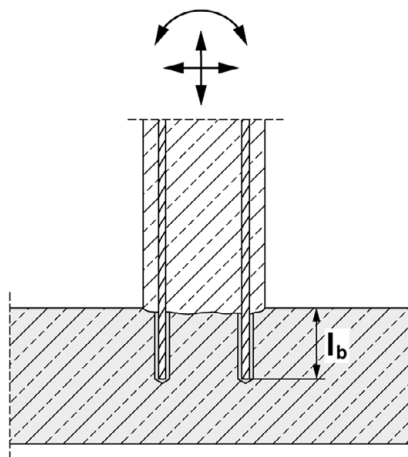
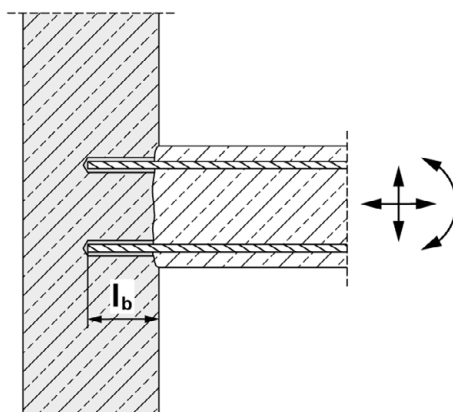


Figure A1.2:

Slab / beam to wall or beam to column



Figures not to scale

fischer injection system FIS EM Plus

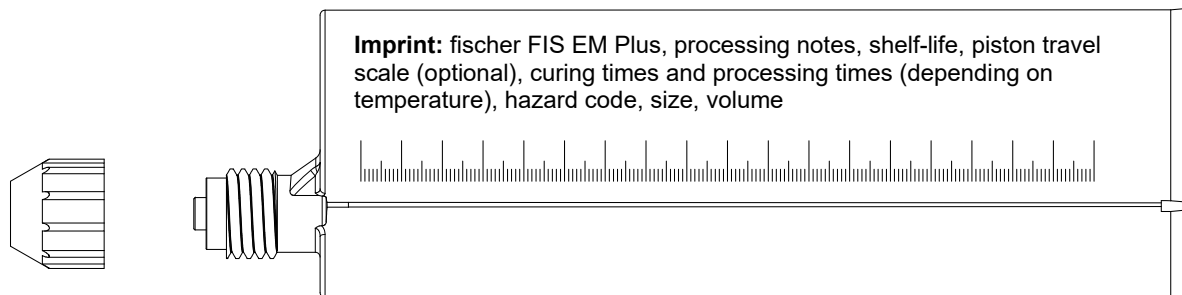
Product description

Installation conditions and application examples reinforcing bars

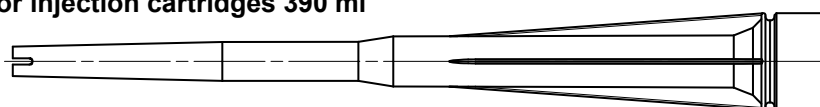
Annex A 1

Overview system components

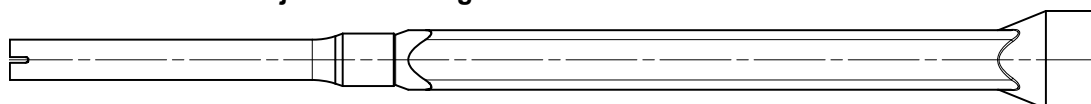
Injection cartridge (shuttle cartridge) FIS EM Plus with sealing cap; Sizes: 390 ml, 585 ml, 1100 ml, 1500 ml



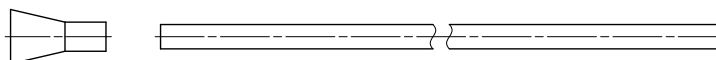
Static mixer FIS MR Plus for injection cartridges 390 ml



Static mixer FIS UMR for injection cartridges ≥ 585 ml



**Injection adapter and extension tube $\varnothing 9$ for static mixer FIS MR Plus;
Injection adapter and extension tube $\varnothing 9$ or $\varnothing 15$ for static mixer FIS UMR**



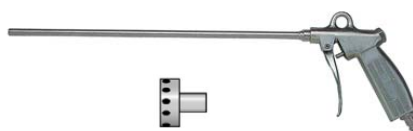
Reinforcing bar (rebar) Sizes: $\varnothing 8$, $\varnothing 10$, $\varnothing 12$, $\varnothing 14$, $\varnothing 16$, $\varnothing 20$, $\varnothing 22$, $\varnothing 24$, $\varnothing 25$, $\varnothing 26$, $\varnothing 28$, $\varnothing 30$, $\varnothing 32$, $\varnothing 34$, $\varnothing 36$, $\varnothing 40$



fischer cleaning brush



Compressed-air cleaning tool with fischer compressed-air nozzle



Figures not to scale

fischer injection system FIS EM Plus

Product description

Overview system components: injection mortar, static mixer, injection adapter, reinforcing bar, cleaning tools

Annex A 2

Properties of reinforcing bars (rebar)

Figure A3.1:



- The minimum value of related rib area $f_{R,min}$ according to EN 1992-1-1:2004+AC:2010
- The maximum outer rebar diameter over the ribs shall be:
 - The nominal diameter of the bar with rib $\phi + 2 \cdot h_{rib}$ ($h_{rib} \leq 0,07 \cdot \phi$)
 - (ϕ : Nominal diameter of the bar; h_{rib} = rib height of the bar)

Table A3.1: Installation conditions for rebars

Nominal diameter of the bar		ϕ	8 ¹⁾		10 ¹⁾		12 ¹⁾		14	16	20	22	24		
Nominal drill hole diameter	d ₀	[mm]	10	12	12	14	14	16	18	20	25	30	30		
Drill hole depth	h ₀		h ₀ ≥ l _b												
Effective embedment depth	l _b = l _v		acc. to static calculation												
Minimum thickness of concrete member	h _{min}		l _b + 30 (≥ 100)					l _b + 2d ₀							
Nominal diameter of the bar		ϕ	25 ¹⁾		26	28	30	32	34	36	40				
Nominal drill hole diameter	d ₀	[mm]	30	35	35	35	40	40	40	45	55				
Drill hole depth	h ₀		h ₀ ≥ l _b												
Effective embedment depth	l _b = l _v		acc. to static calculation												
Minimum thickness of concrete member	h _{min}		l _b + 2d ₀												

¹⁾ Both drill hole diameters can be used

Table A3.2: Materials of rebars

Designation	Reinforcing bar (rebar)
Reinforcing bar EN 1992-1-1:2004+AC:2010, Annex C	Bars and de-coiled rods class B or C with f_{yk} and k according to NDP or NCI of EN 1992-1-1/NA $f_{uk} = f_{tk} = k \cdot f_{yk}$

fischer injection system FIS EM Plus

Product description
Properties and materials of reinforcing bars (rebar)

Annex A 3

Specifications of intended use part 1

Table B1.1: Overview use and performance categories

Fastenings subject to		FIS EM Plus with ...	
		Reinforcing bar 	
Hammer drilling with standard drill bit 		all sizes	
Hammer drilling with hollow drill bit (fischer "FHD", Heller "Duster Expert"; Bosch „Speed Clean“; Hilti "TE-CD, TE-YD", DreBo „D-Plus“, DreBo „D-Max“) 		Nominal drill bit diameter (d ₀) 12 mm to 35 mm	
Use category	I1 dry or wet concrete	all sizes	
	I2 water filled hole	all sizes (not permitted in combination with working life 100 years)	
Characteristic resistance under static and quasi-static loading	in uncracked concrete	all sizes	Tables: C1.1 C1.2 C2.1 C3.1
	in cracked concrete	all sizes	
Seismic performance		all sizes	Tables: C4.1
Installation direction		D3 (downward and horizontal and upwards (e.g. overhead))	
Installation temperature		T _{i,min} = -5 °C to T _{i,max} = +40 °C for the standard variation of temperature after installation	
Service temperature	Temperature range I	-40 °C to +60 °C	(max. short term temperature +60 °C; max. long term temperature +35 °C)
	Temperature range II	-40 °C to +72 °C	(max. short term temperature +72 °C; max. long term temperature +50 °C)
fischer injection system FIS EM Plus			Annex B 1
Intended use Specifications part 1			

Specifications of intended use part 2

Anchorage subject to:

- Static and quasi-static loading: reinforcing bar (rebar) size 8 mm to 40 mm
- Seismic action: reinforcing bar (rebar) size 8 mm to 40 mm

Base materials:

- Compacted reinforced or unreinforced normal weight concrete without fibres according to EN 206:2013+A1:2016.
 - Strength classes C20/25 to C50/60 according to EN 206:2013+A1:2016
 - Maximum chloride content of 0,40 % (CL 0.40) related to the cement content according to EN 206:2013+A1:2016
 - Non-carbonated concrete
- Note: In case of a carbonated surface of the existing concrete structure, the carbonated layer shall be removed in the area of the post-installed rebar connection with a diameter of $\phi + 60$ mm prior to the installation of the new rebar. The depth of concrete to be removed shall correspond to at least the minimum concrete cover in accordance with EN 1992-1-1 :2004+AC:2010. The foregoing may be neglected if building components are new and not carbonated and if building components are in dry conditions.

Design:

- Fastenings are designed under the responsibility of an engineer experienced in fastenings and concrete work.
- Verifiable calculation notes and drawings are prepared taking account of the forces to be transmitted.
- Design under static and quasi-static loading and for seismic actions in accordance with EOTA Technical Report TR 069 June 2021.
- The actual position of the reinforcement in the existing structure shall be determined on the basis of the construction documentation and taken into account when designing.
- The shear force must be transferred via the rough joint; the subsequent reinforcement must not be applied for shear force transfer.

Installation:

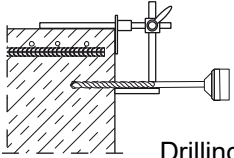
- Rebar installation carried out by appropriately qualified personnel and under the supervision of the person responsible for technical matters of the site.
- Check the position of the existing rebars (if the position of existing rebars is not known, it shall be determined using a rebar detector suitable for this purpose as well as on the basis of the construction documentation and then marked on the building component for the overlap joint).
- Rebars in overhead installation have to be fixed in their position until the injection mortar is cured.

fischer injection system FIS EM Plus

Intended use
Specifications part 2

Annex B 2

Table B3.1: Minimum concrete cover c_{min} ¹⁾ depending on the drilling method and the drilling tolerance²⁾

Drilling method	nominal diameter of reinforcing bar ϕ [mm]	Minimum concrete cover c_{min}		 Drilling aid
		Without drilling aid [mm]	With drilling aid [mm]	
Hammer drilling with standard drill bit	< 25	30 mm + 0,06 $l_b \geq 2 \phi$	30 mm + 0,02 $l_b \geq 2 \phi$	
	≥ 25	40 mm + 0,06 $l_b \geq 2 \phi$	40 mm + 0,02 $l_b \geq 2 \phi$	
Hammer drilling with hollow drill bit (fischer "FHD", Heller "Duster Expert"; Bosch "Speed Clean"; Hilti "TE-CD, TE-YD")	< 25	30 mm + 0,06 $l_b \geq 2 \phi$	30 mm + 0,02 $l_b \geq 2 \phi$	
	≥ 25	40 mm + 0,06 $l_b \geq 2 \phi$	40 mm + 0,02 $l_b \geq 2 \phi$	

¹⁾ Note: The minimum concrete cover as specified in EN 1992-1-1:2004+AC:2010 must be observed.

²⁾ Minimum clear spacing is $a = \max(40 \text{ mm}; 4 \cdot \phi)$

Table B3.2: Dispensers and cartridge sizes corresponding to maximum embedment depth $l_{b,max}$

reinforcing bars (rebar)	Manual dispenser	Pneumatic or cordless dispenser (small)	Pneumatic or cordless dispenser (large)	
	Cartridge size 390 ml, 585 ml	Cartridge size 390 ml, 585 ml	Cartridge size 1500 ml	
ϕ [mm]	$l_{b,max}$ [mm]	$l_{b,max}$ [mm]	$l_{b,max}$ [mm]	
8	1000	1000	1800	
10				
12		1200		
14		1500		
16	700	1300	2000	
20		1000		
22 / 24 / 25	500	700		
26 / 28				
30 / 32 / 34	no performance assessed	500		
36 / 40				

Figures not to scale

fischer injection system FIS EM Plus

Intended use

Minimum concrete cover;
dispenser and cartridge sizes corresponding to maximum embedment depth

Annex B 3

Table B4.1: Conditions for use **static mixer** without an **extension tube**

Nominal drill hole diameter d_0		10	12	14	16	18	20	24	25	28	30	35	40
Drill hole depth h_0 by FIS MR Plus	[mm]	≤ 90		≤ 120	≤ 140	≤ 150	≤ 160	≤ 190	≤ 210				
using FIS UMR		-	-	≤ 90	≤ 160	≤ 180	≤ 190	≤ 220		≤ 250			

Table B4.2: Working times t_{work} and curing times t_{cure}

Temperature at anchoring base [°C]	Maximum processing time ¹⁾ t_{work}	Minimum curing time ²⁾ t_{cure}
-5 to 0	240 min ³⁾	200 h
>0 to 5	150 min ³⁾	90 h
>5 to 10	120 min ³⁾	40 h
>10 to 20	30 min	18 h
>20 to 30	14 min	10 h
>30 to 40	7 min ⁴⁾	5 h

¹⁾ Maximum time from the beginning of the injection to the setting and the final positioning of the rebar

²⁾ For wet concrete the curing time must be doubled

³⁾ If the temperature in the concrete falls below 10 °C the cartridge must be warmed up to +15 °C.

⁴⁾ If the temperature in the concrete exceeds 30 °C the cartridge must be cooled down to +15 °C up to 20 °C

Table B4.3: Installation tools for drilling and cleaning the bore hole and injection of the mortar

reinforcing bars (rebar)	Drilling and cleaning				Injection	
	Nominal drill bit diameter	Diameter of cutting edge	Steel brush diameter	Diameter of cleaning nozzle ³⁾	Diameter of extension tube	Injection adapter
ϕ [mm]	d ₀ [mm]	d _{cut} [mm]	d _b [mm]	[mm]	[mm]	[colour]
8 ¹⁾	10 ²⁾	≤ 10,50	11	---	9	---
	12	≤ 12,50	14	11		nature
10 ¹⁾	12	≤ 12,50	14			15
	14	≤ 14,50	16	red		
12 ¹⁾	14	≤ 14,50	16	19	9 or 15	yellow
	16	≤ 16,50	20			green
14	18	≤ 18,50	20	28		black
16	20	≤ 20,55	25			grey
20	25	≤ 25,55	27	38		grey
22 / 24	30	≤ 30,55	32			brown
25 ¹⁾	30	≤ 30,55	32	38		brown
	35	≤ 35,70	37			red
26 / 28	35	≤ 35,70	37	38		yellow
30 / 32 / 34	40 ²⁾	≤ 40,70	42			nature
36	45 ²⁾	≤ 45,70	47			
40	55 ²⁾	≤ 55,70	58			

¹⁾ Both drill bit diameters can be used

²⁾ Only hammer drilling with standard drill bit

³⁾ Cleaning nozzle and extension is only necessary if bore hole depth is greater than the length of compressed-air cleaning tool

fischer injection system FIS EM Plus

Intended use

Conditions for use static mixer without an extension tube; Working times and curing times; Installation tools for drilling and cleaning the bore hole and injection of the mortar

Annex B 4

Safety regulations

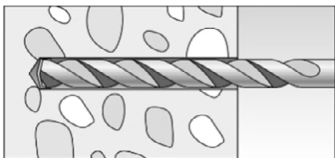
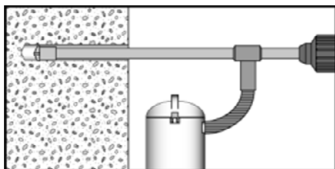
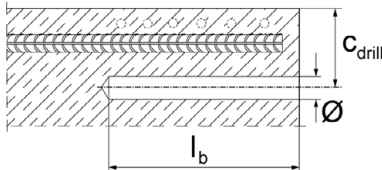
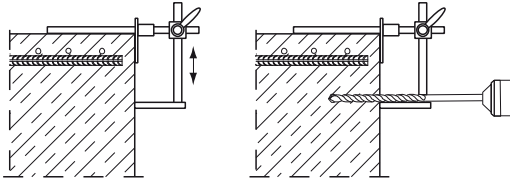


Review the Safety Data Sheet (SDS) before use for proper and safe handling!
Wear well-fitting protective goggles and protective gloves when working with mortar FIS EM Plus.
Important: Observe the instructions for use provided with each cartridge.

Installation instruction part 1

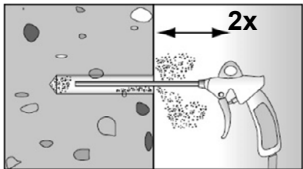
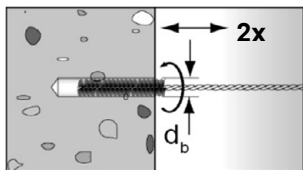
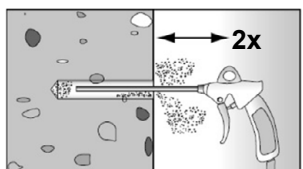
Hole drilling

Note: Before drilling, remove carbonated concrete; clean contact areas (see Annex B 2)
In case of aborted drill holes the drill hole shall be filled with mortar.

1a	Hammer drilling with standard drill bit  Drill the hole to the required embedment depth using a hammer drill with carbide drill bit set in rotation hammer mode. Nominal drill hole diameter d₀ (see table B4.3) and drill hole depth h₀ (see table A3.1).
1b	Hammer drilling with hollow drill bit  Check a suitable hollow drill (see table B1.1) for correct operation of the dust extraction.
	 Drill the hole to the required embedment depth using a hammer drill with hollow drill bit in rotation hammer mode. Dust extraction conditions see drill hole cleaning Annex B 6. Nominal drill hole diameter d₀ (see table B4.3) and drill hole depth h₀ (see table A3.1).
2	 Measure and control concrete cover <i>c</i> ($c_{drill} = c + \varnothing / 2$) Drill parallel to surface edge and to existing rebar. Where applicable use fischer drilling aid.
	 For holes <i>l_b</i> > 20 cm use drilling aid. Three different options can be considered: A) fischer drilling aid B) Slat or spirit level C) Visual check Minimum concrete cover <i>c_{min}</i> see table B3.1.
fischer injection system FIS EM Plus	
Intended use Safety regulations; Installation instruction part 1, hole drilling	Annex B 5

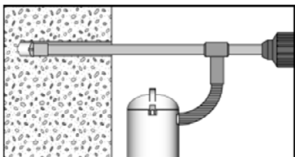
Installation instruction part 2

Drill hole cleaning (hammer drilling with standard drill bit)

3a		Cleaning the drill hole. Blow out the drill hole twice, with oil free compressed air ($p \geq 6$ bar). If the drill hole depth is greater than the length of the compressed-air cleaning tool, an extension and appropriate fischer cleaning nozzle must be used. Corresponding diameters see table B4.3.	
		Brush the drill hole twice. For drill hole diameter ≥ 30 mm use a power drill. For deep holes use an extension. Corresponding brushes see table B4.3.	
		Cleaning the drill hole: Blow out the drill hole twice, with oil free compressed air ($p \geq 6$ bar). If the drill hole depth is greater than the length of the compressed-air cleaning tool, an extension and appropriate fischer cleaning nozzle must be used. Corresponding diameters see table B4.3.	

Go to step 4

Drill hole cleaning (hammer drilling with hollow drill bit)

3b		Use a suitable dust extraction system, e. g. fischer FVC 35 M or a comparable dust extraction system with equivalent performance data. Drill the hole with hollow drill bit. The dust extraction system has to extract the drill dust nonstop during the drilling process and must be adjusted to maximum power. Check the hollow drill for correct operation of the dust extraction. No further cleaning steps necessary.	
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Go to step 4

fischer injection system FIS EM Plus

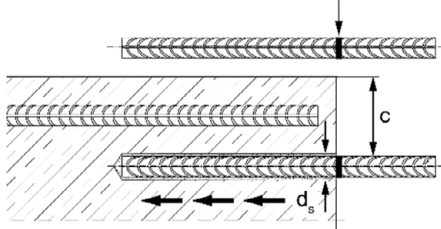
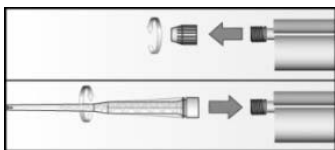
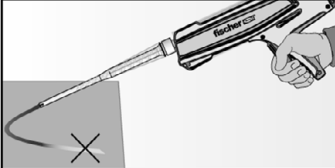
Intended use

Installation instruction part 2, drill hole cleaning

Annex B 6

Installation instruction part 3

Reinforcing bars (rebar) and cartridge preparation

4		Before use, make asure that the rebar is dry and free of oil or other residue. Mark the embedment depth l_b (e.g. with tape) Insert rebar in borehole, to verify drill hole depth and setting depth l_b.
5		Twist off the sealing cap Twist on the static mixer (the spiral in the static mixer must be clearly visible).
6		Place the cartridge into a suitable dispenser.
7		Press out approximately 10 cm of mortar until the resin is permanently grey in colour. Mortar which is not grey in colour will not cure and must be disposed.

Go to step 8

fischer injection system FIS EM Plus

Intended use

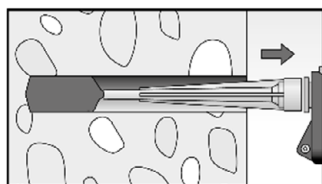
Installation instruction part 3,
reinforcing bars (rebar) and cartridge preparation

Annex B 7

Installation instruction part 4; Installation with FIS EM Plus

Injection of the mortar without extension tube

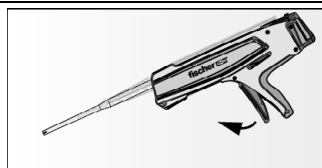
8a



Inject the mortar from the back of the hole towards the front and slowly withdraw the static mixer step by step with each trigger pull. Avoid bubbles.

Fill holes approximately 2/3 (for $h_0 = l_b$) full, to ensure that the annular gap between the rebar and the concrete will be completely filled with adhesive over the entire embedment length. For $h_0 > l_b$ more mortar is needed.

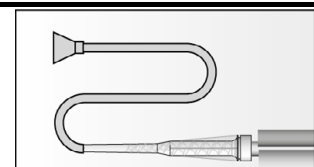
The conditions for mortar injection without extension tube can be found in **table B4.1**



After injecting, release the dispenser. This will prevent further mortar discharge from the static mixer.

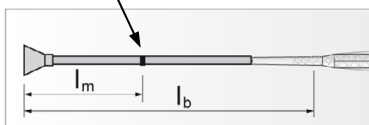
Injection of the mortar with extension tube

8b



Assemble mixing nozzle FIS MR Plus or FIS UMR, extension tube and appropriate injection adapter (see **table B4.3**).

Mortar level mark



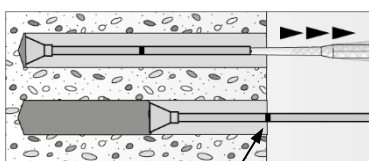
Mark the required mortar level l_m and embedment depth l_b with tape or marker on the injection extension tube.

a) Estimation:

$$l_m = \frac{1}{3} \cdot l_b \text{ [mm]}$$

b) Precise equation for optimum mortar volume:

$$l_m = l_b \cdot \left(1,2 \cdot \frac{d_s^2}{d_0^2} - 0,2 \right) \text{ [mm]}$$

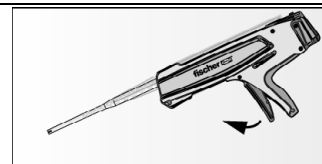


Mortar level mark

Insert injection adapter to back of the hole. Begin injection allowing the pressure of the injected adhesive mortar to push the injection adapter towards the front of the hole. Do not actively pull out!

Fill holes approximately 2/3 (for $h_0 = l_b$) full, to ensure that the annular gap between the rebar and the concrete will be completely filled with adhesive over the embedment length. For $h_0 > l_b$ more mortar is needed. When using an injection adapter continue injection until the mortar level mark l_m becomes visible.

Maximum embedment depth, see **table B3.2**.



After injecting, release the dispenser. This will prevent further mortar discharge from the static mixer.

Go to step 9

fischer injection system FIS EM Plus

Intended use

Installation instruction part 4, mortar injection

Annex B 8

Installation instruction part 5; Installation with FIS EM Plus

Insert rebar

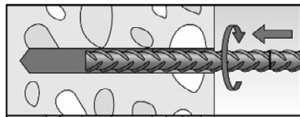
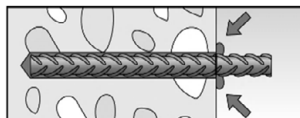
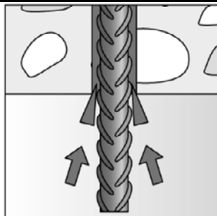
9		Insert the rebar slowly twisted into the borehole until the embedment mark is reached. Recommendation: Rotation back and forth of the reinforcement bar makes pushing easy
10		After installing the rebar the annular gap must be completely filled with mortar. Proper installation • Desired embedment depth is reached l_b: embedment mark at concrete surface • Excess mortar flows out of the borehole after the rebar have been fully inserted up to the embedment mark.
11		For overhead installation, support the rebar and secure it from falling till mortar started to harden, e.g. using wedges.
12		Observe the working time "t_{work}" (see table B4.2), which varies according to temperature of base material. Minor adjustments to the rebar position may be performed during the working time Full load may be applied only after the curing time "t_{cure}" has elapsed (see table B4.2)
fischer injection system FIS EM Plus		Annex B 9
Intended use Installation instruction part 5, insert rebar		

Table C2.1: Characteristic resistance under tension loading for reinforcing bars in hammer drilled holes; uncracked or cracked concrete; working life 50 years

Nominal diameter of the bar			ϕ	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40	
Combined pullout and concrete cone failure																					
Calculation diameter		d	[mm]	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40	
Uncracked concrete																					
Characteristic bond resistance in uncracked concrete C20/25																					
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																					
Tem- perature range	I: 35 °C / 60 °C		$\tau_{Rk,ucr,50}$	[N/mm ²]	16	15	15	14	14	13	13	13	12	12	12	12	12	11	11	11	
	II: 50 °C / 72 °C				15	14	14	13	13	12	12	12	12	11	11	11	11	11	11	10	10
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)																					
Tem- perature range	I: 35 °C / 60 °C		$\tau_{Rk,ucr,50}$	[N/mm ²]	16	16	14	13	12	12	11	11	10	10	10	10	9	9	9	8	8
	II: 50 °C / 72 °C				15	14	13	12	12	11	11	10	10	9	9	9	9	8	8	8	8
Installation factors																					
Dry or wet concrete		γ_{inst}	[-]	1,0																	
Water filled hole				1,4																	
Influence of cracked concrete on combined pullout and concrete cone failure for working life of 50 years																					
Hammer-drilling with standard drill bit or hollow drill bit																					
Factor for influence of cracked concrete		$\Omega_{cr,03}$	[-]	0,91	0,91	0,91	0,91	0,91	0,91	0,92	0,92	0,92	0,92	0,92	0,92	0,92	0,93	0,93	0,93	0,93	
fischer injection system FIS EM Plus														Annex C 2							
Performances Characteristic resistance under tension loading for reinforcing bars; uncracked or cracked concrete; working life 50 years																					

Table C3.1: Characteristic resistance under tension loading for reinforcing bars in hammer drilled holes; uncracked or cracked concrete; working life 100 years

Nominal diameter of the bar		ϕ	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40	
Combined pullout and concrete cone failure																				
Calculation diameter		d	[mm]	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40
Uncracked concrete																				
Characteristic bond resistance in uncracked concrete C20/25 ¹⁾																				
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr,50}$	[N/mm ²]	16	15	15	14	14	13	13	13	12	12	12	12	12	11	11	11	
	II: 50 °C / 72 °C			15	14	14	13	13	12	12	12	12	11	11	11	11	11	11	10	10
Installation factors																				
Dry or wet concrete		γ_{inst}	[-]	1,0																
Tem- perature range	I: 35 °C / 60 °C	$\alpha_{100\text{ years}}$	[-]	0,55	0,75	0,60	0,75	0,60	0,75	0,65	0,75	0,65	0,75	0,65	0,75	0,65	0,75	0,65	0,75	
	II: 50 °C / 72 °C			0,55	0,75	0,60	0,75	0,60	0,75	0,65	0,75	0,65	0,75	0,65	0,75	0,65	0,75	0,65	0,75	
Influence of cracked concrete on combined pullout and concrete cone failure for working life of 100 years																				
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																				
Factor for influence of cracked concrete		$\Omega_{cr,03}$	[-]	0,91	0,91	0,91	0,91	0,91	0,91	0,92	0,92	0,92	0,92	0,92	0,92	0,93	0,93	0,93	0,93	

¹⁾ Calculation of characteristic bond resistance in uncracked concrete $\tau_{Rk,100, ucr}$:

$$\tau_{Rk,100, ucr} = \alpha_{100\text{ years}} \cdot \tau_{Rk,ucr,50}$$

fischer injection system FIS EM Plus

Performances

Characteristic resistance under tension loading for reinforcing bars; uncracked or cracked concrete; working life 100 years

Annex C 3

Table C4.1: Characteristic resistance under tension loading for reinforcing bars (rebars) in concrete under seismic action; working life of 50 and 100 years

Nominal diameter of the bar	ϕ	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40
Resistance to pull-out failure in uncracked concrete under cyclic loading for working life of 50 and 100 years																		
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																		
Reduction factor for pull-out Resistance under seismic action	$\alpha_{eq,p}$	[N/mm ²]	0,76										1,0					
Influence of increased crack width on resistance to pull-out failure for working life of 50 and 100 years																		
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																		
Factor for influence of cracked concrete	$\Omega_{cr,05}^{1)}$	[-]	0,86	0,86	0,86	0,86	0,86	0,86	0,86	0,86	0,86	0,86	0,86	0,86	0,87	0,87	0,87	0,87
	$\Omega_{cr,08}^{1)}$		0,76	0,76	0,76	0,76	0,76	0,76	0,76	0,76	0,76	0,76	0,76	0,76	0,76	0,73	0,70	0,63
Resistance to bond-splitting failure under cyclic loading for working life of 50 and 100 years																		
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																		
Reduction factor for bond- splitting resistance under seismic action	$\alpha_{eq,sp}$	[-]	0,94															

¹⁾ Assumed crack width in accordance with EOTA Technical Report TR 069 June 2021; Section 3.6.

fischer injection system FIS EM Plus

Performances

Characteristic resistance under tension loading for reinforcing bars (rebars) in concrete under seismic action working life of 50 and 100 years

Annex C 4

Approval body for construction products
and types of construction

Bautechnisches Prüfamt

An institution established by the Federal and
Laender Governments



European Technical Assessment

ETA-17/0979
of 17 June 2020

English translation prepared by DIBt - Original version in German language

General Part

Technical Assessment Body issuing the
European Technical Assessment:

Deutsches Institut für Bautechnik

Trade name of the construction product

fischer Injection System FIS EM PLUS

Product family
to which the construction product belongs

Bonded fastener for use in concrete

Manufacturer

fischerwerke GmbH & Co. KG
Otto-Hahn-Straße 15
79211 Denzlingen
DEUTSCHLAND

Manufacturing plant

fischerwerke

This European Technical Assessment
contains

41 pages including 3 annexes which form an integral part
of this assessment

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issued in accordance with Regulation (EU)
No 305/2011, on the basis of

EAD 330499-01-0601

This version replaces

ETA-17/0979 issued on 22 July 2019

European Technical Assessment

ETA-17/0979

English translation prepared by DIBt

Page 2 of 41 | 17 June 2020

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Specific Part

1 Technical description of the product

The "fischer injection system FIS EM Plus" is a bonded fastener consisting of a cartridge with injection mortar fischer FIS EM Plus and a steel element according to Annex A5.

The steel element is placed into a drilled hole filled with injection mortar and is anchored via the bond between metal part, injection mortar and concrete.

The product description is given in Annex A.

2 Specification of the intended use in accordance with the applicable European Assessment Document

The performances given in Section 3 are only valid if the anchor is used in compliance with the specifications and conditions given in Annex B.

The verifications and assessment methods on which this European Technical Assessment is based lead to the assumption of a working life of the anchor of at least 50 or 100 years. The indications given on the working life cannot be interpreted as a guarantee given by the producer, but are to be regarded only as a means for choosing the right products in relation to the expected economically reasonable working life of the works.

3 Performance of the product and references to the methods used for its assessment

3.1 Mechanical resistance and stability (BWR 1)

Essential characteristic	Performance
Characteristic resistance to tension load (static and quasi-static loading)	See Annex B 3 to B 8, C 1 to C 12
Characteristic resistance to shear load (static and quasi-static loading)	See Annex C 1 to C 4
Displacements under short-term and long-term loading	See Annex C 13 and C 14
Characteristic resistance and displacements for seismic performance categories C1 and C2	See Annex C 15 to C 18

3.2 Hygiene, health and the environment (BWR 3)

Essential characteristic	Performance
Content, emission and/or release of dangerous substances	No performance assessed

4 Assessment and verification of constancy of performance (AVCP) system applied, with reference to its legal base

In accordance with the European Assessment Document EAD 330499-01-0601 the applicable European legal act is: [96/582/EC].

The system to be applied is: 1

5 Technical details necessary for the implementation of the AVCP system, as provided for in the applicable European Assessment Document

Technical details necessary for the implementation of the AVCP system are laid down in the control plan deposited at Deutsches Institut für Bautechnik.

Issued in Berlin on 17 June 2020 by Deutsches Institut für Bautechnik

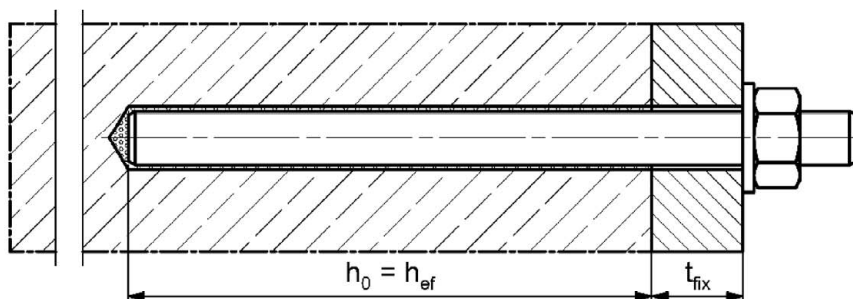
BD Dipl.-Ing. Andreas Kummerow
Head of Department

beglaubigt:
Baderschneider

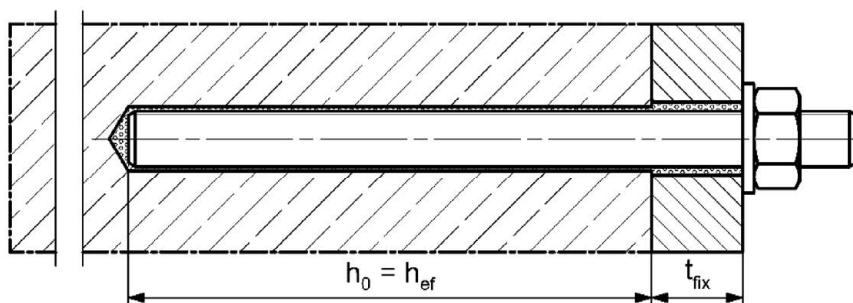
Installation conditions part 1

fischer anchor rod

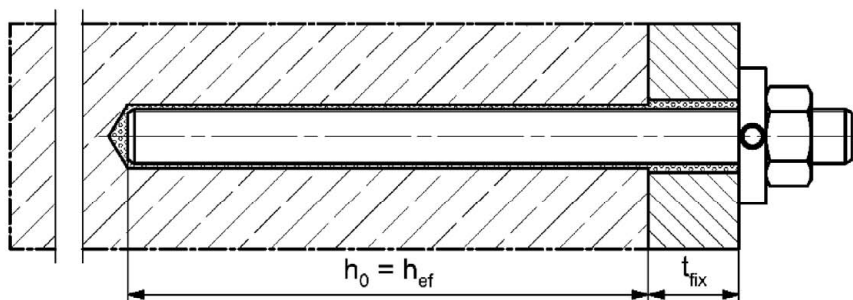
Pre-positioned installation



Push through installation (annular gap filled with mortar)



Pre-positioned or push through installation with subsequently injected fischer filling disc (annular gap filled with mortar)



Figures not to scale

h_0 = drill hole depth

h_{ef} = effective embedment depth

t_{fix} = thickness of fixture

fischer injection system FIS EM Plus

Product description

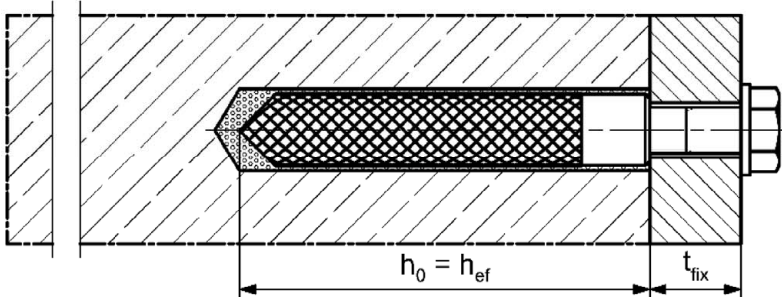
Installation conditions part 1

Annex A 1

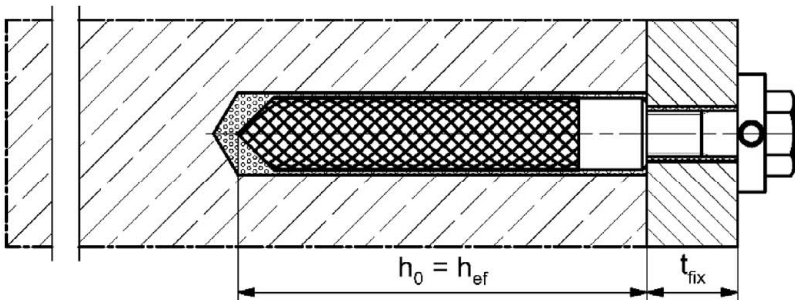
Installation conditions part 2

fischer internal threaded anchor RG MI

Pre-positioned installation



Pre-positioned installation with subsequently injected fischer filling disc (annular gap filled with mortar)



Figures not to scale

h_0 = drill hole depth
 t_{fix} = thickness of fixture

h_{ef} = effective embedment depth

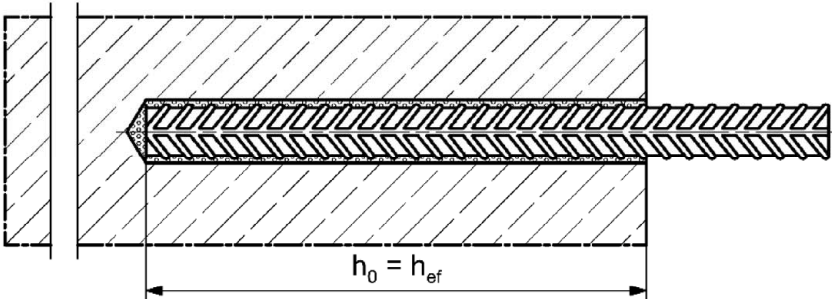
fischer injection system FIS EM Plus

Product description
Installation conditions part 2

Annex A 2

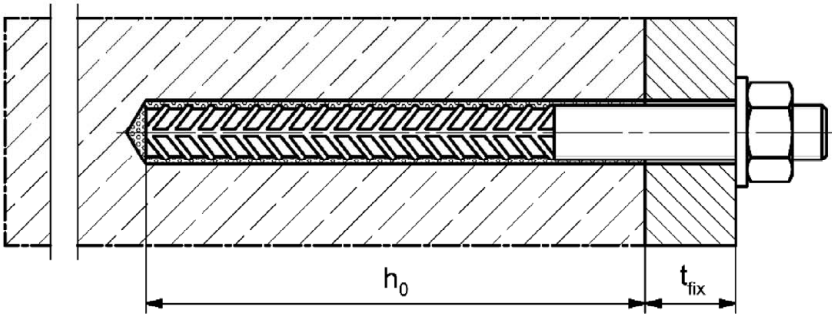
Installation conditions part 3

Reinforcing bar

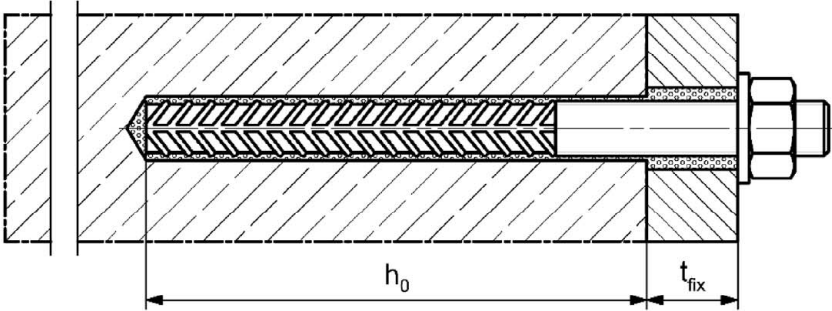


fischer rebar anchor FRA

Pre-positioned installation



Push through installation (annular gap filled with mortar)



Figures not to scale

h_0 = drill hole depth
 t_{fix} = thickness of fixture

h_{ef} = effective embedment depth

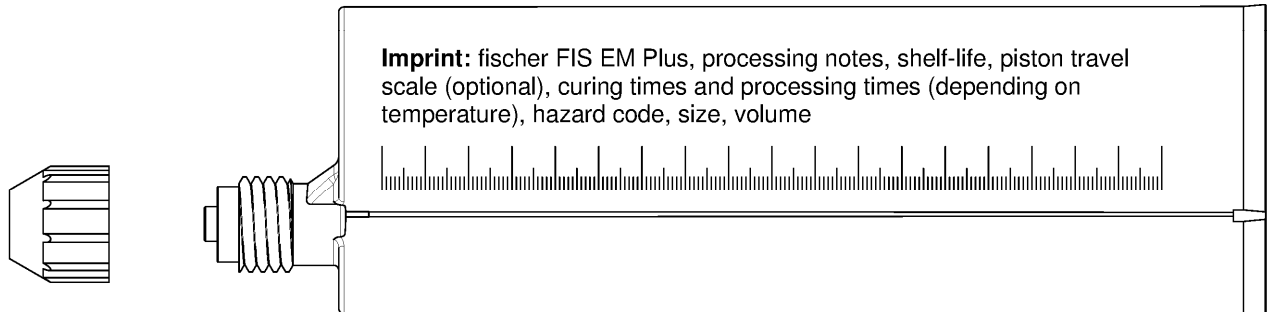
fischer injection system FIS EM Plus

Product description
Installation conditions part 3

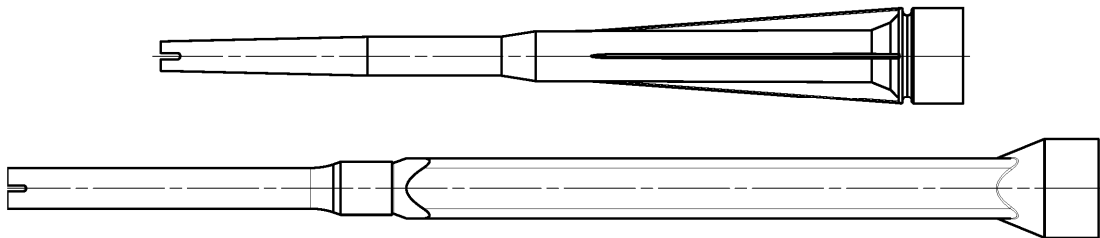
Annex A 3

Overview system components part 1

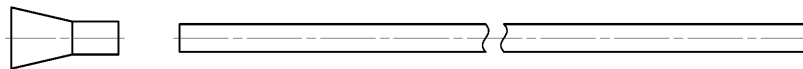
Injection cartridge (shuttle cartridge) with sealing cap; Size: 390 ml, 585 ml, 1100 ml, 1500 ml



Static mixer FIS MR Plus or UMR



Injection adapter and Extension tube for static mixer



Cleaning brush BS / BSB



Blow-out pump ABP



Figures not to scale

fischer injection system FIS EM Plus

Product description

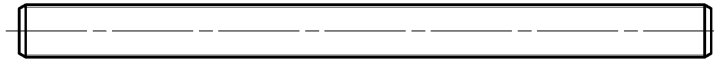
Overview system components part 1;
cartridges / static mixer / accessories

Annex A 4

Overview system components part 2

fischer anchor rod

Size: M8, M10, M12, M14, M16, M20, M22, M24, M27, M30

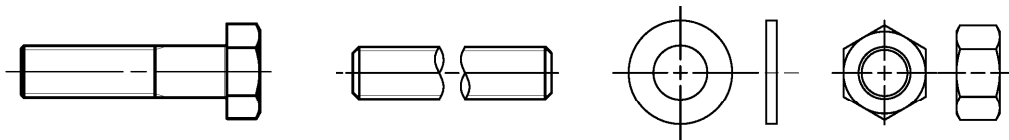


fischer internal threaded anchor RG MI

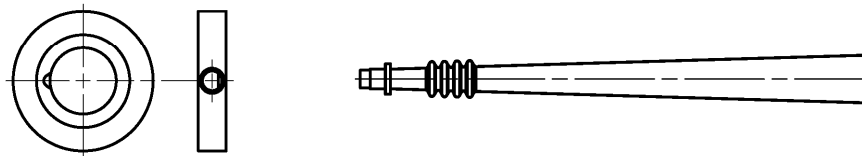
Size: M8, M10, M12, M16, M20



Screw / threaded rod / washer / hexagon nut



fischer filling disc with injection adapter



Reinforcing bar

Nominal diameter: $\phi 8$, $\phi 10$, $\phi 12$, $\phi 14$, $\phi 16$, $\phi 18$, $\phi 20$, $\phi 22$, $\phi 24$, $\phi 25$, $\phi 26$, $\phi 28$, $\phi 30$, $\phi 32$, $\phi 34$, $\phi 36$, $\phi 40$



fischer rebar anchor FRA

Size: M12, M16, M20, M24



Figures not to scale

fischer injection system FIS EM Plus

Product description

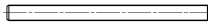
Overview system components part 2;
steel components

Annex A 5

Table A6.1: Materials				
Part	Designation	Material		
1	Injection cartridge	Mortar, hardener, filler		
	Steel grade	Steel	Stainless steel R	High corrosion resistant steel HCR
		zinc plated	acc. to EN 10088-1:2014 Corrosion resistance class CRC III acc. to EN 1993-1-4:2015	acc. to EN 10088-1:2014 Corrosion resistance class CRC V acc. to EN 1993-1-4:2015
2	Anchor rod	Property class 4.8, 5.8 or 8.8; EN ISO 898-1:2013 zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) or hot dip galvanised ≥ 40 µm EN ISO 10684:2004 f _{uk} ≤ 1000 N/mm ² A ₅ > 12% fracture elongation	Property class 50, 70 or 80 EN ISO 3506-1:2009 1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; 1.4062, 1.4662, 1.4462; EN 10088-1:2014 f _{uk} ≤ 1000 N/mm ² A ₅ > 12% fracture elongation	Property class 50 or 80 EN ISO 3506-1:2009 or property class 70 with f _{yk} = 560 N/mm ² 1.4565; 1.4529; EN 10088-1:2014 f _{uk} ≤ 1000 N/mm ² A ₅ > 12% fracture elongation
		Fracture elongation A ₅ > 8 %, for applications without requirements for seismic performance category C2		
3	Washer ISO 7089:2000	zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) or hot dip galvanised ≥ 40 µm EN ISO 10684:2004	1.4401; 1.4404; 1.4578;1.4571; 1.4439; 1.4362; EN 10088-1:2014	1.4565; 1.4529; EN 10088-1:2014
4	Hexagon nut	Property class 4, 5 or 8; EN ISO 898-2:2012 zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) or hot dip galvanised ≥ 40 µm EN ISO 10684:2004	Property class 50, 70 or 80 EN ISO 3506-1:2009 1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; EN 10088-1:2014	Property class 50, 70 or 80 EN ISO 3506-1:2009 1.4565; 1.4529 EN 10088-1:2014
5	fischer internal threaded anchor RG MI	Property class 5.8 ISO 898-1:2013 zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K)	Property class 70 EN ISO 3506-1:2009 1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; EN 10088-1:2014	Property class 70 EN ISO 3506-1:2009 1.4565; 1.4529; EN 10088-1:2014
6	Commercial standard screw or threaded rod for fischer internal threaded anchor RG MI	Property class 5.8 or 8.8; EN ISO 898-1:2013 zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) A ₅ > 8 % fracture elongation	Property class 70 EN ISO 3506-1:2009 1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; EN 10088-1:2014 A ₅ > 8 % fracture elongation	Property class 70 EN ISO 3506-1:2009 1.4565; 1.4529; EN 10088-1:2014 A ₅ > 8 % fracture elongation
7	fischer filling disc similar to DIN 6319-G	zinc plated ≥ 5 µm, ISO 4042:2018/Zn5/An(A2K) or hot dip galvanised ≥ 40 µm EN ISO 10684:2004	1.4401; 1.4404; 1.4578; 1.4571; 1.4439; 1.4362; EN 10088-1:2014	1.4565;1.4529; EN 10088-1:2014
8	Reinforcing bar EN 1992-1-1:2004 and AC:2010, Annex C	Bars and de-coiled rods, class B or C with f _{yk} and k according to NDP or NCL according to EN 1992-1-1/NA f _{uk} = f _{tk} = k · f _{yk}		
9	fischer rebar anchor FRA	Rebar part: Bars and de-coiled rods class B or C with f _{yk} and k according to NDP or NCL of EN 1992-1-1:2004+AC:2010 f _{uk} = f _{tk} = k · f _{yk}	Threaded part: Property class 70 or 80 EN ISO 3506-1:2009 1.4401, 1.4404, 1.4571, 1.4578, 1.4439, 1.4362, 1.4062 acc. to EN 10088-1:2014 Corrosion resistance class CRC III acc. to EN 1993-1-4:2015 1.4565; 1.4529 acc. to EN 10088-1:2014 Corrosion resistance class CRC V acc. to EN 1993-1-4:2015	
fischer injection system FIS EM Plus				Annex A 6
Product description Materials				

Specifications of intended use (part 1)

Table B1.1: Overview use and performance categories

Anchorages subject to		FIS EM Plus with ...							
		Anchor rod		fischer internal threaded anchor RG MI		Reinforcing bar		fischer rebar anchor FRA	
									
Hammer drilling with standard drill bit 		all sizes							
Hammer drilling with hollow drill bit (fischer "FHD", Heller "Duster Expert"; Bosch „Speed Clean“; Hilti "TE-CD, TE-YD", DreBo „D-Plus“, DreBo „D-Max“) 		Nominal drill bit diameter (d ₀) 12 mm to 35 mm							
Diamond drilling 		all sizes							
Static and quasi static load, in	uncracked concrete	all sizes	Tables: C1.1 C4.1 C5.1 C6.1 C13.1	all sizes	Tables: C2.1 C4.1 C7.1 C8.1 C13.2	all sizes	Tables: C3.1 C4.1 C9.1 C10.1 C14.1	all sizes	Tables: C3.2 C4.1 C11.1 C12.1 C14.2
	cracked concrete								
Seismic performance category (only hammer drilling with standard / hollow drill bits)	C1	M10 to M30	Tables: C15.1 C16.2 C17.1	_1)	_1)	_1)	Tables: C16.1 C16.2 C17.2	_1)	_1)
	C2	M12 M16 M20 M24	Tables: C15.1 C16.2 C18.1				_1)		
Use category	I1 dry or wet concrete	all sizes							
	I2 water filled hole	all sizes (not permitted in combined with working life 100 years)							
Installation direction		D3 (downward and horizontal and upwards (e.g. overhead) installation)							
Installation temperature		T _{i,min} = -5 °C to T _{i,max} = +40 °C							
In-service temperature	Temperature range I	-40 °C to +60 °C		(max. short term temperature +60 °C; max. long term temperature +35 °C)					
	Temperature range II	-40 °C to +72 °C		(max. short term temperature +72 °C; max. long term temperature +50 °C)					
_1) no performance assessed									
fischer injection system FIS EM Plus							Annex B 1		
Intended use Specifications (part 1)									

Specifications of intended use (part 2)

Base materials:

- Compacted reinforced or unreinforced normal weight concrete without fibres of strength classes C20/25 to C50/60 according to EN 206:2013+A1:2016

Use conditions (Environmental conditions):

- Structures subject to dry internal conditions (zinc coated steel, stainless steel or high corrosion resistant steel).
- For all other conditions according to EN1993-1-4:2015 corresponding to corrosion resistance classes to Annex A 6 table A6.1.

Design:

- Anchorages have to be designed by a responsible engineer with experience of concrete anchor design.
- Verifiable calculation notes and drawings are to be prepared taking account of the loads to be anchored. The position of the anchor is indicated on the design drawings (e. g. position of the anchor relative to reinforcement or to supports, etc.).
- Anchorages are designed in accordance with:
EN 1992-4:2018 and EOTA Technical Report TR 055, Edition February 2018

Installation:

- Anchor installation is to be carried out by appropriately qualified personnel and under the supervision of the person responsible for technical matters of the site
- In case of aborted hole: The hole shall be filled with mortar
- Anchorage depth should be marked and adhered to on installation
- Overhead installation is allowed

fischer injection system FIS EM Plus

Intended use
Specifications (part 2)

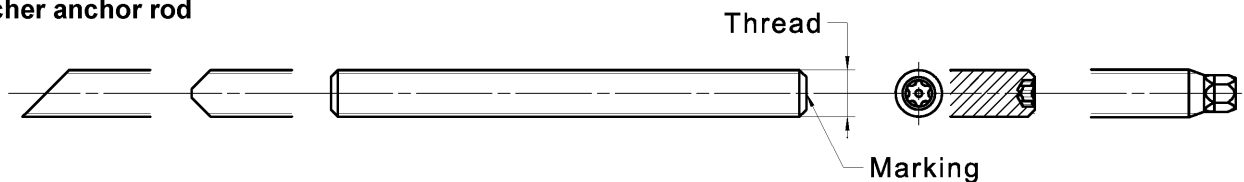
Annex B 2

Table B3.1: Installation parameters for anchor rods

Anchor rods		Thread	M8	M10	M12	M14	M16	M20	M22	M24	M27	M30
Width across flats	SW	[mm]	13	17	19	22	24	30	32	36	41	46
Nominal drill hole diameter	d_0		10	12	14	16	18	22 24 ¹⁾	25	28	30	35
Drill hole depth	h_0		$h_0 = h_{ef}$									
Effective embedment depth	$h_{ef, min}$		60	60	70	75	80	90	93	96	108	120
	$h_{ef, max}$		160	200	240	280	320	400	440	480	540	600
Diameter of the clearance hole of the fixture	pre-positioned installation d_f		9	12	14	16	18	22	24	26	30	33
	push through installation d_f		12	14	16	18	20	26	28	30	33	40
Minimum thickness of concrete member	h_{min}		$h_{ef} + 30$ (≥ 100)			$h_{ef} + 2d_0$						
Maximum installation torque	$\max T_{inst}$	[Nm]	10	20	40	50	60	120	135	150	200	300

¹⁾ Both drill hole diameters can be used

fischer anchor rod



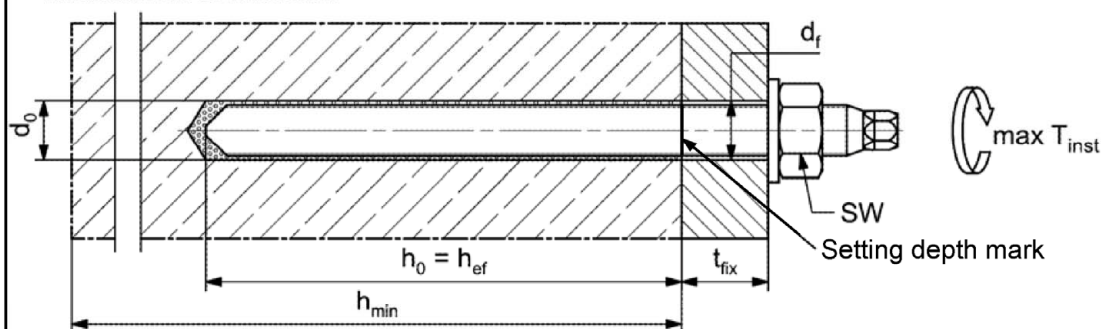
Marking (on random place) fischer anchor rod:

Steel zinc plated PC ¹⁾ 8.8	• or +	Steel hot-dip PC ¹⁾ 8.8	•
High corrosion resistant steel HCR PC ¹⁾ 50	•	High corrosion resistant steel HCR PC ¹⁾ 70	-
High corrosion resistant steel HCR PC ¹⁾ 80	(	Stainless steel R property class 50	~
Stainless steel R property class 80	*		

Alternatively: Colour coding according to DIN 976-1:2016

¹⁾ PC = property class

Installation conditions:



Commercial standard threaded rods, washers and hexagon nuts may also be used if the following requirements are fulfilled

- Materials, dimensions and mechanical properties according to Annex A 6, Table A6.1
- Inspection certificate 3.1 according to EN 10204:2004, the documents have to be stored
- Setting depth is marked

Figures not to scale

fischer injection system FIS EM Plus

Intended use
Installation parameters anchor rods

Annex B 3

Table B4.1: Minimum spacing and minimum edge distance for **anchor rods** and **reinforcing bars**

Anchor rods			M8	M10	M12	M14	M16	-	M20	M22	M24	
Reinforcing bars (nominal diameter)			ϕ	8	10	12	14	16	18	20	22	24
Minimum edge distance												
Uncracked / cracked concrete		Cmin	[mm]	40	45	45	45	50	55	55	55	60
Minimum spacing		Smin		according to Annex B5								
Minimum spacing												
Uncracked / cracked concrete		Smin	[mm]	40	45	55	60	65	85	85	95	105
Minimum edge distance		Cmin		according to Annex B5								
Required projecting area												
Uncracked concrete		A _{sp,req}	[1000 mm²]	8	13	22	23	24	38,5	38,5	39,5	40
Cracked concrete				6,5	10	16,5	17,5	18,5	29,5	29,5	30	30,5

Anchor rods			-	-	M27	-	M30	-	-	-	-	
Reinforcing bars (nominal diameter)			ϕ	25	26	-	28	30	32	34	36	40
Minimum edge distance												
Uncracked / cracked concrete		C _{min}	[mm]	75	75	75	80	80	120	120	135	175
Minimum spacing		S _{min}		according to Annex B5								
Minimum spacing												
Uncracked / cracked concrete		S _{min}	[mm]	120	120	120	140	140	160	160	160	160
Minimum edge distance		C _{min}		according to Annex B5								
Required projecting area												
Uncracked concrete		A _{sp,req}	[1000 mm²]	47,5	47,5	47,5	64	64	64	64	64	64
Cracked concrete				36,5	36,5	36,5	49	49	49	49	49	49

Splitting failure for minimum edge distance and spacing in dependence of the effective embedment depth h_{ef} .

For the calculation of minimum spacing and minimum edge distance of anchors in combination with different embedment depths and thicknesses of concrete members the following equation shall be fulfilled:

$$A_{sp,req} < A_{sp,t}$$

$A_{sp,req}$ = required projecting area

$A_{sp,t} = A_{sp,ef}$ = effective projecting area (according to Annex B5)

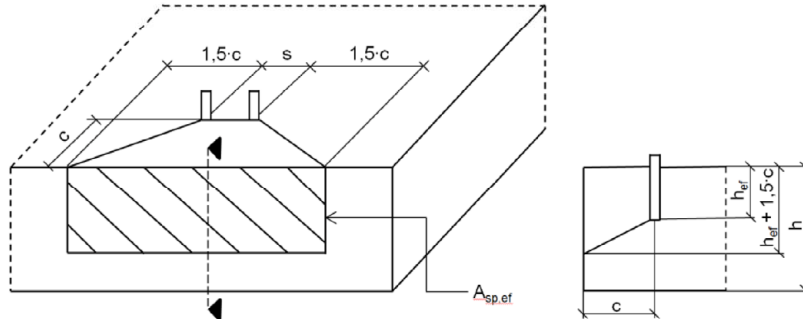
fischer injection system FIS EM Plus

Intended use

Minimum spacing and edge distance for anchor rods and reinforcing bars

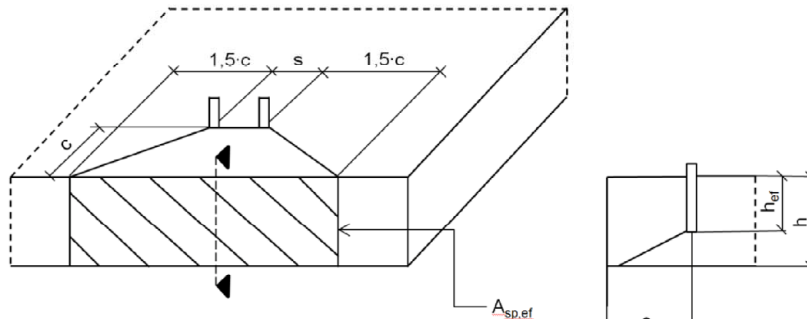
Annex B 4

Table B5.1: Effective projecting area $A_{sp,t}$ with concrete member thickness $h > h_{ef} + 1,5 \cdot c$ and $h \geq h_{min}$



Single anchor	$A_{sp,t} = (3 \cdot c) \cdot (h_{ef} + 1,5 \cdot c)$	[mm ²]	with $c \geq c_{min}$
Group of anchors with $s > 3 \cdot c$	$A_{sp,t} = (6 \cdot c) \cdot (h_{ef} + 1,5 \cdot c)$	[mm ²]	
Group of anchors with $s \leq 3 \cdot c$	$A_{sp,t} = (3 \cdot c + s) \cdot (h_{ef} + 1,5 \cdot c)$	[mm ²]	with $c \geq c_{min}$ and $s \geq s_{min}$

Table B5.2: Effective projecting area $A_{sp,t}$ with concrete member thickness $h \leq h_{ef} + 1,5 \cdot c$ and $h \geq h_{min}$



Single anchor	$A_{sp,t} = 3 \cdot c \cdot \text{existing } h$	[mm ²]	with $c \geq c_{min}$
Group of anchors with $s > 3 \cdot c$	$A_{sp,t} = 6 \cdot c \cdot \text{existing } h$	[mm ²]	
Group of anchors with $s \leq 3 \cdot c$	$A_{sp,t} = (3 \cdot c + s) \cdot \text{existing } h$	[mm ²]	with $c \geq c_{min}$ and $s \geq s_{min}$

Edge distance and axial spacing shall be rounded up to at least 5 mm

Figures not to scale

fischer injection system FIS EM Plus

Intended use

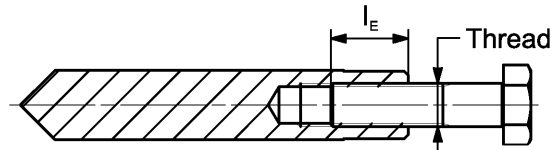
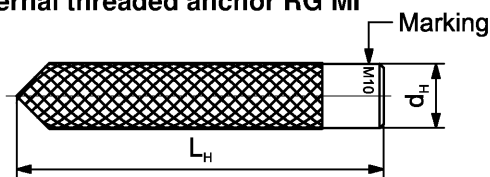
Minimum thickness of concrete member for anchor rods,
minimum spacing and edge distance

Annex B 5

Table B6.1: Installation parameters for **fischer internal threaded anchors RG MI**

Internal threaded anchors RG MI		Thread	M8	M10	M12	M16	M20
Diameter of anchor	$d_{nom} = d_H$	[mm]	12	16	18	22	28
Nominal drill hole diameter	d_0		14	18	20	24	32
Drill hole depth	h_0		$h_0 = h_{ef} = L_H$				
Effective embedment depth ($h_{ef} = L_H$)	h_{ef}		90	90	125	160	200
Minimum spacing and minimum edge distance	S_{min} = C_{min}		55	65	75	95	125
Diameter of clearance hole in the fixture	d_f		9	12	14	18	22
Minimum thickness of concrete member	h_{min}		120	125	165	205	260
Maximum screw-in depth	$l_{E,max}$		18	23	26	35	45
Minimum screw-in depth	$l_{E,min}$		8	10	12	16	20
Maximum installation torque	$\max T_{inst}$	[Nm]	10	20	40	80	120

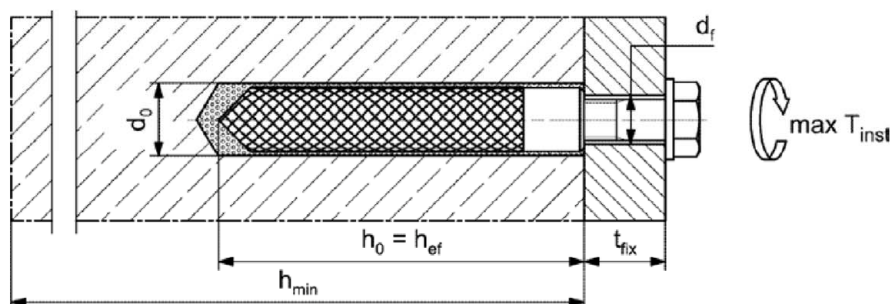
fischer internal threaded anchor RG MI



Marking: Anchor size e. g.: **M10**
Stainless steel → additional **R**; e.g.: **M10 R**
High corrosion resistant steel → additional **HCR**; e.g.: **M10 HCR**

Retaining bolt or threaded rods (including nut and washer) must comply with the appropriate material and strength class of Annex A 6, Table A6.1

Installation conditions:



Figures not to scale

fischer injection system FIS EM Plus

Intended use
Installation parameters internal threaded anchors RG MI

Annex B 6

Table B7.1: Installation parameters for reinforcing bars ¹⁾

Nominal diameter of the bar		ϕ	8 ²⁾		10 ²⁾		12 ²⁾		14	16	18	20	22	24
Nominal drill hole diameter	d ₀	[mm]	10	12	12	14	14	16	18	20	25	25	30	30
Drill hole depth	h ₀		h ₀ = h _{ef}											
Effective	h _{ef,min}		60	60	70	75	80	85	90	94	98			
embedment depth	h _{ef,max}		160	200	240	280	320	360	400	440	480			
Minimum thickness of concrete member	h _{min}		h _{ef} + 30 (≥ 100)				h _{ef} + 2d ₀							

Nominal diameter of the bar ϕ		25	26	28	30	32	34	36	40	-
Nominal drill hole diameter d_0	[mm]	30	35	35	40	40	40	45	55	-
Drill hole depth h_0		$h_0 = h_{ef}$								
Effective embedment depth $h_{ef,min}$		100	104	112	120	128	136	144	160	-
$h_{ef,max}$		500	520	560	600	640	680	720	800	-
Minimum thickness of concrete member h_{min}		$h_{ef} + 2d_0$								

¹⁾ minimum spacing and minimum edge distance see Annex B 4

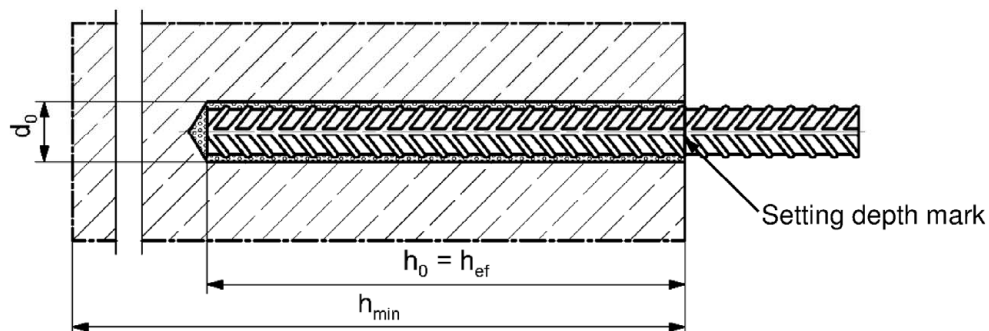
²⁾ Both drill hole diameters can be used

Reinforcing bar



- The minimum value of related rib area $f_{R,min}$ must fulfil the requirements of EN 1992-1-1:2004+AC:2010
- The rib height must be within the range: $0,05 \cdot \phi \leq h_{rib} \leq 0,07 \cdot \phi$
(ϕ = Nominal diameter of the bar, h_{rib} = rib height)

Installation conditions:



Figures not to scale

fischer injection system FIS EM Plus

Intended use
Installation parameters reinforcing bars

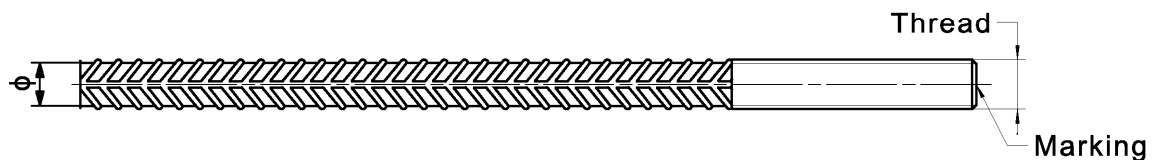
Annex B 7

Table B8.1: Installation parameters for **fischer rebar anchor FRA**

Rebar anchor FRA		Thread	M12 ¹⁾		M16	M20	M24
Nominal diameter of the bar	ϕ	[mm]	12		16	20	25
Width across flats	SW		19		24	30	36
Nominal drill hole diameter	d_0		14	16	20	25	30
Drill hole depth	h_0		$h_{ef} + l_e$				
Effective embedment depth	$h_{ef,min}$		70		80	90	96
	$h_{ef,max}$		140		220	300	380
Distance concrete surface to welded joint	l_e		100				
Minimum spacing and minimum edge distance	$s_{min} = c_{min}$		55		65	85	105
Diameter of clearance hole in the fixture	pre-positioned anchorage $\leq d_f$		14		18	22	26
	push through anchorage $\leq d_f$		18		22	26	32
Minimum thickness of concrete member	h_{min}		$h_0 + 30$	$h_0 + 2d_0$			
Maximum installation torque	$\max T_{inst}$	[Nm]	40		60	120	150

¹⁾ Both drill hole diameters can be used

fischer rebar anchor FRA

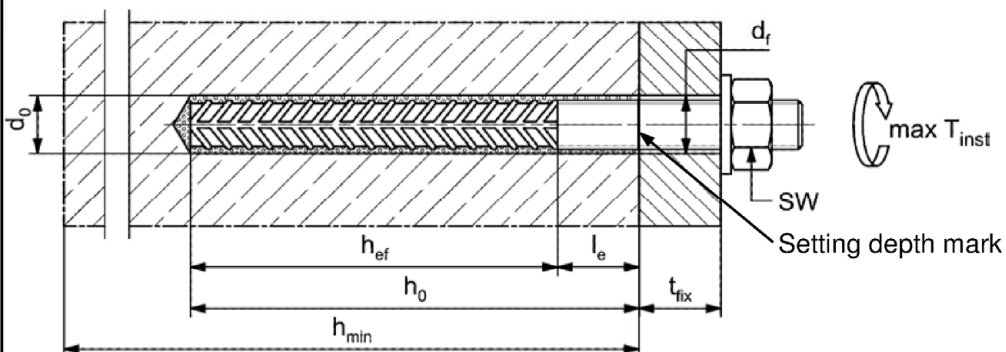


Marking frontal e. g:

 FRA (for stainless steel);

 FRA HCR (for high corrosion resistant steel)

Installation conditions:



Figures not to scale

fischer injection system FIS EM Plus

Intended use
Installation parameters rebar anchor FRA

Annex B 8

Table B9.1: Parameters of the **cleaning brush BS / BSB** (steel brush with steel bristles)

The size of the cleaning brush refers to the drill hole diameter

Nominal drill hole diameter	d_0		10	12	14	16	18	20	24	25	28	30	32	35	40	45	55
Steel brush diameter BS	d_b	[mm]	11	14	16	20		25	26	27	30	40			-	-	-
Steel brush diameter BSB	d_b		-	-	-	-		-	-	-	-	-			42	47	58



Table B9.2 **Maximum processing time** of the mortar and **minimum curing time**
(During the curing time of the mortar the concrete temperature may not fall below the listed minimum temperature)

Temperature at anchoring base [°C]	Maximum processing time t_{work}	Minimum curing time ¹⁾ t_{cure}
-5 to 0 ²⁾	240 min	200 h
> 0 to 5 ²⁾	150 min	90 h
> 5 to 10	120 min	40 h
> 10 to 20	30 min	18 h
> 20 to 30	14 min	10 h
> 30 to 40	7 min	5 h

¹⁾ In wet concrete or water filled holes the curing times must be doubled

²⁾ Minimal cartridge temperature +5°C

fischer injection system FIS EM Plus

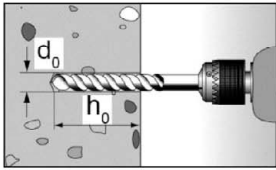
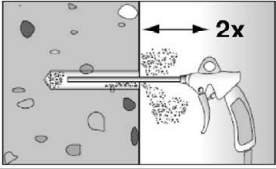
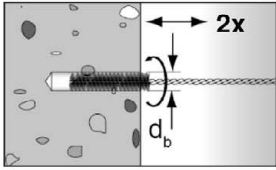
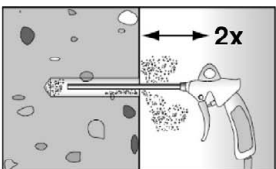
Intended use

Cleaning brush (steel brush)
Processing time and curing time

Annex B 9

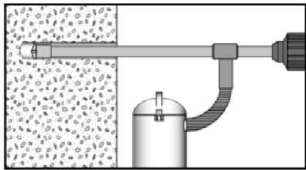
Installation instructions part 1

Drilling and cleaning the hole (hammer drilling with standard drill bit)

1		Drill the hole. Nominal drill hole diameter d_0 and drill hole depth h_0 see tables B3.1, B6.1, B7.1, B8.1	
2		Cleaning the drill hole: Blow out the drill hole twice, with oil free compressed air ($p \geq 6$ bar)	
3		Brush the drill hole twice. For drill hole diameter ≥ 30 mm use a power drill. For deep holes use an extension. Corresponding brushes see table B9.1	
4		Cleaning the drill hole: Blow out the drill hole twice, with oil free compressed air ($p \geq 6$ bar)	

Go to step 6

Drilling and cleaning the hole (hammer drilling with hollow drill bit)

1		Check a suitable hollow drill (see table B1.1) for correct operation of the dust extraction	
2		Use a suitable dust extraction system, e. g. fischer FVC 35 M or a comparable dust extraction system with equivalent performance data Drill the hole with hollow drill bit. The dust extraction system has to extract the drill dust nonstop during the drilling process and must be adjusted to maximum power. Nominal drill hole diameter d_0 and drill hole depth h_0 see tables B3.1, B6.1, B7.1, B8.1	

Go to step 6

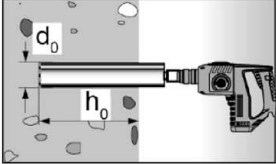
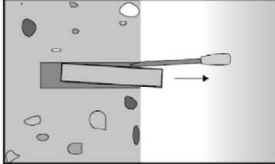
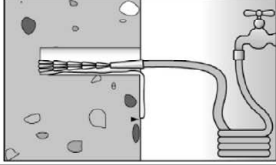
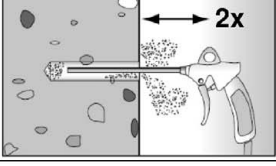
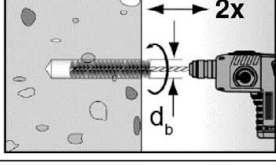
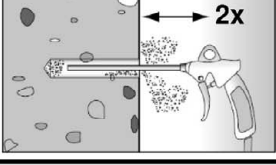
fischer injection system FIS EM Plus

Intended use
Installation instructions part 1

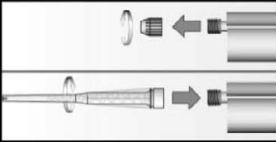
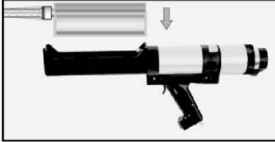
Annex B 10

Installation instructions part 2

Drilling and cleaning the hole (wet drilling with diamond drill bit)

1		Drill the hole. Drill hole diameter d_0 and nominal drill hole depth h_0 see tables B3.1, B6.1, B7.1, B8.1		Break the drill core and remove it
2		Flush the drill hole with clean water until it flows clear		
3		Blow out the drill hole twice, using oil-free compressed air ($p > 6$ bar)		
4		Brush the drill hole twice using a power drill. Corresponding brushes see table B9.1		
5		Blow out the drill hole twice, using oil-free compressed air ($p > 6$ bar)		

Preparing the cartridge

6		Remove the sealing cap Screw on the static mixer (the spiral in the static mixer must be clearly visible)		
7			Place the cartridge into the dispenser	
8			Extrude approximately 10 cm of material out until the resin is evenly grey in colour. Do not use mortar that is not uniformly grey	

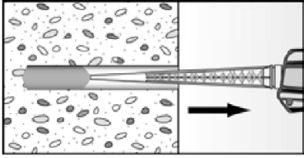
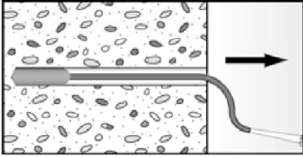
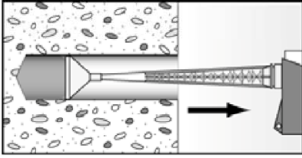
fischer injection system FIS EM Plus

Intended use
Installation instructions part 2

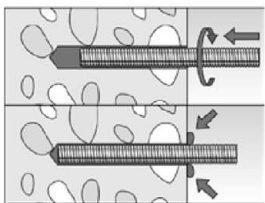
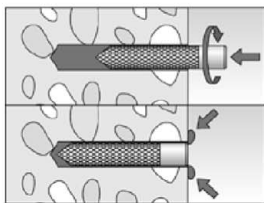
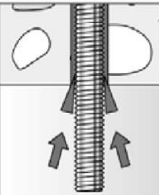
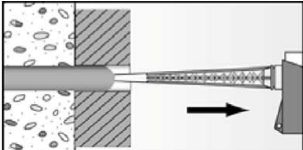
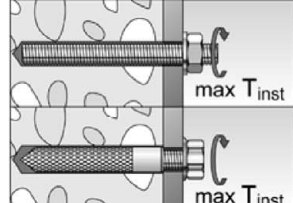
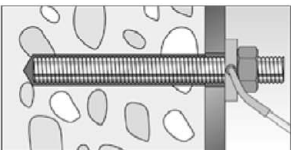
Annex B 11

Installation instructions part 3

Injection of the mortar

9	 Fill approximately 2/3 of the drill hole with mortar. Always begin from the bottom of the hole and avoid bubbles	 For drill hole depth ≥ 150 mm use an extension tube	 For overhead installation, deep holes ($h_0 > 250$ mm) or drill hole diameter ($d_0 \geq 40$ mm) use an injection-adaptor
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Installation of anchor rods or fischer internal threaded anchors RG MI

10	 	Only use clean and oil-free metal parts. Mark the setting depth of the metal parts. Push the anchor rod or fischer internal threaded RG MI anchor down to the bottom of the hole, turning it slightly while doing so. After inserting the metal part, excess mortar must be emerged around the anchor element.
	 For overhead installations support the metal part with wedges (e. g. fischer centering wedges) or fischer overhead clips.	 For push through installation fill the annular gap with mortar
11	 Wait for the specified curing time t_{cure} see table B9.2	12  Mounting the fixture max T_{inst} see tables B3.1 and B6.1
Option		After the minimum curing time is reached, the gap between metal part and fixture (annular clearance) may be filled with mortar via the fischer filling disc. Compressive strength ≥ 50 N/mm² (e.g. fischer injection mortars FIS HB, FIS SB, FIS V, FIS EM Plus) ATTENTION: Using fischer filling disc reduces t_{fix} (usable length of the anchor)

fischer injection system FIS EM Plus

Intended use
Installation instructions part 3

Annex B 12

Installation instructions part 4

Installation reinforcing bars and fischer rebar anchor FRA

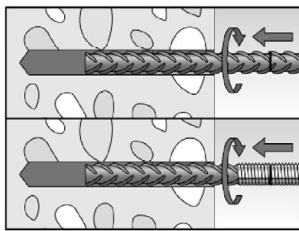
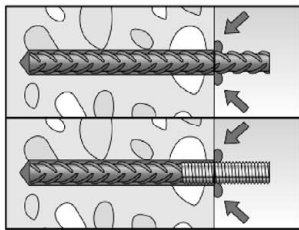
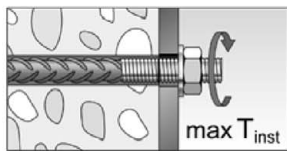
10		Only use clean and oil-free reinforcing bars or fischer FRA. Mark the setting depth. Turn while using force to push the reinforcement bar or the fischer FRA into the filled hole up to the setting depth mark			
		When the setting depth mark is reached, excess mortar must be emerged from the mouth of the drill hole.			
11		Wait for the specified curing time t_{cure} see table B9.2	12		Mounting the fixture max T_{inst} see table B8.1
fischer injection system FIS EM Plus					Annex B 13
Intended use Installation instructions part 4					

Table C1.1: Characteristic values for **steel failure** under tension / shear load of **fischer anchor rods** and **standard threaded rods**

Anchor rod / standard threaded rod				M8	M10	M12	M14	M16	M20	M22	M24	M27	M30	
Bearing capacity under tension load, steel failure ³⁾														
Characteristic resistance $N_{Rk,s}$	Steel zinc plated	Property class	4.8	[kN]	15(13)	23(21)	33	46	63	98	121	141	184	224
			5.8		19(17)	29(27)	43	58	79	123	152	177	230	281
			8.8		29(27)	47(43)	68	92	126	196	243	282	368	449
	Stainless steel R and high corrosion resistant steel HCR		50		19	29	43	58	79	123	152	177	230	281
			70		26	41	59	81	110	172	212	247	322	393
			80		30	47	68	92	126	196	243	282	368	449
Partial factors ¹⁾														
Partial factor $\gamma_{Ms,N}$	Steel zinc plated	Property class	4.8	[-]	1,50									
			5.8		1,50									
			8.8		1,50									
	Stainless steel R and high corrosion resistant steel HCR		50		2,86									
			70		1,50 ²⁾ / 1,87									
			80		1,60									
Bearing capacity under shear load, steel failure ³⁾														
without lever arm														
Characteristic resistance $V_{Rk,s}$	Steel zinc plated	Property class	4.8	[kN]	9(8)	14(13)	20	28	38	59	73	85	110	135
			5.8		11(10)	17(16)	25	34	47	74	91	106	138	168
			8.8		15(13)	23(21)	34	46	63	98	122	141	184	225
	Stainless steel R and high corrosion resistant steel HCR		50		9	15	21	29	39	61	76	89	115	141
			70		13	20	30	40	55	86	107	124	161	197
			80		15	23	34	46	63	98	122	141	184	225
Ductility factor			k ₇	[-]	1,0									
with lever arm														
Charact. resistance $M_{Rk,s}^0$	Steel zinc plated	Property class	4.8	[Nm]	15(13)	30(27)	52	83	133	259	357	448	665	899
			5.8		19(16)	37(33)	65	104	166	324	447	560	833	1123
			8.8		30(26)	60(53)	105	167	266	519	716	896	1333	1797
	Stainless steel R and high corrosion resistant steel HCR		50		19	37	65	104	166	324	447	560	833	1123
			70		26	52	92	146	232	454	626	784	1167	1573
			80		30	60	105	167	266	519	716	896	1333	1797
Partial factors ¹⁾														
Partial factor $\gamma_{Ms,V}$	Steel zinc plated	Property class	4.8	[-]	1,25									
			5.8		1,25									
			8.8		1,25									
	Stainless steel R and high corrosion resistant steel HCR		50		2,38									
			70		1,25 ²⁾ / 1,56									
			80		1,33									
¹⁾ In absence of other national regulations ²⁾ Only admissible for high corrosion resist. steel HCR, with $f_{yk}/f_{uk} \geq 0,8$ and $A_5 > 12\%$ (e.g. fischer anchor rods) ³⁾ Values in brackets are valid for undersized threaded rods with smaller stress area A_s for hot dip galvanised standard threaded rods according to EN ISO 10684:2004+AC:2009														
fischer injection system FIS EM Plus										Annex C 1				
Performances Characteristic values for steel failure under tension / shear load of fischer anchor rods and standard threaded rods														

Table C2.1: Characteristic values for **steel failure** under tension / shear load of **fischer internal threaded anchors RG MI**

fischer internal threaded anchors RG MI				M8	M10	M12	M16	M20	
Bearing capacity under tension load, steel failure									
Charact. resistance with screw	N _{Rk,s}	Property class	5.8	[kN]	19	29	43	79	123
			8.8		29	47	68	108	179
		Property class 70	R		26	41	59	110	172
			HCR		26	41	59	110	172
Partial factors ¹⁾									
Partial factors	γ _{Ms,N}	Property class	5.8	[-]	1,50				
			8.8		1,50				
		Property class 70	R		1,87				
			HCR		1,87				
Bearing capacity under shear load, steel failure									
Without lever arm									
Charact. resistance with screw	V ⁰ _{Rk,s}	Property class	5.8	[kN]	9,2	14,5	21,1	39,2	62,0
			8.8		14,6	23,2	33,7	54,0	90,0
		Property class 70	R		12,8	20,3	29,5	54,8	86,0
			HCR		12,8	20,3	29,5	54,8	86,0
Ductility factor			k ₇	[-]	1,0				
With lever arm									
Charact. resistance with screw	M ⁰ _{Rk,s}	Property class	5.8	[Nm]	20	39	68	173	337
			8.8		30	60	105	266	519
		Property class 70	R		26	52	92	232	454
			HCR		26	52	92	232	454
Partial factors ¹⁾									
Partial factors	γ _{Ms,V}	Property class	5.8	[-]	1,25				
			8.8		1,25				
		Property class 70	R		1,56				
			HCR		1,56				
¹⁾ In absence of other national regulations									
fischer injection system FIS EM Plus								Annex C 2	
Performances Characteristic values for steel failure under tension / shear load of fischer internal threaded anchor RG MI									

Table C3.1: Characteristic values for **steel failure** under tension / shear load of **reinforcing bars**

Nominal diameter of the bar	ϕ	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40
Bearing capacity under tension load, steel failure																		
Characteristic resistance	$N_{Rk,s}$	[kN]	$A_s \cdot f_{uk}^{1)}$															
Bearing capacity under shear load, steel failure																		
Without lever arm																		
Characteristic resistance	$V^0_{Rk,s}$	[kN]	$0,5 \cdot A_s \cdot f_{uk}^{1)}$															
Ductility factor	k_7	[-]	1,0															
With lever arm																		
Characteristic resistance	$M^0_{Rk,s}$	[Nm]	$1,2 \cdot W_{el} \cdot f_{uk}^{1)}$															

¹⁾ f_{uk} or f_{yk} respectively must be taken from the specifications of the reinforcing bar

Table C3.2: Characteristic values for **steel failure** under tension / shear load of **fischer rebar anchors FRA**

fischer rebar anchor FRA			M12	M16	M20	M24
Bearing capacity under tension load, steel failure						
Characteristic resistance	N _{Rk,s}	[kN]	63	111	173	270
Partial factor ¹⁾						
Partial factor	γ _{Ms,N}	[-]	1,4			
Bearing capacity under shear load, steel failure						
Without lever arm						
Characteristic resistance	V ⁰ _{Rk,s}	[kN]	30	55	86	124
Ductility factor	k ₇	[-]	1,0			
With lever arm						
Characteristic resistance	M ⁰ _{Rk,s}	[Nm]	92	233	454	785
Partial factor ¹⁾						
Partial factor	γ _{Ms,V}	[-]	1,56			

¹⁾ In absence of other national regulations

fischer injection system FIS EM Plus

Performances

Characteristic values for steel failure under tension / shear load of reinforcing bars and fischer rebar anchors FRA

Annex C 3

Table C4.1: Characteristic values for concrete failure under tension / shear load

Size			All sizes																		
Tension load																					
Installation factor		γ_{inst}	[-]	See annex C 5 to C 12 and C 17 to C 18																	
Factors for the compressive strength of concrete > C20/25																					
Increasing factor for τ_{Rk}	C25/30	Ψ_c	[-]	1,02																	
	C30/37			1,04																	
	C35/45			1,06																	
	C40/50			1,07																	
	C45/55			1,08																	
	C50/60			1,09																	
Splitting failure																					
Edge distance	$h / h_{ef} \geq 2,0$	$C_{cr,sp}$	[mm]	1,0 h_{ef}																	
	$2,0 > h / h_{ef} > 1,3$			4,6 $h_{ef} - 1,8 h$																	
	$h / h_{ef} \leq 1,3$			2,26 h_{ef}																	
Spacing		$S_{cr,sp}$		2 $C_{cr,sp}$																	
Concrete cone failure																					
Uncracked concrete		$k_{ucr,N}$	[-]	11,0																	
Cracked concrete		$k_{cr,N}$		7,7																	
Edge distance		$C_{cr,N}$	[mm]	1,5 h_{ef}																	
Spacing		$S_{cr,N}$		2 $C_{cr,N}$																	
Factors for sustained tension load																					
Factor		Ψ_{sus}^0	[-]	_1)																	
Shear load																					
Installation factor		γ_{inst}	[-]	1,0																	
Concrete pry-out failure																					
Factor for pry-out failure		k_8	[-]	2,0																	
Concrete edge failure																					
Effective length of fastener in shear loading		l_f	[mm]	for $d_{nom} \leq 24$ mm: min (h_{ef} ; 12 d_{nom}) for $d_{nom} > 24$ mm: min (h_{ef} ; 8 d_{nom} ; 300 mm)																	
Calculation diameters																					
Size				M8	M10	M12	M14	M16	M20	M22	M24	M27	M30								
fischer anchor rods and standard threaded rods		d_{nom}	[mm]	8	10	12	14	16	20	22	24	27	30								
fischer internal threaded anchors RG MI		d_{nom}		12	16	18	_2)	22	28	_2)	_2)	_2)	_2)								
fischer rebar anchor FRA		d_{nom}		_2)	_2)	12	_2)	16	20	_2)	25	_2)	_2)								
Size (nominal diameter of the bar)			ϕ	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40	
Reinforcing bar			d_{nom}	[mm]	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40
¹⁾ No performance assessed ²⁾ Anchor type not part of the ETA																					
fischer injection system FIS EM Plus												Annex C 4									
Performances Characteristic values for concrete failure under tension / shear load																					

Table C5.1: Characteristic values for **combined pull-out** and concrete failure for **fischer anchor rods** and **standard threaded rods** in hammer or diamond drilled holes; **uncracked or cracked concrete; working life 50 years**

Anchor rod / standard threaded rod			M8	M10	M12	M14	M16	M20	M22	M24	M27	M30	
Combined pullout and concrete cone failure													
Calculation diameter		d	[mm]	8	10	12	14	16	20	22	24	27	30
Uncracked concrete													
Characteristic bond resistance in uncracked concrete C20/25													
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)													
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	18	18	18	17	17	16	15	15	15	14
	II: 50 °C / 72 °C			18	17	17	16	16	15	14	14	14	13
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)													
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	16	16	15	13	13	11	11	10	10	9
	II: 50 °C / 72 °C			15	14	14	13	12	11	10	10	9	9
Diamond-drilling (dry or wet concrete as well as water filled hole)													
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	16	15	13	12	12	10	10	10	9	9
	II: 50 °C / 72 °C			15	14	12	11	11	10	9	9	8	8
Installation factors													
Dry or wet concrete		γ_{inst}	[-]	1,0									
Water filled hole				1,4									
Cracked concrete													
Characteristic bond resistance in cracked concrete C20/25													
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)													
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	7,5	7,5	9	8,5	8,5	8,5	8,5	8,5	8,5	8,5
	II: 50 °C / 72 °C			7,5	7,5	9	8,5	8,5	8,5	8,5	8,5	8,5	8,5
Diamond - drilling (dry or wet concrete)													
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	7	7	7	7	6	6	7	7	7	7
	II: 50 °C / 72 °C			7	7	7	7	6	6	7	7	7	7
Hammer-drilling with standard drill bit or hollow drill bit and diamond-drilling (water filled hole)													
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	6	7,5	7,5	7	6	6	6	6	6	6
	II: 50 °C / 72 °C			6	7	7	7	6	6	6	6	6	6
Installation factors													
Dry or wet concrete		γ_{inst}	[-]	1,0									
Water filled hole				1,2				1,4					
fischer injection system FIS EM Plus										Annex C 5			
Performances													
Characteristic values for combined pull-out and concrete failure for fischer anchor rod and standard threaded rods; working life 50 years													

Table C6.1: Characteristic values for **combined pull-out** and concrete failure for **fischer anchor rods** and **standard threaded rods** in hammer or diamond drilled holes; **uncracked or cracked concrete; working life 100 years**

Anchor rod / standard threaded rod			M8	M10	M12	M14	M16	M20	M22	M24	M27	M30		
Combined pullout and concrete cone failure														
Calculation diameter		d	[mm]	8	10	12	14	16	20	22	24	27	30	
Uncracked concrete														
Characteristic bond resistance in uncracked concrete C20/25														
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)														
Temperature range	I: 35 °C / 60 °C		$\tau_{Rk,ucr}$	[N/mm ²]	18	18	18	17	17	16	15	15	15	14
	II: 50 °C / 72 °C				18	17	17	16	16	15	14	14	14	13
Diamond-drilling (dry or wet concrete)														
Temperature range	I: 35 °C / 60 °C		$\tau_{Rk,ucr}$	[N/mm ²]	16	15	13	12	12	10	10	10	9	9
	II: 50 °C / 72 °C				15	14	12	11	11	10	9	9	8	8
Installation factors														
Dry or wet concrete		γ_{inst}	[-]	1,0										
Working life 100 years	I: 35 °C / 60 °C		$\alpha_{100\text{ years}}$	[-]	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75
	II: 50 °C / 72 °C				0,55	0,60	0,60	0,65	0,65	0,65	0,65	0,65	0,65	0,65
Cracked concrete														
Characteristic bond resistance in cracked concrete C20/25														
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)														
Temperature range	I: 35 °C / 60 °C		$\tau_{Rk,cr}$	[N/mm ²]	7,5	7,5	9	8,5	8,5	8,5	8,5	8,5	8,5	8,5
	II: 50 °C / 72 °C				7,5	7,5	9	8,5	8,5	8,5	8,5	8,5	8,5	8,5
Diamond - drilling (dry or wet concrete)														
Temperature range	I: 35 °C / 60 °C		$\tau_{Rk,cr}$	[N/mm ²]	7	7	7	7	6	6	7	7	7	7
	II: 50 °C / 72 °C				7	7	7	7	6	6	7	7	7	7
Installation factors														
Dry or wet concrete		γ_{inst}	[-]	1,0										
Working life 100 years	I: 35 °C / 60 °C		$\alpha_{100\text{ years}}$	[-]	0,60	0,85	0,80	0,65	0,65	0,65	0,65	0,65	0,65	0,65
	II: 50 °C / 72 °C				0,60	0,85	0,80	0,65	0,65	0,65	0,65	0,65	0,65	0,65

1) Calculation of characteristic bond resistance in uncracked concrete $\tau_{Rk,100, ucr}$:

$$\tau_{Rk,100, ucr} = \alpha_{100 \text{ years}} \cdot \tau_{Rk,ucr}$$

2) Calculation of characteristic bond resistance in cracked concrete $\tau_{Rk,100, cr}$:

$$\tau_{Rk,100, cr} = \alpha_{100 \text{ years}} \cdot \tau_{Rk,cr}$$

fischer injection system FIS EM Plus

Performances

Characteristic values for combined pull-out and concrete failure for fischer anchor rod and standard threaded rods; working life 100 years

Annex C 6

Table C7.1: Characteristic values for **combined pull-out** and concrete failure for **fischer internal threaded anchors RG MI** in hammer or diamond drilled holes; **uncracked or cracked concrete; working life 50 years**

Internal threaded anchor RG MI			M8	M10	M12	M16	M20	
Combined pullout and concrete cone failure								
Calculation diameter		d	[mm]	12	16	18	22	28
Uncracked concrete								
Characteristic bond resistance in uncracked concrete C20/25								
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)								
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	15	14	14	13	12
	II: 50 °C / 72 °C			14	13	13	12	11
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)								
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	14	12	12	11	10
	II: 50 °C / 72 °C			13	12	11	10	9
Diamond-drilling (dry or wet concrete as well as water filled hole)								
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	13	12	11	10	9
	II: 50 °C / 72 °C			12	11	10	9	8
Installation factors								
Dry or wet concrete		γ_{inst}	[-]	1,0				
Water filled hole				1,4				
Cracked concrete								
Characteristic bond resistance in cracked concrete C20/25								
Hammer-drilling with standard drill bit or hollow drill bit and diamond-drilling (dry or wet concrete)								
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	7	6	6	7	7
	II: 50 °C / 72 °C			7	6	6	7	7
Hammer-drilling with standard drill bit or hollow drill bit and diamond-drilling (water filled hole)								
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	7	6,5	6	6	6
	II: 50 °C / 72 °C			7	6	6	6	6
Installation factors								
Dry or wet concrete		γ_{inst}	[-]	1,0				
Water filled hole				1,2		1,4		

Table C8.1: Characteristic values for **combined pull-out** and concrete failure for **fischer internal threaded anchors RG MI** in hammer or diamond drilled holes; **uncracked or cracked concrete; working life 100 years**

Internal threaded anchor RG MI			M8	M10	M12	M16	M20	
Combined pullout and concrete cone failure								
Calculation diameter		d	[mm]	12	16	18	22	28
Uncracked concrete								
Characteristic bond resistance in uncracked concrete C20/25								
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)								
Temperature range	I: 35 °C / 60 °C		$\tau_{Rk,ucr}$ [N/mm ²]	15	14	14	13	12
	II: 50 °C / 72 °C			14	13	13	12	11
Diamond-drilling (dry or wet concrete)								
Temperature range	I: 35 °C / 60 °C		$\tau_{Rk,ucr}$ [N/mm ²]	13	12	11	10	9
	II: 50 °C / 72 °C			12	11	10	9	8
Installation factors								
Dry or wet concrete		γ_{inst}	[-]	1,0				
Working life 100 years	I: 35 °C / 60 °C		$\alpha_{100 \text{ years}}$ [-]	0,75	0,75	0,75	0,75	0,75
	II: 50 °C / 72 °C			0,55	0,60	0,60	0,65	0,65
Cracked concrete								
Characteristic bond resistance in cracked concrete C20/25								
Hammer-drilling with standard drill bit or hollow drill bit and diamond-drilling (dry or wet concrete)								
Temperature range	I: 35 °C / 60 °C		$\tau_{Rk,cr}$ [N/mm ²]	7	6	6	7	7
	II: 50 °C / 72 °C			7	6	6	7	7
Installation factors								
Dry or wet concrete		γ_{inst}	[-]	1,0				
Working life 100 years	I: 35 °C / 60 °C		$\alpha_{100 \text{ years}}$ [-]	0,60	0,85	0,80	0,65	0,65
	II: 50 °C / 72 °C			0,60	0,85	0,80	0,65	0,65

1) Calculation of characteristic bond resistance in uncracked concrete $\tau_{Rk,100,ucr}$:

$$\tau_{Rk,100,ucr} = \alpha_{100 \text{ years}} \cdot \tau_{Rk,ucr}$$

2) Calculation of characteristic bond resistance in cracked concrete $\tau_{Rk,100,cr}$:

$$\tau_{Rk,100,cr} = \alpha_{100 \text{ years}} \cdot \tau_{Rk,cr}$$

fischer injection system FIS EM Plus

Performances

Characteristic values for combined pull-out and concrete failure for fischer internal threaded anchors RG MI; working life 100 years

Annex C 8

Table C9.1: Characteristic values for **combined pull-out** and concrete failure for **reinforcing bars** in hammer or diamond drilled holes; **uncracked or cracked concrete; working life 50 years**

Nominal diameter of the bar ϕ			8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40	
Combined pullout and concrete cone failure																				
Calculation diameter		d	[mm]	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40
Uncracked concrete																				
Characteristic bond resistance in uncracked concrete C20/25																				
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	16	15	15	14	14	13	13	13	12	12	12	12	12	11	11	11	
	II: 50 °C / 72 °C			15	14	14	13	13	12	12	12	12	11	11	11	11	11	10	10	
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	16	16	14	13	12	12	11	11	10	10	10	10	9	9	9	8	8
	II: 50 °C / 72 °C			15	14	13	12	12	11	11	10	10	9	9	9	9	8	8	8	8
Diamond-drilling (dry or wet concrete as well as water filled hole)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	16	15	13	12	12	11	10	10	10	9	9	9	9	8	8	8	7
	II: 50 °C / 72 °C			15	14	12	11	11	10	10	9	9	9	8	8	8	8	7	7	7
Installation factors																				
Dry or wet concrete		γ_{inst}	[-]	1,0																
Water filled hole				1,4																
Cracked concrete																				
Characteristic bond resistance in cracked concrete C20/25																				
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	7	7	8	8	8	8	8	8	8	8	8	8	8	8	8	8	
	II: 50 °C / 72 °C			7	7	8	8	8	8	8	8	8	8	8	8	8	8	8	8	
Diamond-drilling (dry or wet concrete)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	7	7	7	7	6	6	6	7	7	7	7	7	5	5	5	5	
	II: 50 °C / 72 °C			7	7	7	7	6	6	6	7	7	7	7	7	5	5	5	5	
Hammer-drilling with standard drill bit or hollow drill bit and diamond-drilling (water filled hole)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	6	7,5	6,5	6,5	6,5	6	6	6	6	6	6	6	5	5	5	5	
	II: 50 °C / 72 °C			6	6,5	6,5	6	6	6	6	6	6	6	6	6	5	5	5	5	
Installation factors																				
Dry or wet concrete		γ_{inst}	[-]	1,0																
Water filled hole				1,2				1,4												
fischer injection system FIS EM Plus													Annex C 9							
Performances Characteristic values for combined pull-out and concrete failure for reinforcing bars; working life 50 years																				

Table C10.1: Characteristic values for **combined pull-out** and concrete failure for **reinforcing bars** in hammer or diamond drilled holes; **uncracked or cracked concrete; working life 100 years**

Nominal diameter of the bar ϕ			8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40	
Combined pullout and concrete cone failure																				
Calculation diameter		d	[mm]	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40
Uncracked concrete																				
Characteristic bond resistance in uncracked concrete C20/25																				
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	16	15	15	14	14	13	13	13	12	12	12	12	12	11	11	11	
	II: 50 °C / 72 °C			15	14	14	13	13	12	12	12	12	11	11	11	11	11	11	10	10
Diamond-drilling (dry or wet concrete)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm ²]	16	15	13	12	12	11	10	10	10	9	9	9	9	8	8	8	7
	II: 50 °C / 72 °C			15	14	12	11	11	10	10	9	9	9	8	8	8	8	7	7	7
Installation factors																				
Dry or wet concrete		γ_{inst}	[-]	1,0																
Working life 100 years	I: 35 °C / 60 °C	$\alpha_{100\text{ years}}$	[-]	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	0,75	
	II: 50 °C / 72 °C			0,55	0,60	0,60	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	
Cracked concrete																				
Characteristic bond resistance in cracked concrete C20/25																				
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	7	7	8	8	8	8	8	8	8	8	8	8	8	8	8	8	
	II: 50 °C / 72 °C			7	7	8	8	8	8	8	8	8	8	8	8	8	8	8	8	
Diamond-drilling (dry or wet concrete)																				
Tem- perature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm ²]	7	7	7	7	6	6	6	7	7	7	7	7	5	5	5	5	
	II: 50 °C / 72 °C			7	7	7	7	6	6	6	7	7	7	7	7	5	5	5	5	
Installation factors																				
Dry or wet concrete		γ_{inst}	[-]	1,0																
Working life 100 years	I: 35 °C / 60 °C	$\alpha_{100\text{ years}}$	[-]	0,60	0,85	0,80	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	
	II: 50 °C / 72 °C			0,60	0,85	0,80	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	0,65	

1) Calculation of characteristic bond resistance in uncracked concrete $\tau_{Rk,100,ucr}$:

$$\tau_{Rk,100,ucr} = \alpha_{100\text{ years}} \cdot \tau_{Rk,ucr}$$

2) Calculation of characteristic bond resistance in cracked concrete $\tau_{Rk,100,cr}$:

$$\tau_{Rk,100,cr} = \alpha_{100\text{ years}} \cdot \tau_{Rk,cr}$$

fischer injection system FIS EM Plus

Performances

Characteristic values for combined pull-out and concrete failure for reinforcing bars;
working life 100 years

Annex C 10

Table C11.1: Characteristic values for **combined pull-out** and concrete failure for **fischer rebar anchors FRA** in hammer or diamond drilled holes; **uncracked or cracked concrete; working life 50 years**

fischer rebar anchor FRA			M12	M16	M20	M24	
Combined pullout and concrete cone failure							
Calculation diameter		d	[mm]	12	16	20	25
Uncracked concrete							
Characteristic bond resistance in uncracked concrete C20/25							
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)							
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm²]	15	14	13	12
	II: 50 °C / 72 °C			14	13	12	12
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)							
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm²]	14	12	11	10
	II: 50 °C / 72 °C			13	12	11	9
Diamond-drilling (dry or wet concrete as well as water filled hole)							
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,ucr}$	[N/mm²]	13	12	10	9
	II: 50 °C / 72 °C			12	11	10	9
Installation factors							
Dry or wet concrete		γ_{inst}	[-]	1,0			
Water filled hole				1,4			
Cracked concrete							
Characteristic bond resistance in cracked concrete C20/25							
Hammer-drilling with standard drill bit or hollow drill bit and diamond-drilling (dry or wet concrete)							
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm²]	8	8	8	8
	II: 50 °C / 72 °C			8	8	8	8
Hammer-drilling with standard drill bit or hollow drill bit and diamond-drilling (water filled hole)							
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,cr}$	[N/mm²]	7	6	6	6
	II: 50 °C / 72 °C			7	6	6	6
Installation factors							
Dry or wet concrete		γ_{inst}	[-]	1,0			
Water filled hole				1,2		1,4	

Table C12.1: Characteristic values for **combined pull-out** and concrete failure for **fischer rebar anchors FRA** in hammer or diamond drilled holes; **uncracked or cracked concrete; working life 100 years**

fischer rebar anchor FRA			M12	M16	M20	M24		
Combined pullout and concrete cone failure								
Calculation diameter		d	[mm]	12	16	20	25	
Uncracked concrete								
Characteristic bond resistance in uncracked concrete C20/25								
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)								
Tem- perature range	I: 35 °C / 60 °C		$\tau_{Rk,ucr}$	[N/mm ²]	15	14	13	12
	II: 50 °C / 72 °C				14	13	12	12
Diamond-drilling (dry or wet concrete)								
Tem- perature range	I: 35 °C / 60 °C		$\tau_{Rk,ucr}$	[N/mm ²]	13	12	10	9
	II: 50 °C / 72 °C				12	11	10	9
Installation factors								
Dry or wet concrete		γ_{inst}	[-]	1,0				
Working life 100 years	I: 35 °C / 60 °C		$\alpha_{100\text{ years}}$	[N/mm ²]	0,75	0,75	0,75	0,75
	II: 50 °C / 72 °C				0,60	0,65	0,65	0,65
Cracked concrete								
Characteristic bond resistance in cracked concrete C20/25								
Hammer-drilling with standard drill bit or hollow drill bit and diamond-drilling (dry or wet concrete)								
Tem- perature range	I: 35 °C / 60 °C		$\tau_{Rk,cr}$	[N/mm ²]	8	8	8	8
	II: 50 °C / 72 °C				8	8	8	8
Installation factors								
Dry or wet concrete		γ_{inst}	[-]	1,0				
Working life 100 years	I: 35 °C / 60 °C		$\alpha_{100\text{ years}}$	[-]	0,80	0,65	0,65	0,65
	II: 50 °C / 72 °C				0,80	0,65	0,65	0,65

1) Calculation of characteristic bond resistance in uncracked concrete $\tau_{Rk,100,ucr}$:

$$\tau_{Rk,100,ucr} = \alpha_{100\text{ years}} \cdot \tau_{Rk,ucr}$$

2) Calculation of characteristic bond resistance in cracked concrete $\tau_{Rk,100,cr}$:

$$\tau_{Rk,100,cr} = \alpha_{100\text{ years}} \cdot \tau_{Rk,cr}$$

fischer injection system FIS EM Plus

Performances

Characteristic values for pull-out and concrete failure for fischer rebar anchors FRA;
working life 100 years

Annex C 12

Table C13.1: Displacements for anchor rods

Anchor rod		M8	M10	M12	M14	M16	M20	M22	M24	M27	M30
Displacement-Factors for tension load ¹⁾											
Uncracked or cracked concrete; Temperature range I, II											
δ _{N0} -Factor	[mm/(N/mm ²)]	0,07	0,08	0,09	0,09	0,10	0,11	0,11	0,12	0,12	0,13
δ _{N∞} -Factor		0,11	0,12	0,13	0,14	0,15	0,16	0,17	0,18	0,19	0,19
Displacement-Factors for shear load ²⁾											
Uncracked or cracked concrete; Temperature range I, II											
δ _{V0} -Factor	[mm/kN]	0,18	0,15	0,12	0,10	0,09	0,07	0,07	0,06	0,05	0,05
δ _{V∞} -Factor		0,27	0,22	0,18	0,16	0,14	0,11	0,10	0,09	0,08	0,07
1) Calculation of effective displacement: δ _{N0} = δ _{N0-Factor} · τ _{Ed} δ _{N∞} = δ _{N∞-Factor} · τ _{Ed} (τ _{Ed} : Design value of the applied tensile stress)						2) Calculation of effective displacement: δ _{V0} = δ _{V0-Factor} · V _{Ed} δ _{V∞} = δ _{V∞-Factor} · V _{Ed} (V _{Ed} : Design value of the applied shear force)					

Table C13.2: Displacements for fischer internal threaded anchors RG MI

Internal threaded anchor RG MI		M8	M10	M12	M16	M20
Displacement-Factors for tension load ¹⁾						
Uncracked or cracked concrete; Temperature range I, II						
δN0-Factor	[mm/(N/mm²)]	0,09	0,10	0,10	0,11	0,13
δN∞-Factor		0,13	0,15	0,16	0,17	0,19
Displacement-Factors for shear load ²⁾						
Uncracked or cracked concrete; Temperature range I, II						
δV0-Factor	[mm/kN]	0,12	0,09	0,08	0,07	0,05
δV∞-Factor		0,18	0,14	0,12	0,10	0,08
1) Calculation of effective displacement: δN0 = δN0-Factor · τEd δN∞ = δN∞-Factor · τEd (τEd: Design value of the applied tensile stress)				2) Calculation of effective displacement: δV0 = δV0-Factor · VEd δV∞ = δV∞-Factor · VEd (VEd: Design value of the applied shear force)		

fischer injection system FIS EM Plus

Performances

Displacements for anchor rods and fischer internal threaded anchors RG MI

Annex C 13

Table C14.1: Displacements for reinforcing bars

Nominal diameter of the bar		ϕ	8	10	12	14	16	18	20	22	24	25	26	28	30	32	34	36	40
Displacement-Factors for tension load ¹⁾																			
Uncracked or cracked concrete; Temperature range I, II																			
δ _{N0} -Factor	[mm/(N/mm ²)]	0,07	0,08	0,09	0,09	0,10	0,10	0,11	0,11	0,12	0,12	0,12	0,13	0,13	0,13	0,14	0,14	0,15	
δ _{N∞} -Factor		0,11	0,12	0,13	0,14	0,15	0,16	0,16	0,17	0,18	0,18	0,18	0,19	0,19	0,20	0,20	0,21	0,22	
Displacement-Factors for shear load ²⁾																			
Uncracked or cracked concrete; Temperature range I, II																			
δ _{V0} -Factor	[mm/kN]	0,18	0,15	0,12	0,10	0,09	0,08	0,07	0,07	0,06	0,06	0,06	0,05	0,05	0,05	0,04	0,04	0,04	
δ _{V∞} -Factor		0,27	0,22	0,18	0,16	0,14	0,12	0,11	0,10	0,09	0,09	0,08	0,08	0,07	0,07	0,06	0,06	0,05	
1) Calculation of effective displacement: δ_{N0} = δ_{N0}-Factor · τ_{Ed} δ_{N∞} = δ_{N∞}-Factor · τ_{Ed} (τ_{Ed}: Design value of the applied tensile stress)																			
2) Calculation of effective displacement: δ_{V0} = δ_{V0}-Factor · V_{Ed} δ_{V∞} = δ_{V∞}-Factor · V_{Ed} (V_{Ed}: Design value of the applied shear force)																			

Table C14.2: Displacements for fischer rebar anchors FRA

fischer rebar anchor FRA		M12	M16	M20	M24
Displacement-Factors for tension load ¹⁾					
Uncracked or cracked concrete; Temperature range I, II					
δ _{N0} -Factor	[mm/(N/mm ²)]	0,09	0,10	0,11	0,12
δ _{N∞} -Factor		0,13	0,15	0,16	0,18
Displacement-Factors for shear load ²⁾					
Uncracked or cracked concrete; Temperature range I, II					
δ _{V0} -Factor	[mm/kN]	0,12	0,09	0,07	0,06
δ _{V∞} -Factor		0,18	0,14	0,11	0,09
1) Calculation of effective displacement: δ _{N0} = δ _{N0-Factor} · τ _{Ed} δ _{N∞} = δ _{N∞-Factor} · τ _{Ed} (τ _{Ed} : Design value of the applied tensile stress)			2) Calculation of effective displacement: δ _{V0} = δ _{V0-Factor} · V _{Ed} δ _{V∞} = δ _{V∞-Factor} · V _{Ed} (V _{Ed} : Design value of the applied shear force)		

fischer injection system FIS EM Plus

Performance

Displacements for reinforcing bars and fischer rebar anchors FRA

Annex C 14

Table C15.1: Characteristic values for **steel failure** under tension / shear load of **fischer anchor rods and standard threaded rods** under seismic action performance category **C1 or C2**

Anchor rod / standard threaded rod				M10	M12	M14	M16	M20	M22	M24	M27	M30	
Bearing capacity under tension load, steel failure ¹⁾													
fischer anchor rods and standard threaded rods, performance category C1 ²⁾													
Characteristic resistance $N_{Rk,s,C1}$	Steel zinc plated	Property class	5.8	[kN]	29(27)	43	58	79	123	152	177	230	281
			8.8		47(43)	68	92	126	196	243	282	368	449
	Stainless steel R and high corrosion resistant steel HCR		50		29	43	58	79	123	152	177	230	281
			70		41	59	81	110	172	212	247	322	393
			80		47	68	92	126	196	243	282	368	449
fischer anchor rods and standard threaded rods, performance category C2 ²⁾													
Characteristic resistance $N_{Rk,s,C2}$	Steel zinc plated	Property class	5.8	[-]	- ⁴⁾	39	- ⁴⁾	72	108	- ⁴⁾	177	- ⁴⁾	- ⁴⁾
			8.8		- ⁴⁾	61	- ⁴⁾	116	173	- ⁴⁾	282	- ⁴⁾	- ⁴⁾
	Stainless steel R and high corrosion resistant steel HCR		50		- ⁴⁾	39	- ⁴⁾	72	108	- ⁴⁾	177	- ⁴⁾	- ⁴⁾
			70		- ⁴⁾	53	- ⁴⁾	101	152	- ⁴⁾	247	- ⁴⁾	- ⁴⁾
			80		- ⁴⁾	61	- ⁴⁾	116	173	- ⁴⁾	282	- ⁴⁾	- ⁴⁾
Bearing capacity under shear load, steel failure without lever arm ¹⁾													
fischer anchor rods, performance category C1 ²⁾													
Characteristic resistance $V_{Rk,s,C1}$	Steel zinc plated	Property class	5.8	[kN]	17(16)	25	34	47	74	91	106	138	168
			8.8		23(21)	34	46	63	98	122	141	184	225
	Stainless steel R and high corrosion resistant steel HCR		50		15	21	29	39	61	76	89	115	141
			70		20	30	40	55	86	107	124	161	197
			80		23	34	46	63	98	122	141	184	225
Standard threaded rods, performance category C1 ²⁾													
Characteristic resistance $V_{Rk,s,C1}$	Steel zinc plated	Property class	5.8	[kN]	12(11)	17	24	33	52	64	74	97	118
			8.8		16(14)	24	32	44	69	85	99	129	158
	Stainless steel R and high corrosion resistant steel HCR		50		11	15	20	27	43	53	62	81	99
			70		14	21	28	39	60	75	87	113	138
			80		16	24	32	44	69	85	99	129	158
fischer anchor rods and standard threaded rods, performance category C2													
Characteristic resistance $V_{Rk,s,C2}$	Steel zinc plated	Property class	5.8	[-]	- ⁴⁾	14	- ⁴⁾	27	43	- ⁴⁾	62	- ⁴⁾	- ⁴⁾
			8.8		- ⁴⁾	22	- ⁴⁾	44	69	- ⁴⁾	99	- ⁴⁾	- ⁴⁾
	Stainless steel R and high corrosion resistant steel HCR		50		- ⁴⁾	14	- ⁴⁾	27	43	- ⁴⁾	62	- ⁴⁾	- ⁴⁾
			70		- ⁴⁾	20	- ⁴⁾	39	60	- ⁴⁾	87	- ⁴⁾	- ⁴⁾
			80		- ⁴⁾	22	- ⁴⁾	44	69	- ⁴⁾	99	- ⁴⁾	- ⁴⁾
Factor for the annular gap		α_{gap}	[-]	0,5 (1,0) ³⁾									
1) Partial factors for performance category C1 or C2 see table C1.1; for fischer anchor rods FIS A / RGM the factor for steel ductility is 1,0 2) Values in brackets are valid for undersized threaded rods with smaller stress area A_s for hot dip galvanised standard threaded rods according to EN ISO 10684:2004+AC:2009. 3) Values in brackets are valid for filled annular gaps between the anchor rod and the through-hole in the attachment. It is necessary to use the fischer filling disc according to Annex A 1 4) No performance assessed													
fischer injection system FIS EM Plus										Annex C 15			
Performances Characteristic values for steel failure under tension / shear load for fischer anchor rods and standard threaded rods under seismic action (performance category C1 / C2)													

Table C16.1: Characteristic values for **steel failure** under tension / shear load for **reinforcing bars (B500B)** under seismic action performance category **C1**

Nominal diameter of the bar	ϕ	10	12	14	16	18	20	22	24	25	26	28	30	32
Bearing capacity under tension load, steel failure¹⁾														
Reinforcing bar B500B acc. to DIN 488-2:2009-08, performance category C1														
Characteristic resistance	$N_{Rk,s,C1}$	[kN]	44	63	85	111	140	173	209	249	270	292	339	443
Bearing capacity under shear load, steel failure without lever arm¹⁾														
Reinforcing bar B500B acc. to DIN 488-2:2009-08, performance category C1														
Characteristic resistance	$V_{Rk,s,C1}$	[kN]	15	22	30	39	49	61	74	88	95	102	119	155

¹⁾ Partial factors for performance category C1 see table C16.2

Table C16.2: Partial factors for **fischer anchor rods, standard threaded rods** and **reinforcing bars (B500B)** under seismic action performance category **C1 or C2**

Anchor rod / standard threaded rod				M10	M12	M14	M16	M20	M22	M24	M27	M30					
Nominal diameter of the bar				φ	10	12	14	16	18	20	22	24	25	26	28	30	32
Tension load, steel failure ¹⁾																	
Partial factor γ _{Ms,N}	Steel zinc plated	Property class	5.8	[-]	1,50												
			8.8		1,50												
	Stainless steel R and high corrosion resistant steel HCR	50	2,86														
		70	1,50 ²⁾ / 1,87														
		80	1,60														
	Reinforcing bar	B500B			1,40												
Shear load, steel failure ¹⁾																	
Partial factor γ _{Ms,V}	Steel zinc plated	Property class	5.8	[-]	1,25												
			8.8		1,25												
	Stainless steel R and high corrosion resistant steel HCR	50	2,38														
		70	1,25 ²⁾ / 1,56														
		80	1,33														
	Reinforcing bar	B500B			1,50												

¹⁾ In absence of other national regulations

²⁾ Only admissible for high corrosion resistant steel HCR, with $f_{yk} / f_{uk} \geq 0,8$ and $A_5 > 12 \%$ (e.g. fischer anchor rods)

fischer injection system FIS EM Plus

Performances

Characteristic values for steel failure under tension/shear load for reinforcing bars under seismic action (performance category C1); partial factors (perform. category C1 / C2)

Annex C 16

Table C17.1: Characteristic values for **combined pull-out** and concrete failure for **fischer anchor rods** and **standard threaded rods** in hammer drilled holes under seismic action performance category **C1**; **working life 50 and 100 years**

Anchor rod / standard threaded rod				M10	M12	M14	M16	M20	M22	M24	M27	M30
Characteristic bond resistance, combined pullout and concrete cone failure												
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)												
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,C1}$	[N/mm ²]	7,0	7,0	6,7	6,0	5,7	6,7	6,7	6,7	6,7
	II: 50 °C / 72 °C			7,0	7,0	6,7	5,7	5,7	6,7	6,7	6,7	6,7
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)												
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,C1}$	[N/mm ²]	7,5	7,5	6,5	5,7	5,7	5,7	5,7	5,7	5,7
	II: 50 °C / 72 °C			6,8	6,8	6,5	5,7	5,7	5,7	5,7	5,7	5,7
Installation factors												
Dry or wet concrete		γ_{inst}	[-]	1,0								
Water filled hole				1,2 ¹⁾				1,4 ¹⁾				

¹⁾ Not permitted in combination with working life 100 years

Table C17.2: Characteristic values for **combined pull-out** and concrete failure for **reinforcing bars** in hammer drilled holes under seismic action performance category **C1**; **working life 50 and 100 years**

Nominal diameter of the bar		ϕ	10	12	14	16	18	20	22	24	25	26	28	30	32
Characteristic bond resistance, combined pullout and concrete cone failure															
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)															
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,C1}$	[N/mm ²]	7,0	7,0	6,7	5,7	5,7	5,7	6,7	6,7	6,7	6,7	6,7	4,8
	II: 50 °C / 72 °C			7,0	7,0	6,7	5,7	5,7	5,7	6,7	6,7	6,7	6,7	6,7	4,8
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)															
Temperature range	I: 35 °C / 60 °C	$\tau_{Rk,C1}$	[N/mm ²]	7,5	6,5	6,5	5,7	5,7	5,7	5,7	5,7	5,7	5,7	5,7	4,8
	II: 50 °C / 72 °C			6,5	6,5	5,8	5,8	5,7	5,7	5,7	5,7	5,7	5,7	5,7	4,8
Installation factors															
Dry or wet concrete		γ_{inst}	[-]	1,0											
Water filled hole				1,2 ¹⁾					1,4 ¹⁾						

¹⁾ Not permitted in combination with working life 100 years

fischer injection system FIS EM Plus

Performances

Characteristic values for combined pull-out and concrete failure under seismic action (C1) for fischer anchor rods, stand. thread. rods and reinf. bars; working life 50 and 100 years

Annex C 17

Table C18.1: Characteristic values for **combined pull-out** and concrete failure for **fischer anchor rods** and **standard threaded rods** in hammer drilled holes under seismic action performance category **C2**; **working life 50 and 100 years**

Anchor rod / standard threaded rod		M12		M16		M20		M24	
Characteristic bond resistance, combined pullout and concrete cone failure									
Hammer-drilling with standard drill bit or hollow drill bit (dry or wet concrete)									
Temperature range	I: 35 °C / 60 °C		$\tau_{Rk,C2}$	[N/mm ²]	3,5	5,8	5,0	3,1	
	II: 50 °C / 72 °C				3,3	5,5	4,7	2,9	
Hammer-drilling with standard drill bit or hollow drill bit (water filled hole)									
Temperature range	I: 35 °C / 60 °C		$\tau_{Rk,C2}$	[N/mm ²]	3,5	5,8	5,0	3,1	
	II: 50 °C / 72 °C				3,3	5,5	4,7	2,9	
Installation factors									
Dry or wet concrete			γ_{inst}	[-]	1,0				
Water filled hole					1,2 ¹⁾		1,4 ¹⁾		
1) Not permitted in combination with working life 100 years									
Displacement-Factors for tension load ¹⁾									
$\delta_{N,C2}$ (DLS)-Factor			[mm/(N/mm ²)]	0,09	0,10	0,11	0,12		
$\delta_{N,C2}$ (ULS)-Factor				0,15	0,17	0,17	0,18		
Displacement-Factors for shear load ²⁾									
$\delta_{V,C2}$ (DLS)-Factor			[mm/kN]	0,18	0,10	0,07	0,06		
$\delta_{V,C2}$ (ULS)-Factor				0,25	0,14	0,11	0,09		
1) Calculation of effective displacement: $\delta_{N,C2} \text{ (DLS)} = \delta_{N,C2} \text{ (DLS)-Factor} \cdot \tau_{Ed}$$\delta_{N,C2} \text{ (ULS)} = \delta_{N,C2} \text{ (ULS)-Factor} \cdot \tau_{Ed}$$(\tau_{Ed}: \text{Design value of the applied tensile stress})$ 2) Calculation of effective displacement: $\delta_{V,C2} \text{ (DLS)} = \delta_{V,C2} \text{ (DLS)-Factor} \cdot V_{Ed}$$\delta_{V,C2} \text{ (ULS)} = \delta_{V,C2} \text{ (ULS)-Factor} \cdot V_{Ed}$$(V_{Ed}: \text{Design value of the applied shear force})$									
fischer injection system FIS EM Plus							Annex C 18		
Performances Characteristic values for combined pull-out and concrete failure under seismic action (C2) for fischer anchor rods and standard threaded rods; working life 50 and 100 years									

Approval body for construction products
and types of construction

Bautechnisches Prüfamt

An institution established by the Federal and
Laender Governments



European Technical Assessment

ETA-17/1056
of 22 January 2021

English translation prepared by DIBt - Original version in German language

General Part

Technical Assessment Body issuing the
European Technical Assessment:

Deutsches Institut für Bautechnik

Trade name of the construction product

Rebar connection with fischer
injection system FIS EM Plus

Product family
to which the construction product belongs

Injection system for post-installed
rebar connections

Manufacturer

fischerwerke GmbH & Co. KG
Otto-Hahn-Straße 15
79211 Denzlingen
DEUTSCHLAND

Manufacturing plant

fischerwerke

This European Technical Assessment
contains

26 pages including 3 annexes which form an integral part
of this assessment

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No 305/2011, on the basis of

EAD 330087-01-0601 Edition 01/2021

This version replaces

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European Technical Assessment

ETA-17/1056

English translation prepared by DIBt

Page 2 of 26 | 22 January 2021

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Specific Part**1 Technical description of the product**

The subject of this European technical assessment is the post-installed connection, by anchoring or overlap connection joint, of reinforcing bars (rebars) in existing structures made of normal weight concrete, using the injection mortar FIS EM Plus in accordance with the regulations for reinforced concrete construction.

Reinforcing bars with a diameter ϕ from 8 to 40 mm or the fischer rebar anchor FRA of sizes M12 to M24 according to Annex A and the fischer injection mortar FIS EM Plus are used for the post-installed rebar connection. The steel element is placed into a drilled hole filled with injection mortar and is anchored via the bond between embedded reinforcing bar, injection mortar and concrete.

The product description is given in Annex A.

2 Specification of the intended use in accordance with the applicable European assessment Document

The performances given in Section 3 are only valid if the anchor is used in compliance with the specifications and conditions given in Annex B.

The verifications and assessment methods on which this European Technical Assessment is based lead to the assumption of a working life of the rebar connections of at least 50 and/or 100 years. The indications given on the working life cannot be interpreted as a guarantee given by the producer, but are to be regarded only as a means for choosing the right products in relation to the expected economically reasonable working life of the works.

3 Performance of the product and references to the methods used for its assessment**3.1 Mechanical resistance and stability (BWR 1)**

Essential characteristic	Performance
Characteristic resistance under static and quasi-static loading	See Annex C 1 and C 2
Characteristic resistance under seismic action	See Annex B 5 and C 3

3.2 Safety in case of fire (BWR 2)

Essential characteristic	Performance
Reaction to fire	Class A1
Resistance to fire	See Annex C 4 and C 5

4 Assessment and verification of constancy of performance (AVCP) system applied, with reference to its legal base

In accordance with European Assessment Document EAD No. 330087-01-0601, the applicable European legal act is: [96/582/EC].

The system to be applied is: 1

5 Technical details necessary for the implementation of the AVCP system, as provided for in the applicable European Assessment Document

Technical details necessary for the implementation of the AVCP system are laid down in the control plan deposited with Deutsches Institut für Bautechnik.

Issued in Berlin on 22 January 2020 by Deutsches Institut für Bautechnik

Dipl.-Ing. Beatrix Wittstock
Head of Section

beglaubigt:
Lange

Installation conditions and application examples reinforcing bars, part 1

Figure A1.1:

Overlap joint with existing reinforcement for rebar connections of slabs and beams

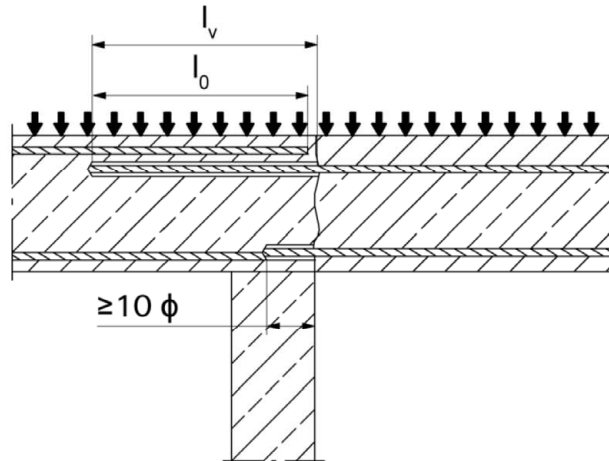


Figure A1.2:

Overlap joint with existing reinforcement at a foundation of a column or wall where the rebars are stressed

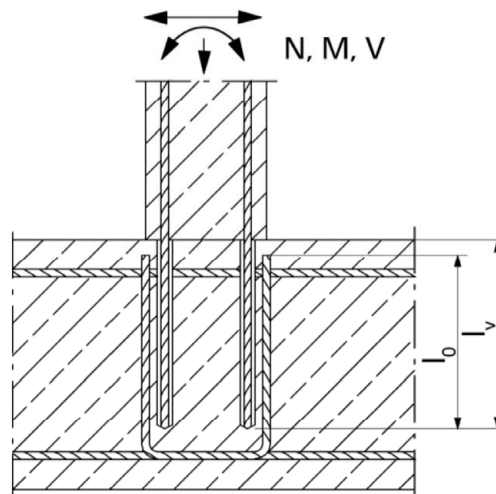
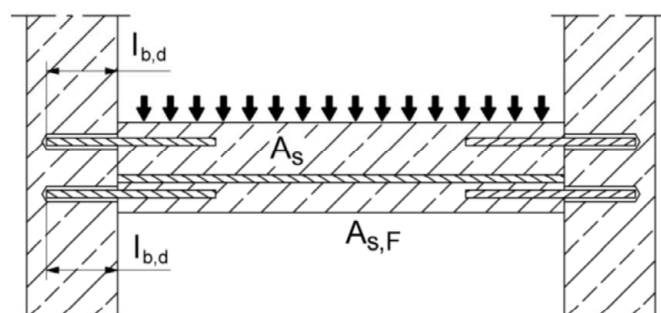


Figure A1.3:

End anchoring of slabs or beams (e.g. designed as simply supported)



Figures not to scale

Rebar connection with fischer injection mortar FIS EM Plus

Product description

Installation conditions and application examples reinforcing bars, part 1

Annex A 1

Installation conditions and application examples reinforcing bars, part 2

Figure A2.1:
Rebar connection for stressed primarily in compression

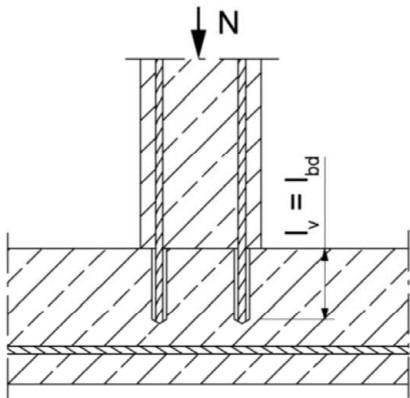
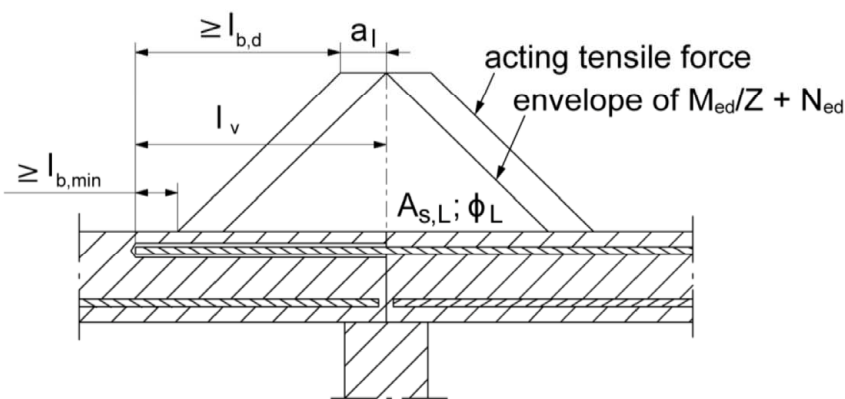


Figure A2.2:
Anchoring of reinforcement to cover the enveloped line of acting tensile force in the bending member



Note to **figure A1.1 to A1.3** and **figure A2.1 to A2.2**

In the figures no traverse reinforcement is plotted, the transverse reinforcement shall comply with EN 1992-1-1: 2004+AC:2010.

Preparing of joints according to **Annex B 2**

Figures not to scale

Rebar connection with fischer injection mortar FIS EM Plus	Annex A 2
Product description Installation conditions and application examples reinforcing bars, part 2	

Installation conditions and application examples fischer rebar anchor, part 3

Figure A3.1:

Lap to a foundation of a column under bending.

1. Shear lug (or fastener loaded in shear)
2. fischer Rebar tension anchor (tension only)
3. Existing stirrup / reinforcement for overlap (lap splice)
4. Slotted hole

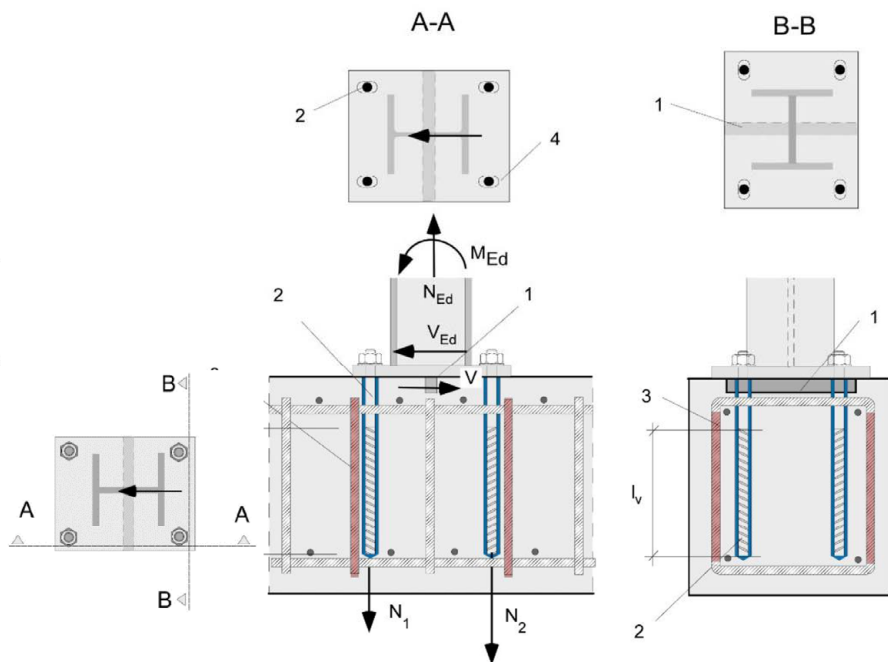
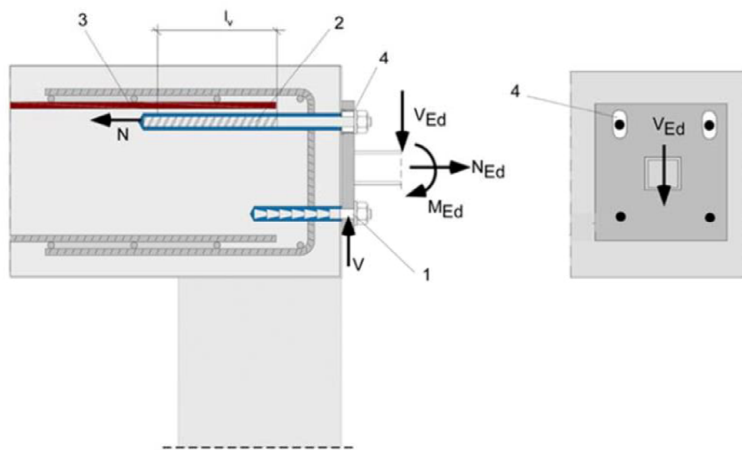


Figure A3.2:

Lap of the anchoring of guardrail posts or anchoring of cantilevered building components.

In the anchor plate, the drill holes for the fischer rebar anchors have to be designed as slotted holes with axial direction to the shear force.

1. Fastener for shear load transfer
2. fischer rebar tension anchor (tension only)
3. Existing stirrup / reinforcement for overlap (lap splice)
4. Slotted hole



The required transverse reinforcement acc. to EN 1992-1-1:2004+AC:2010 is not shown in the figures. **The fischer rebar anchor may be only used for axial tensile force.** The tensile force must be transferred by lap to the existing reinforcement of the building. The transfer of the shear force has to be ensured by suitable measure, e.g. by means of shear force or anchors with European Technical Assessment (ETA).

Figures not to scale

Rebar connection with fischer injection mortar FIS EM Plus

Product description

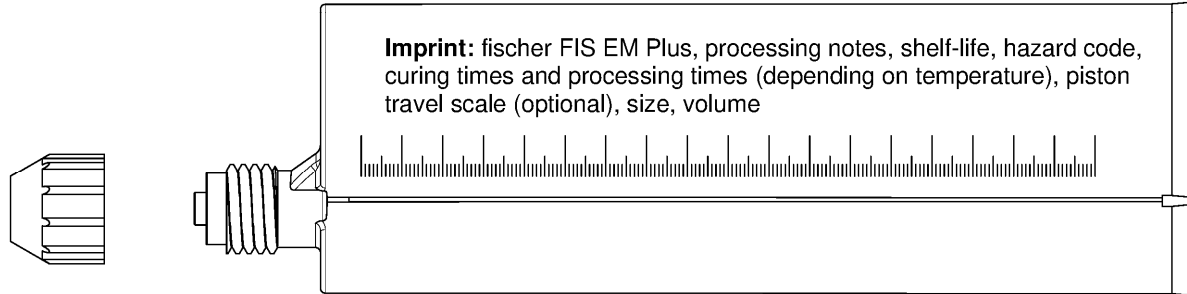
Installation conditions and application examples fischer rebar anchors, part 3

Annex A 3

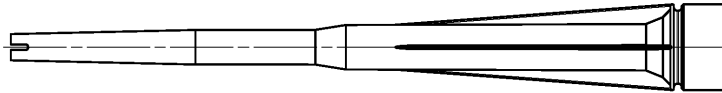
Overview system components

Injection cartridge (shuttle cartridge) FIS EM Plus with sealing cap

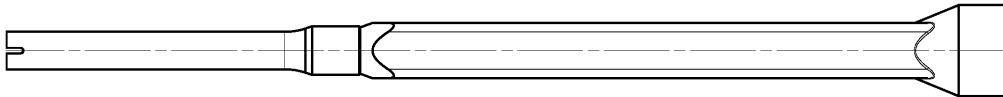
Sizes: 390 ml, 585 ml, 1100 ml, 1500 ml



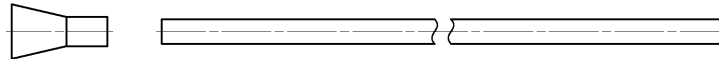
Static mixer FIS MR Plus for injection cartridge 390 ml



Static mixer FIS UMR for injection cartridge 585 ml to 1500 ml

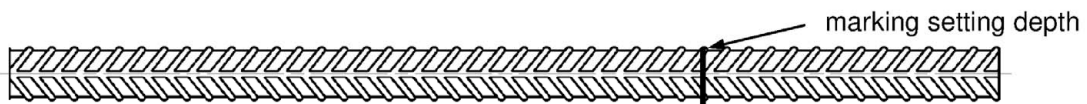


Injection adapter and extension tube Ø 9 for static mixer FIS MR Plus; Injection adapter and extension tube Ø 9 or Ø 15 for static mixer FIS UMR



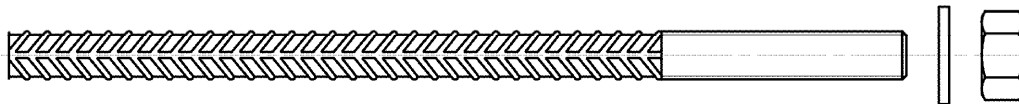
Reinforcing bar (rebar)

Sizes: $\phi 8$, $\phi 10$, $\phi 12$, $\phi 14$, $\phi 16$, $\phi 20$, $\phi 22$, $\phi 24$, $\phi 25$, $\phi 26$, $\phi 28$, $\phi 30$, $\phi 32$, $\phi 34$, $\phi 36$, $\phi 40$



fischer rebar anchor FRA, FRA HCR

Sizes: M12, M16, M20, M24



Blow out pump ABP



Figures not to scale

Rebar connection with fischer injection mortar FIS EM Plus

Product description

Overview system components;
Injection mortar, reinforcing bar, fischer rebar anchor, blow out pump

Annex A 4

Properties of reinforcing bars (rebar)

Figure A5.1:



- The minimum value of related rip area $f_{R,min}$ according to EN 1992-1-1:2004+AC:2010
- The maximum outer rebar diameter over the ribs shall be:
 - The nominal diameter of the rip $\phi + 2 \cdot h$ ($h \leq 0,07 \cdot \phi$)
 - (ϕ : Nominal diameter of the bar; h : rip height of the bar)

Table A5.1: Installation conditions for rebars

Nominal diameter of the bar ϕ		8 ¹⁾		10 ¹⁾		12 ¹⁾		14	16	20	22	24
Nominal drill hole diameter d_0	[mm]	10	12	12	14	14	16	18	20	25	30	30
Drill hole depth h_0		$h_0 = l_v$										
Effective embedment depth l_v		acc. to static calculation										
Minimum thickness of concrete member h_{min}		$l_v + 30$ (≥ 100)					$l_v + 2d_0$					

Nominal diameter of the bar ϕ		25 ¹⁾		26	28	30	32	34	36	40
Nominal drill hole diameter d_0	[mm]	30	35	35	35	40	40	40	45	55
Drill hole depth h_0		$h_0 = l_v$								
Effective embedment depth l_v		acc. to static calculation								
Minimum thickness of concrete member h_{min}		$l_v + 2d_0$								

¹⁾ Both drill hole diameters can be used

Table A5.2: Materials of rebars

Designation	Reinforcing bar (rebar)
Reinforcing bar EN 1992-1-1:2004+AC:2010, Annex C	Bars and de-coiled rods class B or C with f_{yk} and k according to NDP or NCL of EN 1992-1-1/NA $f_{uk} = f_{tk} = k \cdot f_{yk}$

Figures not to scale

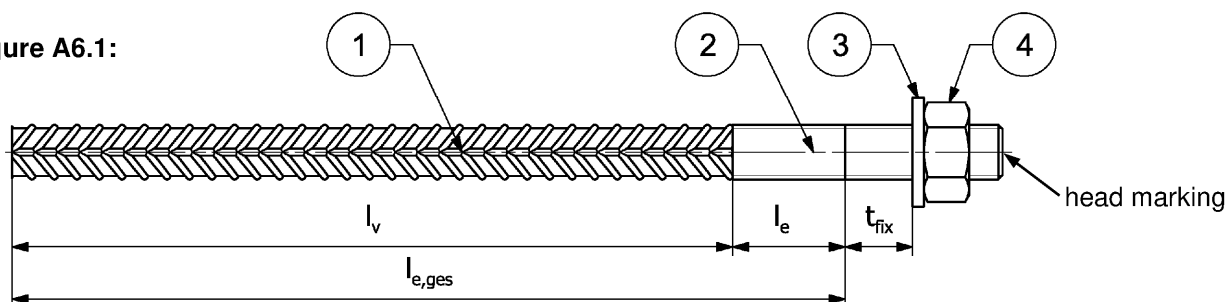
Rebar connection with fischer injection mortar FIS EM Plus

Product description
Properties and materials of reinforcing bars (rebar)

Annex A 5

Properties of fischer rebar anchor

Figure A6.1:



Head marking e.g.:  FRA (for stainless steel)

 FRA HCR (for high corrosion-resistant steel)

Table A6.1: Installation conditions for fischer rebar anchors

Threaded diameter		M12 ²⁾		M16	M20	M24 ²⁾	
Nominal diameter	ϕ [mm]	12		16	20	25	
Width across flat	SW [mm]	19		24	30	36	
Nominal drill bit diameter	d_0 [mm]	14	16	20	25	30	35
Drill hole depth ($h_0 = l_{ges}$)	$l_{e,ges}$ [mm]	$l_v + l_e$					
Effective embedment depth	l_v [mm]	acc. to static calculation					
Distance concrete surface to welded joint	l_e [mm]	100					
Diameter of clearance hole in the fixture ¹⁾	Pre-positioned $\leq d_f$ [mm]	14		18	22	26	
	Push through $\leq d_f$ [mm]	16	18	22	26	32	40
Minimum thickness of concrete member	h_{min} [mm]	$h_0 + 30$ (≥ 100)		$h_0 + 2d_0$			
Maximum torque moment for attachment of the fixture	$\max T_{fix}$ [Nm]	50		100	150	150	

¹⁾ For bigger clearance holes in the fixture see EN 1992-4:2018

²⁾ Both drill bit diameters can be used

Table A6.2: Materials of fischer rebar anchors

Part	Description	Materials	
		FRA Corrosion resistance class CRC III acc. to EN 1993-1-4:2015	FRA HCR Corrosion resistance class CRC V acc. to EN 1993-1-4:2015
1	Reinforcing bar	Bars and de-coiled rods class B or C with f_{yk} and k according to NDP or NCL of EN 1992-1-1:NA; $f_{uk} = f_{tik} = k \cdot f_{yk}$	
2	Round bar with partial or full thread	Stainless steel, strength class 80, according to EN 10088-1:2014	High corrosion-resistant steel, strength class 80, according to EN 10088-1:2014
3	Washer ISO 7089:2000	Stainless steel, according to EN 10088-1:2014	High corrosion-resistant steel, according to EN 10088-1:2014
4	Hexagon nut	Stainless steel, strength class 80, acc. to EN ISO 3506:2009	High corrosion-resistant steel, strength class 80, acc. to EN ISO 3506:2009

Figures not to scale

Rebar connection with fischer injection mortar FIS EM Plus

Product description
Properties and materials of fischer rebar anchors

Annex A 6

Specifications of intended use (part 1)

Table B1.1: Overview use and performance categories

Anchorages subject to		FIS EM Plus with ...			
		Reinforcing bar 		fischer rebar anchor 	
Hammer drilling with standard drill bit 		all sizes			
Hammer drilling with hollow drill bit (fischer "FHD", Heller "Duster Expert"; Bosch „Speed Clean“; Hilti "TE-CD, TE-YD") 		Nominal drill bit diameter (d ₀) 12 mm to 35 mm			
Diamond drilling 		all sizes			
Static and quasi static load, in	uncracked concrete	all sizes	Tables: C1.1 C1.2 C1.3 C2.1	all sizes	Tables: C1.1 C1.2 C1.3 C1.4 C2.1
	cracked concrete				
Seismic action (only hammer drilling with standard / hollow drill bits)		all sizes	Tables: C3.1 C3.2 C3.3	no performance assessed	
Installation temperature		T _{i,min} = -5 °C to T _{i,max} = +40 °C			
Resistance to fire		all sizes	Annex C5	all sizes	Annex C4
Rebar connection with fischer injection mortar FIS EM Plus					Annex B 1
Intended use Specifications (part 1)					

Specifications of intended use (part 2)

Anchorage subject to:

- Static, quasi-static and seismic loads: reinforcing bar (rebar) size 8 mm to 40 mm
- Resistance to fire

Base materials:

- Compacted reinforced or unreinforced normal weight concrete without fibres according to EN 206:2013+A1:2016
- Strength classes C12/15 to C50/60 according to EN 206:2013+A1:2016
- Maximum chloride content of 0,40 % (CL 0.40) related to the cement content according to EN 206:2013+A1:2016
- Non-carbonated concrete

Note: In case of a carbonated surface of the existing concrete structure the carbonated layer shall be removed in the area of the post-installed rebar connection with a diameter of $\phi + 60$ mm prior to the installation of the new rebar. The depth of concrete to be removed shall correspond to at least the minimum concrete cover in accordance with EN 1992-1-1 :2004+AC:2010. The foregoing may be neglected if building components are new and not carbonated and if building components are in dry conditions

Temperature Range:

- - 40°C to +80°C (max. short term temperature +80°C and max long term temperature +50°C).

Installation temperature:

- -5 °C to +40 °C

Use conditions (Environmental conditions) for fischer rebar anchors:

- For all conditions according to EN1993-1-4:2015 corresponding to corrosion resistance classes to Annex A 6 table A6.2

Design:

- Anchorages are designed under the responsibility of an engineer experienced in anchorages and concrete work.
- Verifiable calculation notes and drawings are prepared taking account of the forces to be transmitted.
- Design according to EN 1992-1-1:2004+AC:2010 and Annex B 3 and B 4.
- The actual position of the reinforcement in the existing structure shall be determined on the basis of the construction documentation and taken into account when designing.

Installation:

- Dry or wet concrete
- It must not be installed in water filled holes
- Hole drilling by hammer drill, hollow drill, compressed air drill or diamond drill mode
- Overhead installation allowed
- The installation of post-installed rebar respectively fischer rebar anchor shall be done only by suitable trained installer and under Supervision on site; the conditions under which an installer may be considered as suitable trained and the conditions for Supervision on site are up to the Member States in which the installation is done.
- Check the position of the existing rebars (if the position of existing rebars is not known, it shall be determined using a rebar detector suitable for this purpose as well as on the basis of the construction documentation and then marked on the building component for the overlap joint).

Rebar connection with fischer injection mortar FIS EM Plus

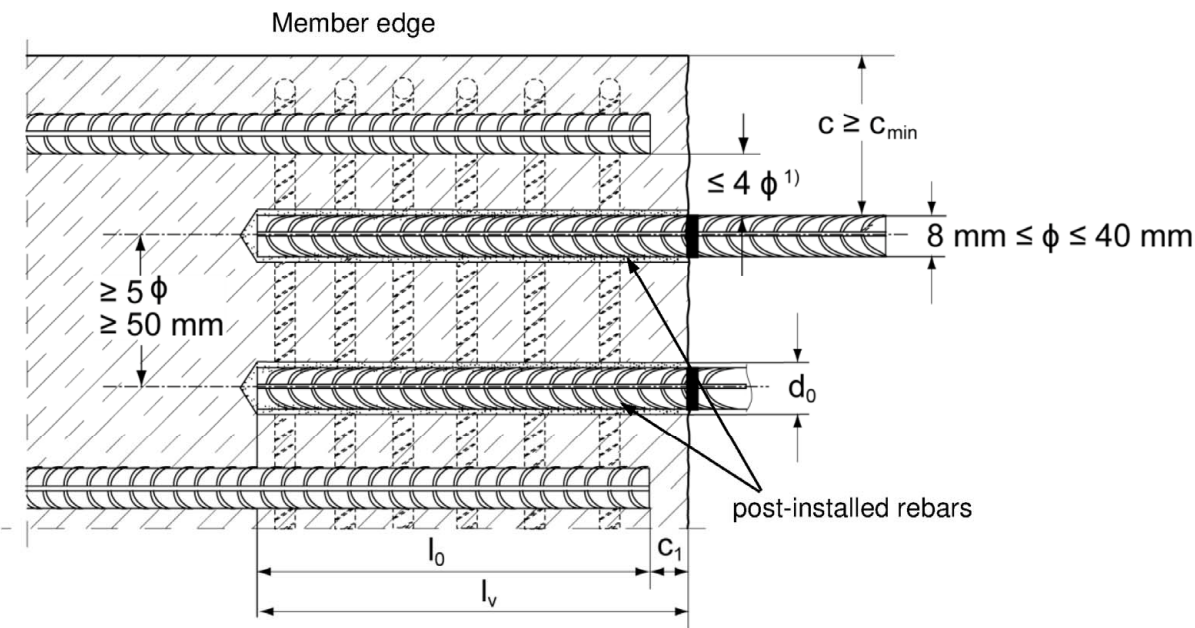
Intended use
Specifications (part 2)

Annex B 2

General construction rules for post-installed rebars

Figure B3.1:

- Only tension forces in the axis of the rebar may be transmitted
- The transfer of shear forces between new concrete and existing structure shall be designed additionally according to EN 1992-1-1:2004+AC:2010.
- The joints for concreting must be roughened to at least such an extent that aggregate protrude.



1) If the clear distance between lapped bars exceeds 4ϕ then the lap length shall be increased by the difference between the clear bar distance and 4ϕ

- c concrete cover of post-installed rebar
 c_1 concrete cover at end-face of existing rebar
 c_{min} minimum concrete cover according to table B5.1 and to EN 1992-1-1:2004+AC:2010, Section 4.4.1.2
 ϕ nominal diameter of reinforcing bar
 l_0 lap length, according to EN 1992-1-1:2004+AC:2010 for static loading and according to EN 1998-1:2004, section 5.6.3 for seismic loading
 l_v effective embedment depth, $\geq l_0 + c_1$
 d_0 nominal drill bit diameter, see Annex B 6

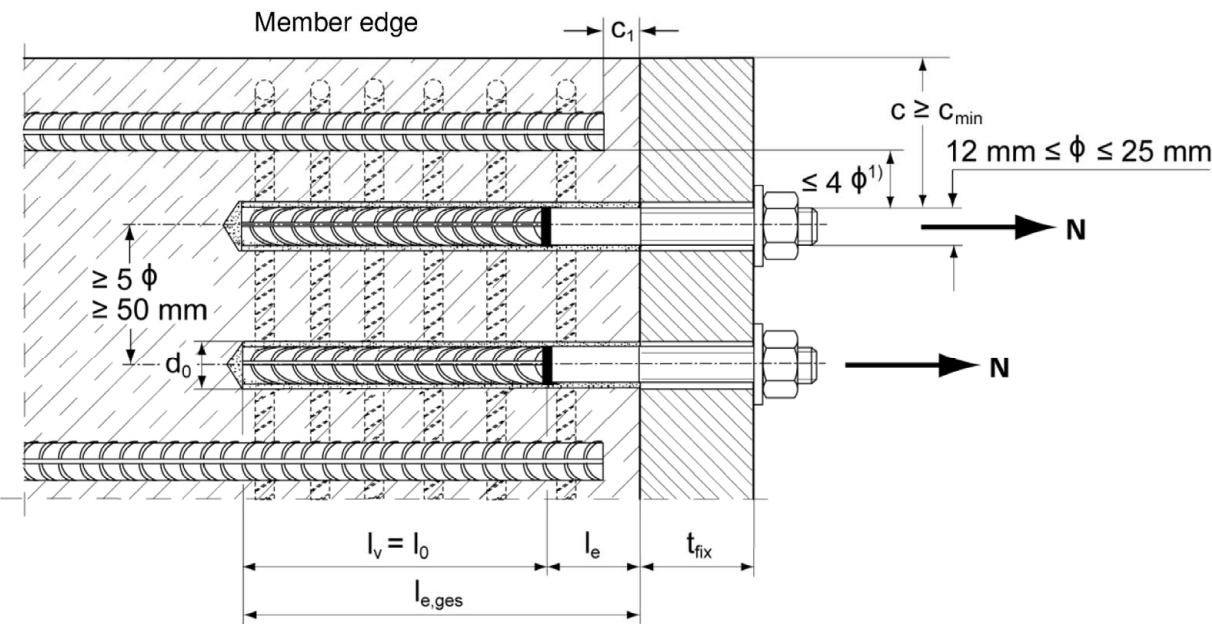
Figures not to scale

Rebar connection with fischer injection mortar FIS EM Plus	Annex B 3
Intended use General construction rules for for post-installed rebars	

General construction rules for post-installed fischer rebar anchors

Figure B4.1:

- Only tension forces in the axis of the fischer rebar anchor may be transmitted.
- The tension force must be transferred via an overlap joint to the reinforcement in the building part.
- The transmission of the shear load shall be ensured by appropriate additional measures, e.g. by shear lugs or by anchors with a European Technical Assessment (ETA).
- In the anchor plate, the holes for the tension anchor shall be executed as slotted holes with the axis in the direction of the shear force.



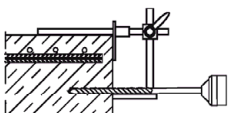
1) If the clear distance between lapped bars exceeds 4ϕ then the lap length shall be increased by the difference between the clear bar distance and 4ϕ .

c	concrete cover of post-installed fischer rebar anchor
c ₁	concrete cover at end-face of existing rebar
c _{min}	minimum concrete cover according to table B5.1 and to EN 1992-1-1:2004+AC:2010, Section 4.4.1.2
φ	nominal diameter of reinforcing bar
l ₀	lap length, according to EN 1992-1-1:2004+AC:2010, Section 8.7.3
l _{e,ges}	overall embedment depth, $\geq l_0 + l_e$
d ₀	nominal drill bit diameter, see Annex B 6
l _e	length of the bonded in threaded part
t _{fix}	thickness of the fixture
l _v	effective embedment depth

Figures not to scale

Rebar connection with fischer injection mortar FIS EM Plus	Annex B 4
Intended use General construction rules for post-installed fischer rebar anchors	

Table B5.1: Minimum concrete cover $c_{min} = c_{min,seis}$ ¹⁾ depending of the drilling method and the drilling tolerance

Drilling method	nominal diameter of reinforcing bar ϕ [mm]	Minimum concrete cover $c_{min} = c_{min,seis}$		 Drilling aid
		Without drilling aid [mm]	Without drilling aid [mm]	
Hammer drilling with standard drill bit	< 25	30 mm + 0,06 $l_v \geq 2 \phi$	30 mm + 0,02 $l_v \geq 2 \phi$	
	≥ 25	40 mm + 0,06 $l_v \geq 2 \phi$	40 mm + 0,02 $l_v \geq 2 \phi$	
Hammer drilling with hollow drill bit (fischer "FHD", Heller "Duster Expert"; Bosch „Speed Clean“; Hilti "TE-CD, TE-YD")	< 25	30 mm + 0,06 $l_v \geq 2 \phi$	30 mm + 0,02 $l_v \geq 2 \phi$	
	≥ 25	40 mm + 0,06 $l_v \geq 2 \phi$	40 mm + 0,02 $l_v \geq 2 \phi$	
Compressed air drilling	< 25	50 mm + 0,08 l_v	50 mm + 0,02 l_v	
	≥ 25	60 mm + 0,08 $l_v \geq 2 \phi$	60 mm + 0,02 $l_v \geq 2 \phi$	
Diamond drilling	< 25	30 mm + 0,06 $l_v \geq 2 \phi$	30 mm + 0,02 $l_v \geq 2 \phi$	
	≥ 25	40 mm + 0,06 $l_v \geq 2 \phi$	40 mm + 0,02 $l_v \geq 2 \phi$	

¹⁾ See Annex B3, figure B3.1 and Annex B4, figure B4.1

Note: The minimum concrete cover as specified in EN 1992-1-1:2004+AC:2010 must be observed.

Table B5.2: Dispensers and cartridge sizes corresponding to maximum embedment depth $l_{v,max}$

reinforcing bars (rebar)	fischer rebar anchor	Manual dispenser	Accu and pneumatic dispenser (small)	Pneumatic dispenser (large)
		Cartridge size 390 ml, 585 ml	Cartridge size 390 ml, 585 ml	Cartridge size 1500 ml
ϕ [mm]	[-]	l _{v,max} / l _{e,ges,max} [mm]	l _{v,max} / l _{e,ges,max} [mm]	l _{v,max} / l _{e,ges,max} [mm]
8	---	1000	1000	1800
10	---		1200	
12	FRA M12 FRA HCR M12			
14	---			
16	FRA M16 FRA HCR M16	700	1500	2000
20	FRA M20 FRA HCR M20		1300	
22 / 24 / 25	FRA M24 FRA HCR M24		1000	
26 / 28	---	500	700	
30 / 32 / 34	---	no performance assessed	500	
36	---			
40	---			

Rebar connection with fischer injection mortar FIS EM Plus

Intended use

Minimum concrete cover;
dispenser and cartridge sizes corresponding to maximum embedment depth

Annex B 5

Table B6.1: Working times t_{work} and curing times t_{cure}

Temperature in the anchorage base [°C]	Maximum working time ¹⁾ t_{work} FIS EM Plus	Minimum curing time ²⁾ t_{cure} FIS EM Plus
-5 to 0	240 min ³⁾	200 h
>0 to 5	150 min ³⁾	90 h
>5 to 10	120 min ³⁾	40 h
>10 to 20	30 min	18 h
>20 to 30	14 min	10 h
>30 to 40	7 min ⁴⁾	5 h

¹⁾ Maximum time from the beginning of the injection to rebar / fischer rebar anchor setting and positioning

²⁾ For wet concrete the curing time must be doubled

³⁾ If the temperature in the concrete falls below 10°C the cartridge has to be warmed up to +15°C.

⁴⁾ If the temperature in the concrete exceeds 30 °C the cartridge has to be cooled down to +15°C up to 20°C

Table B6.2: Installation tools for drilling and cleaning the bore hole and injection of the mortar

reinforcing bars (rebar)	fischer rebar anchor	Drilling and cleaning				Injection		
		Nominal drill bit diameter	Diameter of cutting edge	Steel brush diameter	Diameter of cleaning nozzle	Diameter of extension tube	Injection adapter	
ϕ [mm]	[-]	d ₀ [mm]	d _{cut} [mm]	d _b [mm]	[mm]	[mm]	[colour]	
8 ¹⁾	---	10	≤ 10,50	11,0	---	9	---	
		12	≤ 12,50	12,5	11		nature	
10 ¹⁾	---	12	≤ 12,50	12,5			15	blue
		14	≤ 14,50	15				red
12 ¹⁾	FRA M12 ¹⁾ FRA HCR M12 ¹⁾	14	≤ 14,50	15	19	9 or 15	yellow	
		16	≤ 16,50	17			green	
14	---	18	≤ 18,50	19	28		black	
16	FRA M16 FRA HCR M16	20	≤ 20,55	21,5			grey	
20	FRA M20 FRA HCR M20	25	≤ 25,55	26,5			grey	
22 / 24	---	30	≤ 30,55	32	38		brown	
25	FRA M24 ¹⁾ FRA HCR M24 ¹⁾	30	≤ 30,55	32			brown	
		35	≤ 35,70	37				red
26 / 28	---	35	≤ 35,70	37	38		yellow	
30 / 32 / 34	---	40	≤ 40,70	42			nature	
36	---	45	≤ 45,70	47				
40	---	55	≤ 55,70	58				

¹⁾ Both drill bit diameters can be used

Rebar connection with fischer injection mortar FIS EM Plus

Intended use

Working times and curing times;

Installation tools for drilling and cleaning the bore hole and injection of the mortar

Annex B 6

Safety regulations



Review the Material Safety Data Sheet (SDS) before use for proper and safe handling!

Wear well-fitting protective goggles and protective gloves when working with mortar FIS EM Plus

Important: Observe the instructions for use provided with each cartridge.

Installation instruction part 1; Installation with FIS EM Plus

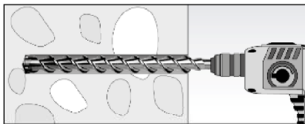
Hole drilling

Note: Before drilling, remove carbonized concrete; clean contact areas (see Annex B 2)

In case of aborted drill holes the drill hole shall be filled with mortar.

1a

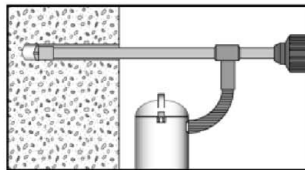
Hammer drilling or compressed air drilling



Drill the hole to the required embedment depth using a hammer drill with carbide drill bit set in rotation hammer mode or a pneumatic drill.
Drill bit sizes see table B6.2

1b

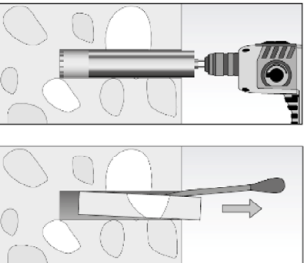
Hammer drilling with hollow drill bit



Drill the hole to the required embedment depth using a hammer drill with hollow drill bit in rotation hammer mode.
Dust extraction conditions see drill hole cleaning annex B 8.
Drill bit sizes see table B6.2

1c

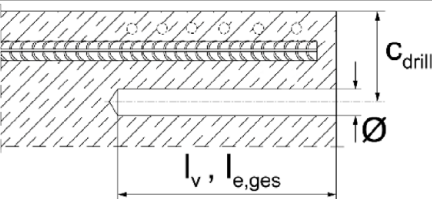
Diamond drilling



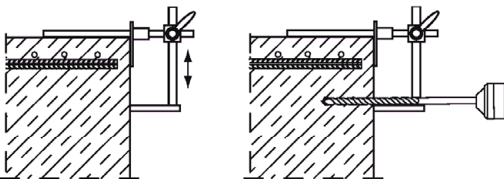
Drill the hole to the required embedment depth using a diamond drill in rotation mode.
Drill bit sizes see table B6.2

Break away the drill core and remove it

2



Measure and control concrete cover c
($c_{\text{drill}} = c + \varnothing / 2$)
Drill parallel to surface edge and to existing rebar.
Where applicable use fischer drilling aid.



For holes $l_v > 20$ cm use drilling aid.
Three different options can be considered:
A) fischer drilling aid
B) Slat or spirit level
C) Visual check

Minimum concrete cover c_{min} see table B5.1

Rebar connection with fischer injection mortar FIS EM Plus

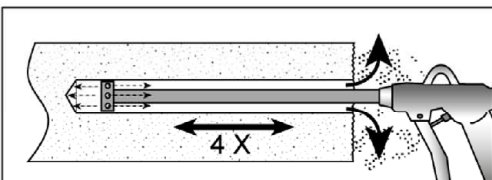
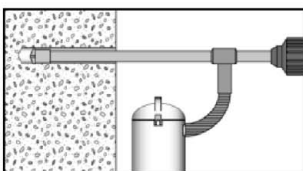
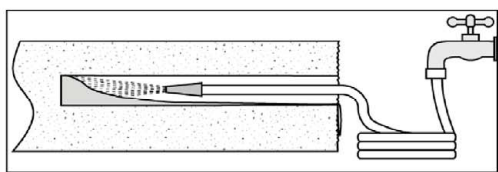
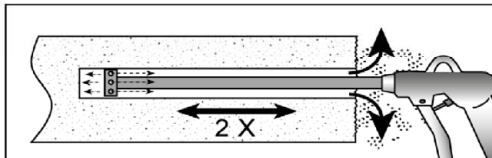
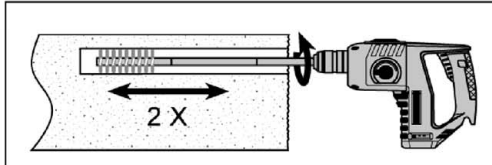
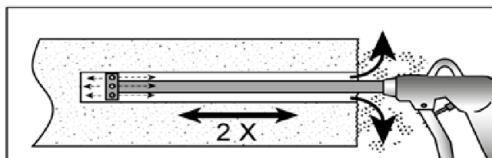
Intended use

Safety regulations; Installation instruction part 1, hole drilling

Annex B 7

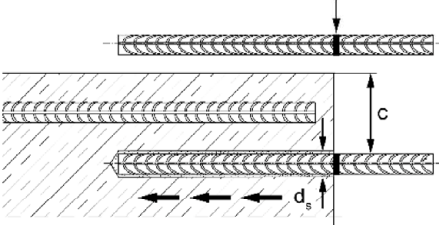
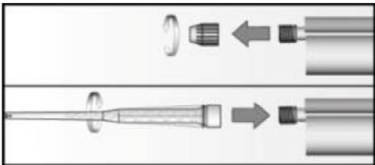
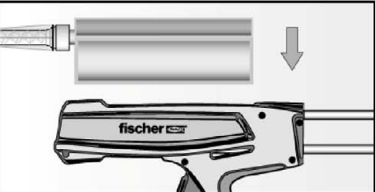
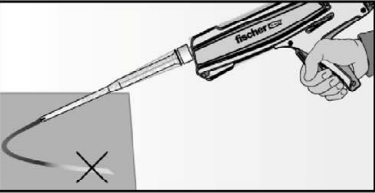
Installation instruction part 2; Installation with FIS EM Plus

Drill hole cleaning

	Hammer or compressed air drilling	
3a		Blowing four times from the back of the hole with the appropriate nozzle (oil-free compressed air ≥ 6 bar) until return air stream is free of noticeable dust. Personal protective equipment must be used (see regulations Annex B 7).
	Hammer drilling with hollow drill bit	
3b		Use a suitable dust extraction system, e. g. fischer FVC 35 M or a comparable dust extraction system with equivalent performance data. Drill the hole with hollow drill bit. The dust extraction system has to extract the drill dust nonstop during the drilling process and must be adjusted to maximum power. No further drill hole cleaning necessary
	Diamond drilling	
		Flush the bore hole until the water comes clear
3c		Blowing twice from the back of the hole with the appropriate nozzle (oil-free compressed air ≥ 6 bar) until return air stream is free of noticeable dust. Personal protective equipment must be used (see regulations Annex B 7).
		Check steel brush with brush control template. Fix an adequate steel brush with an extension into a drilling machine and brush the bore hole twice
		Blowing twice from the back of the hole with the appropriate nozzle (oil-free compressed air ≥ 6 bar) until return air stream is free of noticeable dust. Personal protective equipment must be used (see regulations Annex B 7).
Rebar connection with fischer injection mortar FIS EM Plus		Annex B 8
Intended use Installation instruction part 2, hole cleaning		

Installation instruction part 3; Installation with FIS EM Plus

reinforcing bars (rebar) / fischer rebar anchor and cartridge preparation

4		Before use, make asure that the rebar or the fischer rebar anchor is dry and free of oil or other residue. Mark the embedment depth l_v on the rebar (e.g. with tape) Insert rebar in borehole, to verify drill hole depth and setting depth l_v resp. $l_{e,ges}$
5		Twist off the sealing cap Twist on the static mixer (the spiral in the static mixer must be clearly visible).
6		Place the cartridge into a suitable dispenser.
7		Press out approximately 10 cm of mortar until the resin is permanently grey in colour. Mortar which is not grey in colour will not cure and must be disposed.

Rebar connection with fischer injection mortar FIS EM Plus

Intended use

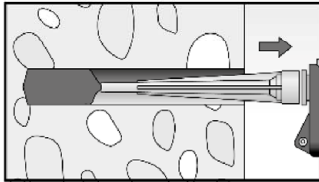
Installation instruction part 3,
reinforcing bars (rebar) / fischer rebar anchor and cartridge preparation

Annex B 9

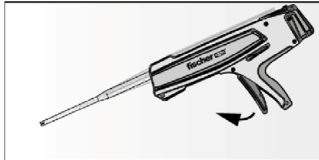
Installation instruction part 4; Installation with FIS EM Plus

Injection of the mortar; borehole depth ≤ 250 mm

8a



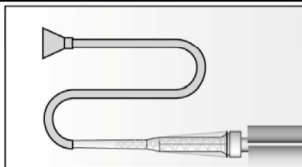
Inject the mortar from the back of the hole towards the front and slowly withdraw the mixing nozzle step by step with each trigger pull. Avoid bubbles. Fill holes approximately 2/3 full, to ensure that the annular gap between the rebar and the concrete will be completely filled with adhesive over the entire embedment length.



After injecting, release the dispenser. This will prevent further mortar discharge from the mixing nozzle.

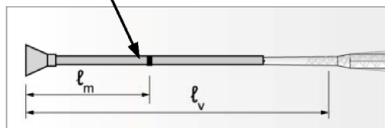
Injection of the mortar; borehole depth > 250 mm

8b



Assemble mixing nozzle FIS MR Plus or FIS UMR, extension tube and appropriate injection adapter (see table B 6.2)

Mortar level mark



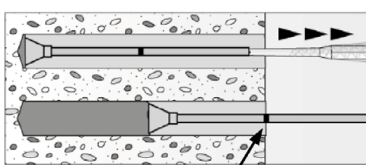
Mark the required mortar level l_m and embedment depth l_v resp. $l_{e,ges}$ with tape or marker on the injection extension tube.

a) Estimation:

$$l_m = \frac{1}{3} * l_v \text{ resp. } l_m = \frac{1}{3} * l_{e,ges} \text{ [mm]}$$

b) Precise equation for optimum mortar volume:

$$l_m = l_v \text{ resp. } l_{e,ges} \left((1,2 * \frac{d_s^2}{d_0^2} - 0,2) \right) \text{ [mm]}$$



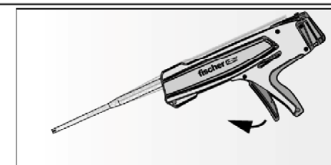
Mortar level mark

Insert injection adapter to back of the hole. Begin injection allowing the pressure of the injected adhesive mortar to push the injection adapter towards the front of the hole. Do not actively pull out!

Fill holes approximately 2/3 full, to ensure that the annular gap between the rebar and the concrete will be completely filled with adhesive over the embedment length.

When using an injection adapter continue injection until the mortar level mark l_m becomes visible.

Maximum embedment depth see table B5.2



After injecting, release the dispenser. This will prevent further mortar discharge from the mixing nozzle.

Rebar connection with fischer injection mortar FIS EM Plus

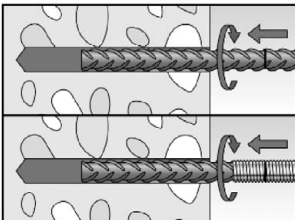
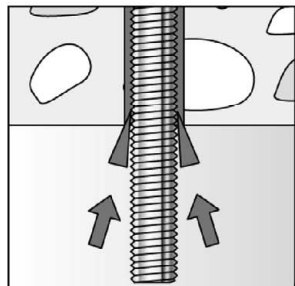
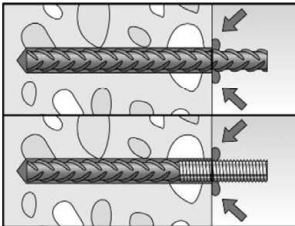
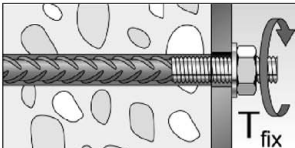
Intended use

Installation instruction part 4, mortar injection

Annex B 10

Installation instruction part 5; Installation with FIS EM Plus

Insert rebar / fischer rebar anchor

9		Insert the rebar / fischer rebar anchor slowly twisted into the borehole until the embedment mark is reached.
10		For overhead installation, support the rebar / fischer rebar anchor and secure it from falling till mortar started to harden, e.g. using wedges.
11		After installing the rebar or fischer rebar anchor the annular gap must be completely filled with mortar. Proper installation • Desired embedment depth is reached l_v: embedment mark at concrete surface. • Excess mortar flows out of the borehole after the rebar has been fully inserted up to the embedment mark.
12		Observe the working time "t_{work}" (see table B6.1), which varies according to temperature of base material. Minor adjustments to the rebar / fischer rebar anchor position may be performed during the working time Full load may be applied only after the curing time "t_{cure}" has elapsed (see table B 6.1)
13		Mounting the fixture, max T_{fix} see table A6.1

Rebar connection with fischer injection mortar FIS EM Plus

Intended use

Installation instruction part 5, insert rebar / fischer rebar anchor

Annex B 11

Minimum anchorage length and minimum lap length

The minimum anchorage length $l_{b,min}$ and the minimum lap length $l_{0,min}$ according to EN 1992-1-1: 2004+AC:2010 shall be multiply by the relevant amplification factor $\alpha_{lb} = \alpha_{lb,100y}$ according to table C1.1.

Table C1.1: Amplification factor $\alpha_{lb} = \alpha_{lb,100y}$ related to concrete strength class and drilling method with a service life of 50 or 100 years

Hammer drilling, hollow drilling and compressed air drilling									
Rebar / fischer rebar anchor ϕ [mm]	Amplification factor $\alpha_{lb} = \alpha_{lb,100y}$								
	Concrete strength class								
	C12/15	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8 to 25	1,0								
26 to 40	1,0								

Diamond drilling									
8 to 12	1,0		1,04	1,08	1,13	1,17	1,21	1,25	
14 to 25	1,0		1,04	1,08	1,13	1,17	1,21	1,25	
26 to 40	1,0		1,08	1,17	1,25	1,33	1,42	1,50	

Table C1.2: Bond efficiency factor $k_b = k_{b,100y}$ for hammer drilling, hollow drilling and compressed air drilling with a service life of 50 or 100 years

Hammer drilling, hollow drilling and compressed air drilling									
Rebar / fischer rebar anchor ϕ [mm]	Bond efficiency factor $k_b = k_{b,100y}$								
	Concrete strength class								
	C12/15	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8 to 25	1,0								0,98
26 to 40	1,0								0,98

Table C1.3: Bond efficiency factor $k_b = k_{b,100y}$ for diamond drilling with a service life of 50 or 100 years

Diamond drilling									
Rebar / fischer rebar anchor ϕ [mm]	Bond efficiency factor $k_b = k_{b,100y}$								
	Concrete strength class								
	C12/15	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8 to 12	1,0								0,95
14 to 25	1,0								0,95
26 to 40	1,0					0,96	0,87	0,81	0,76

Table C1.4: Characteristic values for **steel failure** under tension load of **fischer rebar anchors**

fischer rebar anchor FRA / FRA HCR	M12	M16	M20	M24
Bearing capacity under tension load, steel failure				
Characteristic resistance $N_{Rk,s}$ [kN]	63	111	173	270
Partial factor				
Partial factor $\gamma_{Ms,N}$ [-]	1,4			

Rebar connection with fischer injection mortar FIS EM Plus

Performance

Amplification factor $\alpha_{lb} = \alpha_{lb,100y}$ bond efficiency factor $k_b = k_{b,100y}$

Annex C 1

Table C2.1: Design values of the bond strength $f_{bd,PIR} = f_{bd,PIR,100y}$ in N/mm² for hammer drilling, hollow drilling, compressed air drilling and diamond drilling with a service life of 50 or 100 years

$$f_{bd,PIR} = k_b \cdot f_{bd}$$

$$f_{bd,PIR,100y} = k_{b,100y} \cdot f_{bd}$$

f_{bd} : Design value of the bond strength in N/mm² considering the concrete strength classes and the rebar diameter for good bond condition (for all other bond conditions multiply the values by $\eta_1 = 0,7$) and recommended partial factor $\gamma_c = 1,5$ according to EN 1992-1-1: 2004+AC:2010

k_b Bond efficiency factor according to table C1.2 and C1.3

$k_{b,100y}$ Bond efficiency factor according to table C1.2 and C1.3

Hammer drilling, hollow drilling and compressed air drilling

Rebar / fischer rebar anchor ϕ [mm]	bond strength $f_{bd,PIR} = f_{bd,PIR,100y}$ [N/mm ²]								
	Concrete strength class								
	C12/15	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8-32	1,6	2,0	2,3	2,7	3,0	3,4	3,7	4,0	4,2
34	1,6	2,0	2,3	2,6	2,9	3,3	3,6	3,9	4,1
36	1,5	1,9	2,2	2,6	2,9	3,3	3,6	3,8	4,0
40	1,5	1,8	2,1	2,5	2,8	3,1	3,4	3,7	3,9

Diamond drilling

Rebar / fischer rebar anchor ϕ [mm]	bond strength $f_{bd,PIR} = f_{bd,PIR,100y}$ [N/mm ²]								
	Concrete strength class								
	C12/15	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8-12	1,6	2,0	2,3	2,7	3,0	3,4	3,7	4,0	4,1
14-25						3,4	3,7	4,0	4,1
26-32						3,2	3,2	3,2	3,2
34	1,6	2,0	2,3	2,6	2,9	3,1	3,1	3,1	3,1
36	1,5	1,9	2,2	2,6	2,9	3,1	3,1	3,1	3,1
40	1,5	1,8	2,1	2,5	2,8	2,9	2,9	2,9	2,9

Rebar connection with fischer injection mortar FIS EM Plus

Performance

Design values of the bond strength $f_{bd,PIR} = f_{bd,PIR,100y}$

Annex C 2

Minimum anchorage length and minimum lap length under seismic conditions

The minimum anchorage length $l_{b,min}$ and the minimum lap length $l_{0,min}$ according to EN 1992-1-1: 2004+AC:2010 shall be multiply by the relevant amplification factor $\alpha_{lb,seis}$ according to table C3.1.

Table C3.1: Amplification factor $\alpha_{lb,seis} = \alpha_{lb,seis,100y}$ related to concrete strength class and drilling method

Hammer drilling, hollow drilling and compressed air drilling

Rebar ϕ [mm]	Amplification factor $\alpha_{lb,seis} = \alpha_{lb,seis,100y}$							
	Concrete strength class							
	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8 to 25	1,0							
26 to 40	1,0							

Table C3.2: Bond efficiency factor $k_{b,seis} = k_{b,seis,100y}$ for hammer drilling, hollow drilling and compressed air drilling with a service life of 50 or 100 years

Hammer drilling, hollow drilling and compressed air drilling

Rebar ϕ [mm]	Bond efficiency factor $k_{b,seis} = k_{b,seis,100y}$							
	Concrete strength class							
	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8 to 25	1,00							0,98
26 to 40	1,00							0,98

Table C3.3: Design values of the bond strength $f_{bd,PIR,seis} = f_{bd,PIR,seis,100y}$ in N/mm² for hammer drilling, hollow drilling and compressed air drilling **under seismic action** and for good bond conditions with a service life of 50 or 100 years

$$f_{bd,PIR,seis} = k_{b,seis} \cdot f_{bd}$$

$$f_{bd,PIR,seis,100y} = k_{b,seis,100y} \cdot f_{bd}$$

Hammer drilling, hollow drilling and compressed air drilling

Rebar ϕ [mm]	bond strength $f_{bd,PIR,seis} = f_{bd,PIR,seis,100y}$ [N/mm ²]							
	Concrete strength class							
	C16/20	C20/25	C25/30	C30/37	C35/45	C40/50	C45/55	C50/60
8-32	2,0	2,3	2,7	3,0	3,4	3,7	4,0	4,2
34	1,6	2,0	2,3	2,6	2,9	3,3	3,6	3,9
36	1,5	1,9	2,2	2,6	2,9	3,3	3,6	3,8
40	1,5	1,8	2,1	2,5	2,8	3,1	3,4	3,7

Rebar connection with fischer injection mortar FIS EM Plus

Performance

Amplification factor $\alpha_{lb,seis} = \alpha_{lb,seis,100y}$, bond efficiency factor $k_{b,seis} = k_{b,seis,100y}$, Design values of the bond strength $f_{bd,PIR,seis} = f_{bd,PIR,seis,100y}$

Annex C 3

Table C4.1: Essential characteristics to **steel failure** for **fischer rebar anchors** under fire exposure
concrete strength classes C12/C15 to C50/60, according to EN 1992-4:2018

fischer rebar anchor FRA / FRA HCR				M12	M16	M20	M24
Characteristic resistance to steel failure	R30	$N_{Rk,s,fi}$	[kN]	1,7	2,5	4,7	7,4
	R60			1,5	2,1	3,9	6,1
	R90			1,2	1,7	3,1	4,9
	R120			0,9	1,3	2,5	3,9

Rebar connection with fischer injection mortar FIS EM Plus

Performance

Characteristic resistance to steel failure $N_{Rk,s,fi}$ under fire exposure for fischer rebar anchor

Annex C 4

Bond strength $f_{bk,fi} = f_{bk,fi,100y}$ at increased temperature for concrete strength classes C12/15 to C50/60 (all drilling methods)

The bond strength $f_{bk,fi} = f_{bk,fi,100y}$ at increased temperature has to be calculated by the following equation:

$$f_{bk,fi} = f_{bk,fi,100y} = k_{fi}(\theta) \cdot f_{bd,PIR} \cdot \frac{\gamma_c}{\gamma_{M,fi}}$$

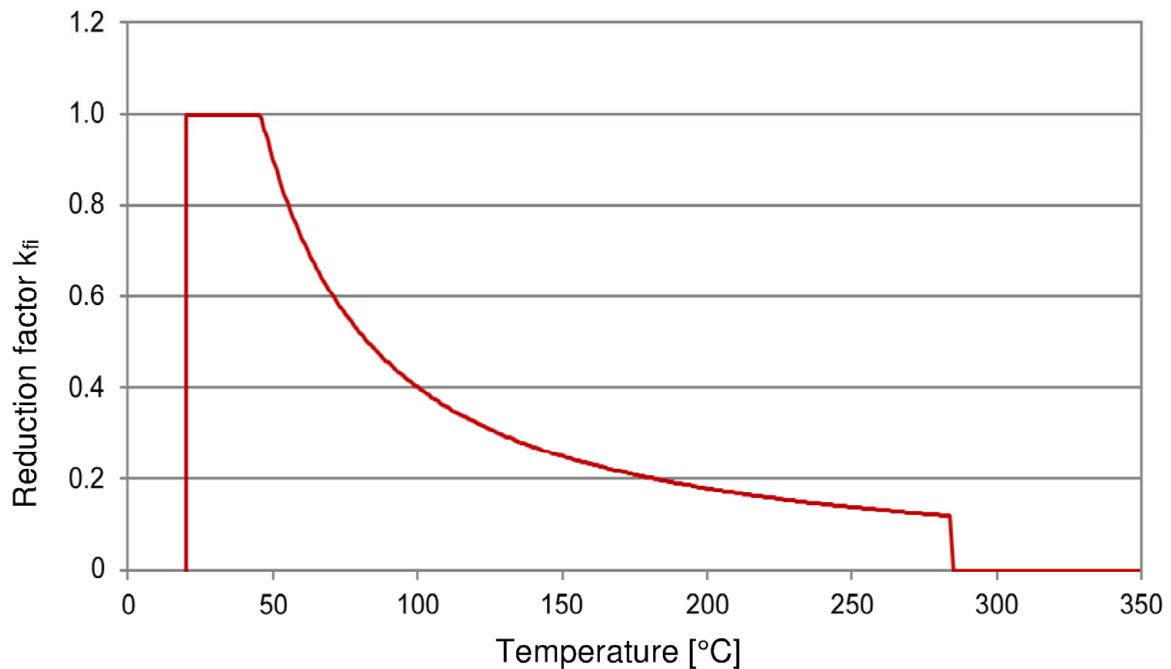
If: $\theta > 46 \text{ °C}$ $k_{fi}(\theta) = \frac{862,3 \cdot \theta^{-1,166}}{f_{bd,PIR} \cdot 4,3} \leq 1,0$

If: $\theta > \theta_{max} (284 \text{ °C})$ $k_{fi}(\theta) = 0$

- $f_{bk,fi}$ = Bond strength at increased temperature in N/mm² for service life 50 years
- $f_{bk,fi,100y}$ = Bond strength at increased temperature in N/mm² for service life 100 years
- (θ) = Temperature in °C in the mortar layer
- $k_{fi}(\theta)$ = Reduction factor at increased temperature
- $= k_{fi,100y}(\theta)$
- $f_{bd,PIR}$ = Design value of the bond strength in N/mm² in cold condition according to table C2.1 considering the concrete classes, the rebar diameter, the drilling method and the bond conditions according to EN 1992-1-1:2004+AC:2010
- γ_c = Partial factor according to EN 1992-1-1:2004+AC:2010
- $\gamma_{M,fi}$ = Partial factor according to EN 1992-1-2:2004+AC:2008

For evidence at increased temperature the anchorage length shall be calculated according to EN 1992-1-1:2004+AC:2010 Equation 8.3 using the temperature-dependent ultimate bond strength $f_{bk,fi}$.

Figure C5.1: Example graph of reduction factor $k_{fi}(\theta)$ for concrete class C20/25 for good bond conditions



Rebar connection with fischer injection mortar FIS EM Plus

Performance

Bond strength $f_{bk,fi} = f_{bk,fi,100y}$ at increased temperature

Annex C 5

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VOC TEST REPORT

VOC Content

18 February 2019

1 Sample Information

Sample name	FIS EM Plus 390 S
Sample no.	392-2019-00052801
Production date	13.12.2018
Batch No.	12/21B88B78
Sample reception	07/02/2019

2 Brief Evaluation of the Results

Regulation or protocol	Conclusion	Version of regulation or protocol
SCAQMD Rule 1168	Pass	January 2005
LEED v4 (VOC Content)	Pass	

Full details based on the testing and direct comparison with limit values are available in the following pages



Morten Sielemann
Analytical Chemist

3 Applied Test Methods

3.1 General Test References

Regulation, protocol or standard	Scope	Version
SCAQMD Rule 1168	Adhesive and sealant applications	January 2005

3.2 Specific Laboratory Sampling and Analyses

Test	Regulation, protocol or standard	Version	Internal SOP	Limit of detection [g/L]	Uncertainty U _m
Solids Content	ASTM D2369	2010	71 M 544830	1	10
VOC	ASTM D2369	2010	71 M 544830	1	10

4 Results

	Remarks on the test results	Results	Unit
Density *	Supplied by the Customer	1.5	g/mL
Water Content *	Supplied by the Customer	0	% (w/w)
Exempt compounds *	Assumed to be 0	0	% (w/w)
Solids Content	Tested by the lab	100	% (w/w)
VOC content (less water)	Calculated based on the results above	< 1	g/L
VOC content (less water)	Calculated based on the results above	< 0.1	% (w/w)

4.1 Comparison with Limit Values of VOC Content (less Water)

Parameter	Results [g/L]	Product type	Regulation or protocol	VOC limit [g/L]
VOC content	< 1	Multipurpose Construction Adhesives	SCAQMD Rule 1168	70

5 Appendices

5.1 How to Understand the Results

5.1.1 Acronyms Used in the Report

- < Means less than
- > Means bigger than
- * Not a part of our accreditation
- Please see section regarding uncertainty in the Appendices.
- 1 Analysed by another Eurofins laboratory

5.2 Description of VOC Content Test

5.2.1 Testing of VOC

Volatile content of the sample was determined gravimetrically by heating to 110 °C in 60 minutes. Multicomponent products are mixed according to the manufacturer's instructions and allowed to cure before heating.

The result is the average of two replicates. The result was calculated as:

$$VOC = \frac{([g \text{ All Volatiles}] - [g \text{ Water}] - [g \text{ Exempt Compounds}])}{([liter \text{ Material}] - [liter \text{ Water}] - [liter \text{ Exempt Compounds}])}$$

5.3 Uncertainty of the Test Method

The relative standard deviation of the overall analysis is 10%. The expanded uncertainty U_m equals 2 x RSD. For further information please visit www.eurofins.dk/uncertainty.

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VOC EMISSION TEST REPORT

CDPH

22 March 2019

1 Sample Information

Sample name	FIS EM Plus 390 S
Batch no.	12/21B88B79
Production date	13/12/2018
Product type	Chemical anchor
Sample reception	24/01/2019

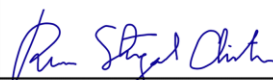
2 Brief Evaluation of the Results

Regulation or protocol	Conclusion	Version of regulation or protocol
CDPH	Pass	CDPH/EHLB/Standard Method V1.2. (January 2017)

Full details based on the testing and direct comparison with limit values are available in the following pages



Malene Laugesen
Analytical Service Manager



Rasmus Stengaard Christensen
Analytical Service Manager, MSc in Chemistry

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3 Applied Test Methods

3.1 General Test References

Regulation, protocol or standard	Version	Reporting limit VOC [$\mu\text{g}/\text{m}^3$]	Calculation of TVOC	Combined uncertainty ^a [RSD(%)]
EN 16516	October 2017	5	Toluene equivalents	22%
ISO 16000 -3 -6 -9 -11	2006-2011 depending on part	2	Toluene equivalents	22%
ASTM D5116-10	2010	-	-	-
CDPH	CDPH/EHLB/Standard Method V1.2. (January 2017)	2	Toluene equivalents	22%

3.2 Specific Laboratory Sampling and Analyses

Procedure	External Method	Internal SOP	Quantification limit / sampling volume	Analytical principle	Uncertainty ^a [RSD(%)]
Sample preparation	ISO 16000-11:2006, EN16402:2013, CDPH, AgBB, EMICODE	71M549810	-	-	-
Emission chamber testing	ISO 16000-9:2006, EN 16516:2017	71M549811	-	Chamber and air control	-
Sampling of VOC	ISO 16000-6:2011, EN 16516:2017	71M549812	5 L	Tenax TA	-
Analysis of VOC	ISO 16000-6:2011, EN 16516:2017	71M542808B	1 $\mu\text{g}/\text{m}^3$	ATD-GC/MS	10%
Sampling of aldehydes	ISO 16000-3:2011, EN 16516:2017	71M549812	35 L	DNPH	-
Analysis of aldehydes	ISO 16000-3:2011, EN 717-1, EN 16516:2017	71M548400	3-6 $\mu\text{g}/\text{m}^3$	HPLC-UV	10%
Sampling on Charcoal tubes	ISO 16200-1:2001	71M546081	60 L	charcoal	-
Analysis of Charcoal tubes	ISO-16200-1:2001	71M546081	20 $\mu\text{g}/\text{m}^3$	Headspace-GC/MS	10%

4 Test Parameters, Sample Preparation and Deviations

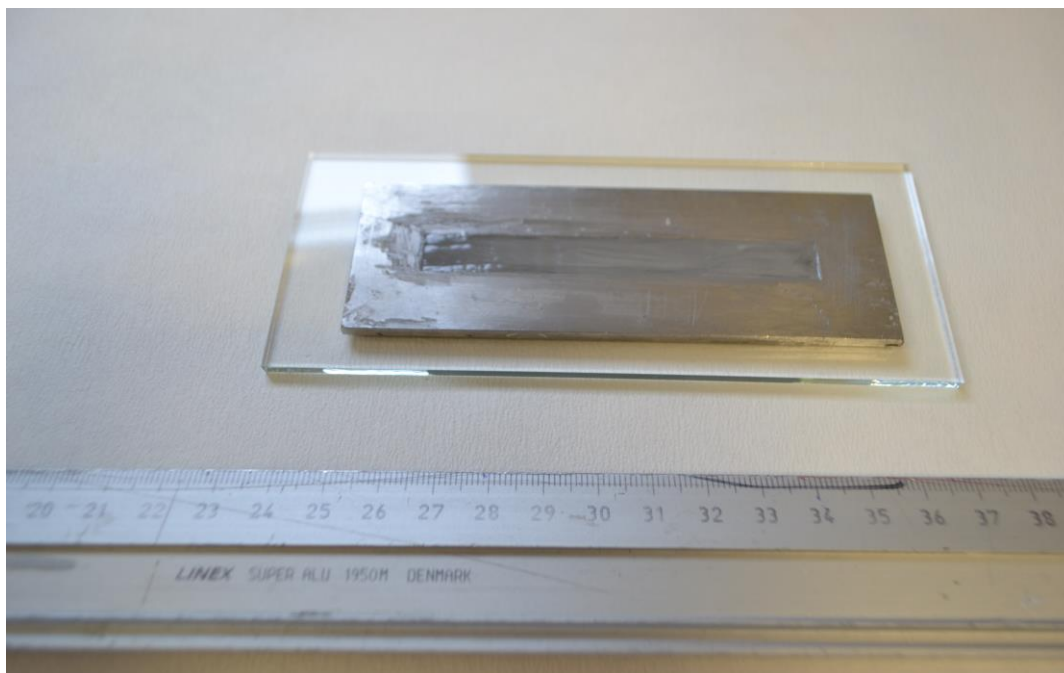
4.1 VOC Emission Chamber Test Parameters

Parameter	Value	Parameter	Value
Chamber volume, V[L]	119	Preconditioning period	-
Air Change rate, $n[h^{-1}]$	0.5	Test period	13/02/2019 - 13/03/2019
Relative humidity of supply air, RH [%]	50 ± 3	Area specific ventilation rate, q [m/h or m ³ /m ² /h]	71
Temperature of supply air, T [°C]	23 ± 1	Loading factor [m ² /m ³]	0.007
		Test scenario	Very small area

4.2 Preparation of the Test Specimen

The sample was applied onto a glass plate and drawn off over a model giving a 3 mm thick and uniform layer with a broadness of 10 mm.

4.3 Picture of Sample



4.4 Deviations from Referenced Protocols and Regulations

The air change rate was 0.5/h (not 1.0/h) during testing; the results were calculated back to normative air exchange rates for class room and small office.

The loading factor was less than the lowest factor of 0.3 m²/m³ that CDPH method specifies for testing; CDPH method does not specify a clear loading factor in any model room. Instead, the loading factor as specified in EN 16516 was applied both during testing and for calculation of the air concentration in office and classroom.

5 Results

5.1 VOC Emission Test Results after 11 Days

	CAS No.	Specific Conc. [µg/m³]	Specific SER [µg/(m²·h)]	Toluene eq. [µg/m³]	Toluene SER [µg/(m²·h)]
TVOC (C5-C17)tol. eq.		-	-	< 2	< 200
Aldehydes					
Formaldehyde	50-00-0	< 3	< 300	-	-
Acetaldehyde	75-07-0	< 3	< 300	-	-

5.2 VOC Emission Test Results after 12 Days

	CAS No.	Specific Conc. [µg/m³]	Specific SER [µg/(m²·h)]	Toluene eq. [µg/m³]	Toluene SER [µg/(m²·h)]
TVOC (C5-C17)tol. eq.		-	-	4.4	310
Aldehydes					
Formaldehyde	50-00-0	< 3	< 300	-	-
Acetaldehyde	75-07-0	< 3	< 300	-	-

5.3 VOC Emission Test Results after 14 Days

	CAS No.	Retention time [min]	ID-Cat	SER [µg/(m²·h)]	Classroom Conc. [µg/m³]	Office Conc. [µg/m³]	½ CREL [µg/m³]
VOC (C5-C17)							
n-Heptane	142-82-5	2.85	1	160	1.4	0.17	
TVOC (C5-C17) tol. eq.				< 200	< 2	< 1	
Aldehydes							
Formaldehyde	50-00-0		1	< 300	< 3	< 1	9
Acetaldehyde	75-07-0		1	< 300	< 3	< 1	70

6 Summary and Evaluation of the Results

6.1 Comparison with Limit Values of CDPH

Parameter	Test after 14 days			
	CAS No. Single compounds	Concentration in Classroom [µg/m³]	Concentration in Office Room [µg/m³]	½ CREL [µg/m³]
TVOC (C5-C17)tol. eq.	-	< 2	< 1	-
Single compounds (with defined CREL values)				
None determined	-	-	-	-
Formaldehyde	50-00-0	< 3	< 1	≤ 9
Acetaldehyde	75-07-0	< 3	< 1	≤ 70

6.1.1 Conversion of Emission Rates to CDPH Reference Room Concentrations

The CDPH method requires calculation of the measured emission rates into concentrations in given reference rooms. The equation and parameters figured below have been applied to calculate the concentrations in an office room or a classroom as required in the CDPH. The area used in the calculation varies depending on the expected usage of the product and therefore several entries can be found. Small and Very Small areas are not provided within the CDPH but are adapted from definitions given in EN 16516 and ISO 16000-9.

$$C_{\text{Calculated}} = \frac{SER_A \cdot A}{n \cdot V}$$

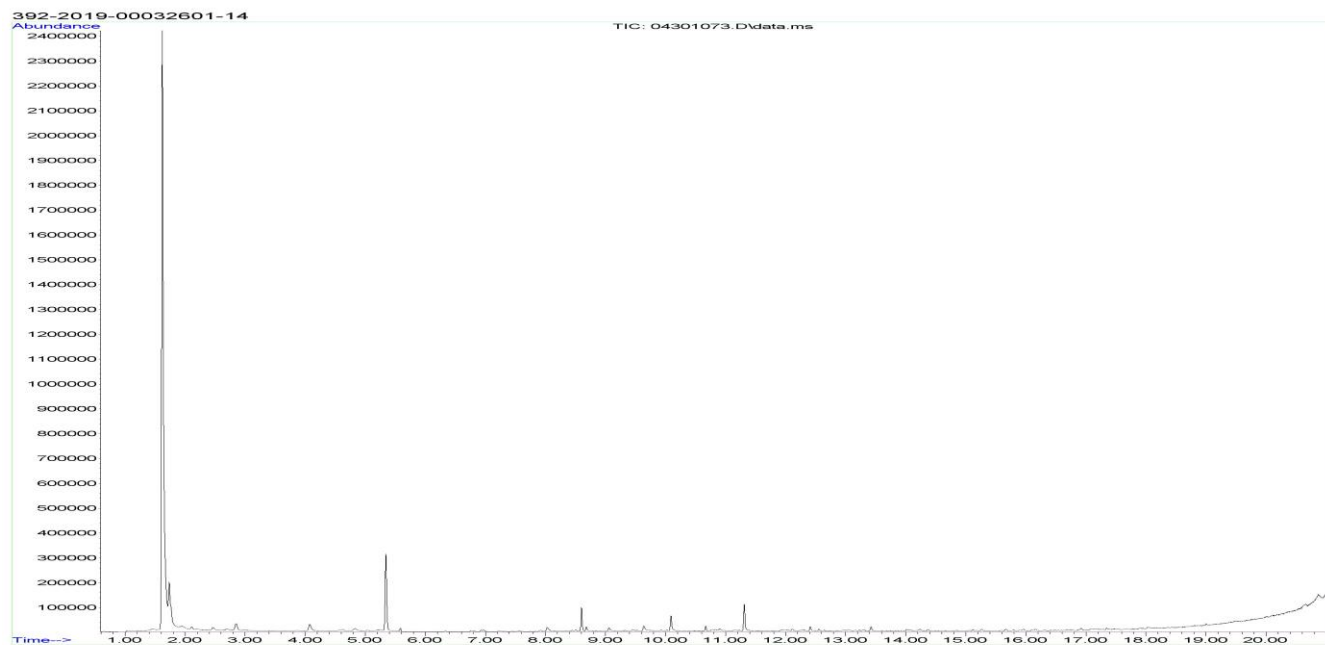
		Classroom parameters	Office Room parameters
SER	Area specific emission rate, µg/(m²h)	As tested	As tested
n	Air change, h ⁻¹	0.82	0.68
V	Volume of reference room, m³	231	30.6
A	Floor area, m²	89.2	11.1
	Walls area, m²	94.3	33.4
	Ceiling and Wall, m²	183.8	N/A
	Door and Millwork, m²	1.89	1.89
	Desk or Chair, units	27	1
	Very Small areas, m²	1.62	0.021
	Small areas, m²	11.55	1.53

The results are only valid for the tested sample(s).

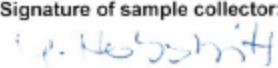
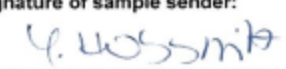
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7 Appendices

7.1 Chromatogram of VOC Emissions after 14 Days



7.2 Chain of Custody

Name of the product: FIS EM Plus		Type of product: Injection mortar	
Model / Program / Series: -		Batch N°.: 12/21B88B78	
Article N°.: 544154		Date of batch production: 13.12.2018	
Name of the manufacturer at the place of sampling (address / stamp): fischerwerke GmbH & Co. KG, Otto-Hahn-Str. 15, 79211 Denzlingen, Germany		Manufacturer (if deviating from company's name at the place of sampling):	
Sample collector (Name, company, telephone): Yvonne Herbstritt, fischerwerke GmbH & Co. KG, 0049 7666 902 2901		Signature of sample collector: 	
Sample is taken from	☒ the ongoing production stocks	Date of sampling: 13.12.2018	
Number of Samples	3 cartridges	Time: 14:50	
Where had the product been stored prior to sampling?	☒ Production ☐ Store ☐ Miscellaneous	How had the product been stored prior to sampling?	☐ open ☒ in the stack ☐ wrapped up
	Place of storage: No storage		Packing material: PE cartridge
Further links in chain of custody (Name, function, company, telephone) Yvonne Herbstritt, Chemical Compliance, fischerwerke GmbH & Co. KG, 0049 7666 902 2901		Signature 	
Further links in chain of custody (Name, function, company, telephone)		Signature	
Sample sender (Name, company, telephone): Yvonne Herbstritt, fischerwerke GmbH & Co. KG, 0049 7666 902 2901		Signature of sample sender: 	
Date and time of sending:		Shipment details/Carrier:	
Where had the product sample been stored prior to sending?	☐ Production ☐ Store ☒ Miscellaneous	How had the product sample been stored prior to sending?	☒ open ☐ in the stack ☐ wrapped up
	Place of storage: goods issuing department		Packing material: PP cartridge
Laboratory receiving details (date, condition of package and sample, assigned lab no.):			
Receptionist, Eurofins Product Testing A/S:		Signature of receptionist:	

The results are only valid for the tested sample(s).

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7.3 How to Understand the Results

7.3.1 Acronyms Used in the Report

- < Means less than
- > Means bigger than
- * Not a part of our accreditation
- Please see section regarding uncertainty in the Appendices.
- § Deviation from method. Please see deviation section
- a The method is not optimal for very volatile compounds. For these substances smaller results and a higher measurement uncertainty cannot be ruled out.
- b The component originates from the wooden panels and is thus removed.
- c The results have been corrected by the emission from wooden panels.
- d Very polar organic compounds are not suitable for reliable quantification using tenax TA adsorbent and HP-5 GC column. A high degree of uncertainty must be expected.
- e The component may be overestimated due to contribution from the system
SER Specific Emission Rate.

7.3.2 Explanation of ID Category

Categories of Identity:

- 1: Identified by comparison with a mass spectrum obtained from library and supported by other information and quantified through specific calibration.
- 2: Identified by comparison with a mass spectrum obtained from library and supported by other information. Quantified as toluene equivalent.
- 3: Identified with a lower match by comparison with a mass spectrum obtained from a library. Quantified as toluene equivalent.
- 4: Not identified, quantified as toluene equivalent.

7.4 Description of VOC Emission Test

7.4.1 Test Chamber

The test chamber is made of stainless steel. A multi-step air clean-up is performed before loading the chamber, and a blank check of the empty chamber is performed.

The chamber operation parameters are as described in the test method section. (EN 16516, ISO 16000-9, internal method no.: 71M549811).

7.4.2 Expression of the Test Results

All test results are calculated as specific emission rate, and as extrapolated air concentration in the European Reference Room (EN 16516, AgBB, EMICODE, M1 and Indoor Air Comfort).

7.4.3 Testing of Carcinogenic VOCs

The emission of carcinogens (EU Categories C1A and C1B, as per European law) is tested by drawing sample air from the test chamber outlet through Tenax TA tubes after the specified duration of storage in the ventilated test chamber. Analysis is performed by ATD-GC/MS (automated thermal desorption coupled with gas chromatography and mass spectroscopy using 30 m HP-5 (slightly polar) column with 0.25 mm ID and 0.25 μ m film, Agilent) (EN 16516, ISO 16000-6, internal methods no.: 71M549812 / 71M542808B).

All identified carcinogenic VOCs are listed; if a carcinogenic VOC is not listed then it has not been detected. Quantification is performed using the TIC signal and authentic response factors, or the relative response factors relative to toluene for the individual compounds.

This test only covers substances that can be adsorbed on Tenax TA and can be thermally desorbed. If other emissions occur, then these substances cannot be detected (or with limited reliability only).

7.4.4 Testing of VOC

The emissions of volatile organic compounds are tested by drawing sample air from the test chamber outlet through Tenax TA tubes after the specified duration of storage in the ventilated test chamber. Analysis is performed by ATD-GC/MS using HP-5 column (30 m, 0.25mm ID, 0.25 μ m film).

This test only covers substances which can be adsorbed on Tenax TA and can be thermally desorbed. If emissions of substances outside these specifications occur then these substances cannot be detected (or with limited reliability only).

7.4.5 Testing of Aldehydes

The presence of aldehydes is tested by drawing air samples from the test chamber outlet through DNPH-coated silicagel tubes after the specified duration of storage in the ventilated test chamber. Analysis is performed by solvent desorption and subsequently by HPLC and UV-/diode array detection.

The absence of formaldehyde and other aldehydes is stated if UV detector response at the specific wavelength is lacking at the specific retention time in the chromatogram. Otherwise it is checked whether the reporting limit is exceeded. In this case the identity is finally checked by comparing full scan sample UV spectra with full scan standard UV spectra.

7.4.6 Testing of Charcoal tubes

The presence of low boiling VOC is tested by drawing air samples from the test chamber outlet through charcoal tubes after the specified duration of storage in the ventilated test chamber. Analysis is performed by solvent desorption and subsequently by HS-GC/MS using a stabilwax column. This test only covers substances which has a CREL value and are not possible to sample on Tenax tubes.

7.5 Quality Assurance

Before loading the test chamber, a blank check of the empty chamber is performed and compliance with background concentrations in accordance with EN 16516 / ISO 16000-9 is determined.

Air sampling at the chamber outlet and subsequent analysis is performed in duplicate. Relative humidity, temperature and air change rate in the chambers is logged every 5 minutes and checked daily. A double determination is performed on random samples at a regular interval and results are registered in a control chart to ensure the uncertainty and reproducibility of the method.

The stability of the analytical system is checked by a general function test of device and column, and by use of control charts for monitoring the response of individual substances prior to each analytical sequence.

7.6 Accreditation

The testing methods described above are accredited on line with EN ISO/IEC 17025 by DANAK (no. 522). This accreditation is valid worldwide due to mutual approvals of the national accreditation bodies (ILAC/IAF, see also www.eurofins.com/galten.aspx#accreditation).

Not all parameters are covered by this accreditation. The accreditation does not cover parameters marked with an asterisk (*), however analysis of these parameters is conducted at the same level of quality as for the accredited parameters.

7.7 Uncertainty of the Test Method

The relative standard deviation of the overall analysis is 22%. The expanded uncertainty U_m equals 2 x RSD. For further information please visit www.eurofins.dk/uncertainty.